12. The Requirements

What the mechanism forces, what the Moon has, and what remains unresolved

Zohar Duran · ORCID 0009-0005-3194-9387 · 15 October 2026

This section defines the dynamical conditions, the physical parameters and the mass balances that make up the formation environment of the Moon in the LEF (Lunar Elevation & Fission) model.

The environment from which the requirements are derived

Every mechanical requirement of the model is derived from the physical picture of the time of the Moon's formation. The table below collects the numerical and dynamical baselines that form the common ground for all the geophysical mechanisms; where it is stated that "the mechanism requires" a given condition, this means that the condition follows directly from these parameters.

The model draws a strict distinction between the conjectured formation environment and the metrics of the modern Moon as they are measured today. Observations of a later state may not be used as initial conditions without the mediation of a model that physically links the two states. The present Moon has undergone billions of years of cooling, residual volcanic activity and surface impact metabolism; modern morphological properties may be an ancient product of the moment of formation or an independent later addition, and the two must be kept fully apart.

Table of parameters, values and methodological status of the scenario environment

Physical domain	Scenario environment and firm numerical parameters	Methodological status and role in the model
Time and chronology	Start of the scenario about 4.5 billion years ago (Ga); a time window of up to about 600 million years (Ma) is kept for testing. In the series' rough timeline exactly 600 million years pass from the start of the bulge's development	Scenario assumption. Geological age differences between the crust and the basalt are not a measurement of the duration of the mechanical construction; a basalt age dates a later

Physical domain	Scenario environment and firm numerical parameters	Methodological status and role in the model
	to the release of the Moon 3.873 billion years ago: about 400 million years for the development of the bulge and the pear, about 100 million years for the flattening stage, and about 100 million more years up to the moment of physical release.	volcanic eruption, not the formation of the primary sialic crust lying beneath it.
Initial parent-body mass	$M_0 = M_E + M_L = 6.0455 \times 10^{24}$ ק"ג The "parent body" is defined as the Earth before the separation of the Moon, so its mass is exactly equal to the mass of the modern Earth (M_E) plus the mass of the modern Moon (M_L).	Firm input. Before the dynamical transfer of material begins, the mass of the future Moon is wholly and indistinguishably contained in the mass of the parent body.
Reference masses after the transfer	$M_E = 5.972 \times 10^{24}$ ק"ג $M_L = 7.35 \times 10^{22}$ ק"ג	Series inputs. Along the mass-transfer path the mass of the mechanical connection and the components of material lost to space must be tracked quantitatively; at intermediate stages part of the physical mass is no longer in the Earth, yet is not yet gathered in the Moon.
Parent-body radius	- Volumetric radius (sphere of equal volume): 6,517 km. - Equatorial radius in the initial-structure example: 6,721 km. The equatorial radius is larger than the volumetric one because a body spinning at high speed bulges centrifugally at its equator.	Different physical values. These quantities cannot be exchanged or substituted for one another within the same dynamical equation.
Rotation rate (day length)	5.5 hours at the start in the reference example; exactly 5.77 hours at the moment of physical release. The lengthening of the pear body mechanically slows the rotation (an increase in the moment of inertia), while the concentration and sinking of the core speeds it up (Paper 2).	A variable value within the admissible range of Paper 2. The rotation rate changes as a function of the change in angular momentum (which stays constant in the absence of an external torque) and of the change in the moment of inertia (a measure of the distribution of mass relative to the axis). The mechanism is analogous to an ice skater: mass moved outward

Physical domain	Scenario environment and firm numerical parameters	Methodological status and role in the model
		(the lengthening of the pear) slows the rotation, and mass drawn inward (heavy iron sinking to the core) speeds it up.
Distance of the corotation line	- At a rotation of 5.5 hours: 15,818 km. - At the moment of release (5.77 hours): 16,331 km (for a massless test particle), 16,399 km (in a two-body calculation that allows for the Moon's mass), and 17,287 km along the release latitude of 22.5° (Papers 2 and 4).	Dynamical states and definitions from Paper 2. The Moon is released when its centre of mass reaches exactly the corotation line (the distance at which a body turning synchronously with the Earth moves at free orbital speed; inside it the body is not fast enough to orbit and must be held up by a mechanical connection). With a day of 5.77 hours the line lay considerably closer to the surface, at a distance of 16,399 km (about 2.6 Earth radii). Corotation is not a sufficient condition for release: the mechanical connection must give way, and the released body must hold together structurally against the Earth's tidal forces.
Gravity field	The dynamical field depends locally on geographic position, on the mass of material transferred and on the internal density structure.	Initial equatorial estimate: about 8.25 m/s^2 on the parent body. This value does not represent the internal gravity on the Moon; it is lower than the surface gravity of the modern Earth because the equator of the parent body lay farther from its centre of mass, and the extreme rotation rate centrifugally offset part of the gravitational pull.
Dimensions of the early lunar body	- A radius of about 2,270 km at a mean density of 1,500 kg/m^3. - A radius of about 1,890 km at a mean density of 2,600 kg/m^3 (for the full target mass).	Illustrative geometric states from Paper 3. The values represent the theoretical radius of the body at each density level; the hotter, lighter and richer in volatile gaseous components the body

Physical domain	Scenario environment and firm numerical parameters	Methodological status and role in the model
		is, the larger its volume and dimensions. The modern Moon is denser and smaller than both, and these data must not be read as a continuously computed contraction history.
Internal structural organisation	A continuous, ongoing process of differentiation of metal and silicate phases: the heavy iron sinks gravitationally toward the centre, and the light silicate floats upward, both within the volume of the Earth and within the developing lunar body.	Consistency constraint. This mechanism must be reconciled with the values of the moment of inertia and with the chronometric data of the radioactive clocks. The existence of a small modern lunar core is not a direct measure of the structural state of the Earth's core at that early point in time.
Source of mass and material	Simultaneous feed from two terrestrial sources: the deep mantle reservoir, and in parallel the outer sialic shell layer (SiAl) of the parent body, which rises together with the body and serves as the surface of the lunar dome.	One body and a unified mass balance (Paper 3). Sial is the light outer layer of the Earth, rich in silicon and aluminium, lying above the denser sima layer (SiMa).
Ejection mechanism	Deep mantle material rising vertically through dedicated openings and conduits in the lithospheric shell.	Mechanical definition. The physical ejection is not defined or computed as material torn or removed from the upper shell itself.
Volatile components and materials	A broad and massive ancient inventory of volatiles held within the volume of the Earth. Volatiles are defined as substances that pass easily into the gas phase — water ($\mathrm{H_2O}$) above all — and whose presence physically lowers the melting point and the mechanical stiffness of the material that carries them.	No verified concentration. There is no verified concentration value for the geological system as a whole; the state of the phase — dissolved, dense supercritical fluid or gas phase — is set solely as a function of lithostatic pressure, heat and chemical composition.
Atmosphere and thermal cooling	Conductive and convective heat transfer outward to space; the existence of an internal dynamical cycle of rise, cooling and renewed sinking of the volatile components.	Energy-balance constraint. The cooling of the body requires a net emission of heat (Net Radiation) to outer space; an internal cycle of material does not by

Physical domain	Scenario environment and firm numerical parameters	Methodological status and role in the model
		itself change the total energy balance, since heat moved from one thermal level to another within the boundaries of the system is not defined as heat lost by it.
Rheology and viscosity	Rheology defines the way in which geological material flows or mechanically resists deformation (Strain). The lithospheric shell is cold and stiff at the surface, but mobile and viscous at its base; the working interface with the interior is the layer of lowest mechanical stiffness.	Structure-preservation constraint (Paper 3). The model requires an outer shell stiff enough to preserve the shape of its craters structurally — with a stiff viscosity of $\gtrsim 10^{23}$ שנייה · פסקל — above a mobile, viscous interior. The unit of viscosity ($\mathrm{Pa{\cdot}s}$) measures resistance to flow; at this threshold value the crust does not flow appreciably over the relevant time spans, so the crater keeps its geometric morphology.
Parameters of the lithospheric shell	A physical thickness of about 18 km, a material density of 2,800 kg/m³, spread over the surface of a hemisphere whose geometric base is 6,000 km (a total area of 5.65×10^{13} מ"ר).	Crustal mass balance. The mass of the crust is about 2.85×10^{21} ק"ג (exactly 3.9% of the total mass of the modern Moon). This crust closes onto the far hemisphere (Far Side) as a thickened crust 50–60 km thick (Paper 3). The thickening is defined as a mechanism of change of area: the shell envelope of the original large dome was gathered and contracted onto the smaller area of the far hemisphere of the finished Moon, which increased its linear thickness about threefold.
The connected geometry (the pear)	The lunar dome is the tip region of the pear, made of the Earth's own material, with a surface width of about 9,000 km. The base of the pear and its internal conduits lie near Kiritimati Island (1.9° N, 157.5° W) in the central Pacific Ocean, directly above the LLVP	Properties of the connection interface. The geodynamic connection is broad and at a high temperature; the contact disc becomes a cap located on the near side (Near Side) of the Moon, whose geometric dimensions

Physical domain	Scenario environment and firm numerical parameters	Methodological status and role in the model
	(Large Low-Shear-Velocity Province) anomaly — the large low-velocity region in the lowermost mantle. The base of the dome faces the parent body, and at the moment of release the geometric contact area is a disc 3,000 km across (equal to one quarter of the area of the original base).	correspond to an angular size of about 41° (the edge of the structure lies 41° from the centre of the near side) in the earlier calibration rules, and to an angular size of exactly 31° in the updated rule of Paper 3 (§B.8). The lunar body peels mechanically from the sial layer on which it sat and carries it with it as part of its structure.
Stratigraphic order of events (the scenario)	The magmatic ejection and the rise of the mechanical dome take place in overlap and simultaneously; the ejection flow decays and stops before the dome envelope closes completely; the structural closure of the dome is completed before the physical detachment of the body.	Chronological constraint of Paper 3. The durations of the dynamical transition stages cannot be measured directly.
The surface of the modern Moon	Composed of the primary anorthositic crust, impact basins (Impact Basins), the basaltic fill of the maria (Mare Basalts) and later tectonic changes.	Separation criterion. The model requires a strict separation between geological properties originating in the formation mechanism and residual structures preserved or added at the surface over the later history.

Summary of the geodynamic picture of the model

The data collected in this table set the physical stage for the following requirements of the model:

1. An early Earth spinning at high speed (a day length of 5.5–5.77 hours) and heavier by a mass equal to the mass of one modern Moon.

2. An asymmetric, pear-like geodynamic structure made entirely of the Earth's own physical material, whose western tip region consolidates and closes as the body of the Moon.

3. A stiff, highly viscous lithospheric shell carried vertically upward and serving as the outer surface of this tip.

4. Release and free gravitational separation taking place exactly within the dynamical range of the corotation line.

Every mechanical requirement presented later in the paper is intended to test and analyse what physical traces this dynamical stage left on the surface of the Moon and on the surface of the Earth.

Abstract

- **The geodynamic construction mechanism:** In the LEF model the Moon was built directly from the Earth's material while still attached to it through a physical interface. Deep mantle material was fed continuously into the developing lunar body, and in parallel the outer lithospheric shell layer of the parent body that lay above the region rose vertically together with the body as the surface of the dome structure, and consolidated into the lunar crust.
- **Definition of the requirement and test criterion:** A requirement is defined as a geophysical or geometric property that the Moon (or the Earth) must display if it formed by this mechanism. Each requirement is set out in three firm parts: the mechanical condition the model must meet, the quantitative measure that tests it, and the measurement result that would not fit the requirement (a testable condition). The analysis rests on separate requirements rather than on a cumulative agreement score, since several agreements may represent one physical fact observed in several ways, and a clear-cut mismatch in one requirement is examined on its own merits and concerns that requirement and its mechanism, independently of the number of other agreements.
- **The range of the seven core requirements:** The paper examines seven fundamental requirements:
  1. A full isotopic affinity to terrestrial sources.
  2. A transfer of metallic phase in an amount limited by the existence of a small lunar core.
  3. A stratigraphic asymmetry of crustal thickness.
  4. A morphological facing direction tied directly to the side of the connection.
  5. A chronology of cooling and of the preservation of crustal structures.
  6. The availability of volatiles at the time of the mass transfer.
  7. A mathematical correlation between the distribution of crustal mass and the offset of the centre of mass.

- **Limits on the interpretation of modern data:** Present-day geological data support the existence of the general morphological properties, but do not by themselves confirm the mechanism of their formation:
  - The ages of the mare basalts do not fix the age of the anorthosite crust lying beneath them.
  - The water content measured in sampled magma does not directly represent the total volatile inventory at the time of construction.
  - The existence of synchronous rotational locking today (one revolution per rotation) is not proof of continuous synchronicity since the moment of detachment.
- **The model crustal balance and the centre-of-mass offset:** The Moon's centre of mass is offset relative to its geometric centre. A light crust of increased thickness on one side shifts the centre of mass toward the opposite side. In a calibrated numerical example, a computed offset of 1.24–3.11 km is obtained for density contrasts (the density difference between the crustal material and the mantle material beneath it) in the range 300–750 kg/m³. This is a mathematical sensitivity range and not an independent prediction of the empirical offset.
- **Division of the crustal mass between the sides of the body:** A mass balance shows that the shell of the dome, whose linear thickness grew threefold during the structural closure, fully supplies the crustal mass of the far side (Far Side). This mass is equal to about 4% of the total lunar mass, a value that matches exactly the gravity findings of the GRAIL mission (NASA). The crust of the near side (Near Side) is a separate member: it is made of the sial material of the terrestrial base that peeled off together with the body, was fully melted above the contact zone, and was thermally smoothed by the deep heat; this crust displays its minimum thickness directly beneath the mare basins, which are defined as structural openings.

`What this paper proposes`

The scientific theses and predictions

- **Isotopic agreement of the source material:** The material of the Moon must be identical to the material of the Earth. Empirical measurements of the oxygen isotopes ($\delta^{18}\mathrm{O}$ and $\delta^{17}\mathrm{O}$) show full agreement within a strict precision range of $-1 \pm 5\ \mathrm{ppm}$ (parts per million).
- **The core-volume limit:** The mass flow rising along the pear carried light silicates and only a small amount of metallic phase. The lunar seismic core has a measured radius

of only 330 km, amounting to about 1.4% of the total lunar mass.

- **Crustal-thickness asymmetry:** The crust must be thicker opposite the old region of mechanical connection, at the location where the crown of the dome carried the shell and the far side froze first; the data of the GRAIL mission confirm that the far-side crust is considerably thicker.
- **Fixing of the surface (preservation of orientation):** The lunar body must keep its original side of connection facing directly toward the Earth.
- **Chronology of thermal closure:** The far-side crust must undergo structural closure and be preserved structurally before the hot feed flow of the near side decayed.
- **Presence of volatile components:** The process of mass feeding and the geodynamic flow needed the presence of volatiles to lower the viscosity. Melt inclusions (Melt Inclusions) — tiny droplets of magma sealed hermetically inside crystals before the volcanic eruption — show a water concentration of 615–1,410 ppm.
- **The empirical centre-of-mass offset:** The Moon's centre of mass is in fact offset by 2 km toward the near side. The crustal shape derived from the model's formulas yields a theoretical offset of 1.2–3.1 km for standard density-contrast values.

Organisation of the data, the mechanisms and the full numerical parameters

Subject	Finding	Source	Status in the model
Isotopic composition	Complete agreement of the oxygen isotopes within a range of -1 ± 5 ppm.	Laboratory measurements of samples	Requirement 1 (confirmed)
Dimensions of the lunar core	Firm seismic radius: 330 km. Core mass: about 1.4% of the total mass of the body.	Seismic data from the Apollo networks	Requirement 2 (confirmed, evidence of silicate flow)
Crustal asymmetry	The far side is considerably thicker than the near side; the far-side crustal mass is about 4% of the lunar mass.	Global gravity map (GRAIL / NASA)	Requirement 3 (confirmed, a dome crown that froze)

Subject	Finding	Source	Status in the model
Buried volatile concentration	Melt inclusions inside closed crystals contain a water concentration of 615–1,410 ppm.	Chemical analysis of volcanic crystals	Requirement 6 (confirmed, evidence of volatiles at the source)
Centre-of-mass offset (measured)	A linear deviation of 2 km toward the near side relative to the geometric centre.	Satellite altimetry and gravity measurements	Requirement 7 (confirmed)
Centre-of-mass offset (model)	A computed theoretical value of 1.2–3.1 km (exact sensitivity range: 1.24–3.11 km) for density contrasts of 300–750 kg/m³.	Mass-balance equations of the dome	Mathematical consistency of the model
Concentric zonal pattern	The lunar surface is arranged in three concentric zones around the centre of the near side: the contact (exposed last), the base ring, and the far side (the old surface of the dome).	Map of the surface gravity gradient (GRAIL)	Requirement 8 (an eighth property, new in Paper 12)
Residual pear mechanisms	Seven geodynamic properties: the self-gravity of the pear, the return mechanism, the Pacific LLVP, the KREEP component on the near side, the terrestrial core as a gauge, the absence of foreign material, and an increased crater density at the poles that depends on crater diameter.	Combined morphological and seismic analysis	Presented together in section C.6
Opening mechanisms and the moons of Mars	Three dynamical properties: 34 mare basins defined as structural openings and a source of the returning material, two material outlets beneath the South Pole–Aitken basin with a dome to their south, and the two moons of Mars (Phobos and Deimos) fired from a single ejection conduit on the Earth's equator.	Ejection models and dynamical orbits	Presented together in section C.7

> **Structural integration of the requirements**
>
> The model does not treat the seven lunar properties as seven separate and independent statistical proofs. The crustal thickness, the centre-of-mass offset, the mare basins and part of the record of geological ages are defined physically as a single structural phenomenon reflected in four different modes of observation.
>
> Each morphological agreement is presented together with the dedicated calculation protocol needed to turn it into a firm quantitative test (such as multi-system isotopic balances, rates of metallic mass transfer, and combined crust–offset fits). These equations, together with their exact physical mismatch thresholds, are set out in section D.3. The central task facing the model is to show that a single, fixed mathematical set of parameters is able to produce all these outcomes at once.

Introduction — from the mechanism to the traces

Methodological characterisation and aim of the analysis

- **Linking the physics to the data:** Once the construction path and the geodynamic mechanism have been defined in Papers 2–3, the central role of Paper 12 is to link the physics of the scenario to empirical data that can be tested independently. The process of the Moon's formation must leave defined physical and morphological signs: in its chemical and isotopic composition, in its internal structure, in the asymmetry of its crustal thickness, in the orientation in which it faces its planet (the Earth), and in the chronological order in which its surface underwent cooling and thermal closure.
- **Separating the Mars case:** Paper 10 is a separate and independent test applied to the Mars system. A mathematical success or a physical difficulty arising within Paper 10 does not change the firm parameters set for the terrestrial case in the LEF model.

Definition of the structural requirement tool

The theoretical tool that guides the paper is the requirement. Each requirement is defined firmly on the basis of three components: the geophysical property that must hold if the Moon formed by the LEF mechanism, the exact empirical measurement that tests it, and the numerical or structural result that would not fit the requirement.

The requirements are collected separately, without weighting a cumulative mathematical score, for two firm methodological reasons:

1. Mutual dependence of structures (a single distribution of material): The morphological properties of the Moon are not independent of one another. A crust thickened on one side (the far side), a centre of mass offset toward the opposite side (the near side), the concentration of mare basins on the thin side, and part of the record of geological ages — all of these are, in the model, a direct result of a single mechanical distribution of material. Summing them as separate points in a cumulative score would mean counting one physical fact twice and repeatedly.

2. The principle of separate testing (each requirement is testable): A weighted score allows many agreements and chance correlations to outweigh a decisive disagreement in a single parameter. By contrast, a requirement defined with its own mismatch condition is examined on its own merits: a clear mismatch in one requirement is recorded against that requirement, and is not hidden by the other agreements. Part B reports the result of each test separately (section B.9).

Chronological classification of the model's outcomes

The order of presentation in the paper places the definition of the mechanical requirement first, and the empirical observation after it. The structure is meant to help the reader examine the logical consistency, but it does not turn known data into predictions (Predictions) recorded before they were discovered. To preserve scientific accuracy, the model distinguishes firmly between three kinds of outcome:

- **Derived outcome:** A value or trend derived directly from the physical equations under defined base assumptions of the scenario.
- **Postdiction (Postdiction):** A fit of the model's parameters to existing empirical observations made in science.
- **Prediction:** An independent structural or numerical claim of the model that has not yet been tested against empirical data.

Consistency-check constraint: A measurement made by outside researchers in the scientific literature, and which was used at the same time as input for choosing a free parameter within the model, is defined as an internal consistency check only (Consistency Check), and must not be regarded as independent empirical support for the mechanism.

Structure of the four parts of the paper

The table below defines the central mechanical question that each part of Paper 12 firmly settles.

Part of the paper	The structural question the part settles
PART A	What exactly does the geodynamic scenario require, and on which physical assumptions does each requirement rest?
PART B	What has actually been measured in the empirical data, and what can be inferred from each geophysical measurement?
PART C	Which further signs and mechanisms deserve examination, without counting them as existing, confirmed outcomes of the model?
PART D	Which outcomes would not fit the requirements, what remains open, and what is the novel element of the work?

PART A

The seven requirements and their assumptions

Part A sets out what the construction path of the LEF model requires of the Moon. In the model the Moon was built while attached to the Earth: deep material was fed upward into a growing body, and the outer shell of the Earth above that region rose together with it as the surface of a dome. If the Moon formed in this way, it cannot look like just any body; it must carry particular traces. Each of the seven sections below names one such trace (R1–R7), explains why the model needs it, and describes what kind of measurement will test it. The measurements themselves are examined in Part B, in the same order, so that every section of Part A has a parallel section in Part B.

The problem and the method of derivation

- **Limits of the spatial geometry:** The geometry of a connected body does not by itself fix the chemical composition, the rate of thermal cooling or the morphological details of the crust. A body attached to the Earth through a broad, hot connection has a near side and a

far side, a fed region and a closed region; but parameters such as the mass of metal transferred, the freezing rate of each wall, or the volume of water that moved with the feed flow — are not derived from the structural shape alone.

- **Methodology of physical integration:** Each mechanical requirement is accompanied by the complementary physical assumption needed to derive it quantitatively. This approach makes it possible to separate firmly the components that follow directly from the geometry from those derived from the additional dynamical assumptions. An empirical test that is not passed points precisely to the specific component of the model with which it does not fit.
- **The range of measurements (R1–R7):** The seven requirements cover seven separate fields of measurement: isotopes (R1), core size (R2), crustal asymmetries (R3), the orientation of the Moon toward the Earth (R4), the order of cooling and preservation (R5), the volatile inventory (R6), and the mathematical relation between the distribution of the crust and the centre-of-mass offset (R7).

A.1 R1 — Affinity to terrestrial material

What it requires

- **Integration with terrestrial reservoirs:** In the geodynamic scenario the source of all the material that builds the Moon is the Earth. For this reason the model requires full chemical and structural agreement with the terrestrial reservoirs from which the material came, after quantitative allowance for the processes of melting, phase separation, evaporation and geological differentiation.
- **Definition of a reservoir (Reservoir):** A large body of material with its own distinctive character and chemical composition — such as the deep mantle of the Earth (Mantle), or its outer sialic shell (SiAl).
- **The terrestrial-mixture constraint:** Since the model derives the lunar material from these two defined terrestrial reservoirs (the deep feed flow and the rising shell), the composition of the final lunar body must be mathematically attainable as a direct mixture of these two sources.

Why the model needs it

- **The obligation of a chemical signature:** This property is the most direct physical consequence of the connection mechanism: if the mass of material was moved from the Earth, it must carry the chemical fingerprint of the parent body.

- **Isotopes as a label of origin:** The main system used for this identification is isotope ratios. Isotopes are polymorphic forms of the same chemical element that differ from one another only in the number of neutrons in the nucleus, and hence in their atomic mass. Isotope ratios show a firm variation between different planetary bodies in the Solar System, and cannot be changed significantly after the stage at which the body consolidated. For this reason they are an absolute label of origin.

What it does not require

- **No absolute equality of concentrations:** A common geological source does not dictate mathematical equality in every concentration of a chemical element or in every single isotope ratio. Differences that are allowed and expected under the scenario include:
  - **Radiogenic isotopes:** Isotopes formed as a result of natural radioactive decay. Two reservoirs that have been physically separated and hold different amounts of the parent element (Parent isotope) will accumulate, over geological time, different amounts of the daughter element (Daughter isotope), even if the moment of separation took place from an entirely identical isotopic composition.
  - **Mass-dependent fractionation (Mass-dependent fractionation):** Mechanical sorting and sifting of light isotopes against heavy isotopes during processes of evaporation, melting or crystallisation (the light isotopes escape from the system or move within it more easily, much as the light phase of an evaporating liquid evaporates first).

How it is tested

- **The multi-system balance criterion:** The test determines whether a quantitative balance of contributions from the two terrestrial sources is able to explain at the same time several different isotopic systems together with the bulk chemical composition (Bulk composition).
- **Limitation of a single measure:** A measurement of oxygen isotopes alone is not sufficient confirmation of the overall balance. The oxygen value is the best-known empirical agreement in science — the isotopic difference between the Earth and the Moon is $-1 \pm 5\ \mathrm{ppm}$ (parts per million; as set out in section B.1) — but the mechanical requirement R1 will be defined as confirmed only if a single mathematical mixture of the two terrestrial reservoirs, under the allowed physical processes, explains at once the oxygen values, the other isotopic systems and the global chemistry of the body.

A.2 R2 — A small metallic core

What it requires

- **A limited volume-and-mass constraint:** The Moon has a small, restricted metallic core: its seismic radius is only about 330 km, amounting to about 1.4% of the total lunar mass (as set out in section B.2). This figure is defined as an unusually small fraction for a planetary body of its dimensions.
- **The feed-flow constraint:** In the model the Moon was fed from below mainly with silicate material ($\mathrm{SiO_2}$), containing only a small amount of metallic phase.
- **The two-sided bounds of the requirement:** The internal transfer mechanism must carry a mass of material able to supply the metal inventory needed to build the existing core (lower bound), without transferring a larger fraction of metal than allowed, which would increase the volume of the core beyond the observed value (upper bound).

What the amount of metal depends on

The volume of metallic mass transferred through the geodynamic medium is set as a function of four firm physical parameters at the source:

1. The fraction of free metal: The concentration of unbound metallic phase present within the feeding material.

2. The dimensions of the metal droplets (Droplet Rheology): The physical size of the metal droplets within the flow. Massive, heavy droplets sink gravitationally back through a slow rising flow, while tiny droplets are carried upward together with the flow (the physical mechanism is analogous to fine grains of sand that stay suspended and in motion in a fast-flowing stream, while heavy pebbles sink to the bed).

3. The speed of the vertical flow: The rise speed of the silicate flow against the hydrodynamic settling speed of the metal droplets.

4. The oxidation state (Oxidation State): Determines the proportion of iron (Fe) locked chemically within the crystal structure of silicate minerals, instead of being present as a free metallic phase. Iron bound in minerals moves together with the silicate phase and cannot separate physically and join the volume of the metallic core.

Mass-balance equations of the core

In the physical case in which the contribution of metallic phase from the rising lithospheric shell is defined as negligible (since the sialic shell is a light material poor in free metals), the mass

balance of the core reads from left to right as a mathematical chain of three components:

$$f_{\text{core}} = f_{\text{int}} X_{\text{met}} \epsilon_{\text{core}}$$

where the parameters are firmly defined as follows:

- f_{core}**:** The linear fraction of the core volume out of the total mass of the modern Moon.
- f_{int}**:** The mass fraction of deep (internal) mantle material supplied in the feed, out of the total mass of the body.
- X_{met}**:** The fraction of free metal present within the internal feed material.
- ϵ_{core}**:** The hydrodynamic efficiency with which the metal sinks and joins the core at the centre of the body.

Numerical derivation of the balance: Taking the existing empirical core value of 1.4% ($f_{\text{core}} = 0.014$), and substituting the deep-mantle-material parameter from Paper 3, equal to 95% of the total lunar mass ($f_{\text{int}} = 0.95$; a model range of 94%–96%), the equation dictates a firm product value of:

$$X_{\text{met}} \epsilon_{\text{core}} \simeq 1.47\%$$

This means that exactly 1.47% of the deep mantle material fed upward must consist of metallic phase that actually reached the centre of the body. If only part of the metal supplied in the flow managed to sink to the core, then the initial feed material must carry a correspondingly higher metal concentration. This equation translates the qualitative term "a little metal" into a measurable physical quantity that the mechanism of Paper 3 must explain.

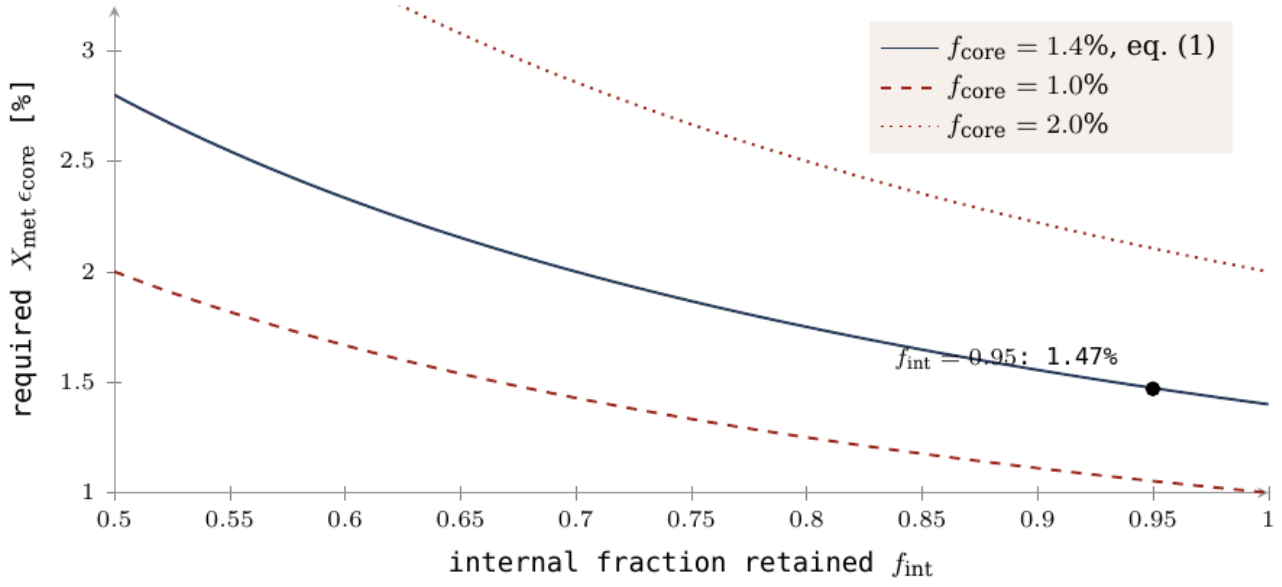


Figure 4. Requirement R2 as a number: the product of the effective metal fraction in the fed material and the fraction of it that reached the core, needed to build a core of 1.0–2.0% of the lunar mass, as a function of the internal fraction of the Moon. “A little metal” becomes a quantity that Paper 3 must explain.

Details of Figure 4 · The R2 inventory requirement as a quantitative measure

- **Horizontal axis (X):** Represents the internal mass fraction of the feed material (f_{int}).
- **Vertical axis (Y):** Represents the required product value $X_{\mathrm{met}}\epsilon_{\mathrm{core}}$ for building the core.
- **The set of model curves:**
  - **Solid dark-blue line:** Represents the core value confirmed in the model, equal to 1.4%.
  - **Dashed red line:** Represents a lower core bound of 1.0%.
  - **Dotted red line:** Represents an upper core bound of 2.0%.
- **Mathematical behaviour:** All the curves show a monotonic decline from left to right; the larger the fraction of the lunar mass made of deep material, the lower the metal concentration each unit of its volume needs to carry. Halving the parameter f_{int} exactly doubles the required product value on the vertical axis.
- **Black reference point:** Marks the values of the firm reference case of Paper 3: $f_{\mathrm{int}} = 0.95$ and a product value of 1.47%. Since the model places 94%–96% of the lunar mass in the deep material, its geometric position lies at the right-hand edge of the figure, which represents the point of minimum requirement of the system.

How it is tested

- **The verification protocol:** The test is defined as a quantitative prediction, derived from the mass-transfer and geological-differentiation equations of Paper 3, that sets the range of core fractions the mechanism is physically able to produce. This prediction is placed directly beside the empirical core measures obtained by lunar seismology (as set out in section B.2).
- **Definition of differentiation (Differentiation):** The mechanical process by which a planetary body separates into distinct layers on the basis of density contrasts — a heavy metallic phase sinks gravitationally to the centre, and a light silicate phase floats and remains above it.

A.3 R3 — An asymmetry tied to the opening

What it requires

- **Asymmetry of crustal thickness:** The lunar crust must be thicker on the far side (Far Side) than on the near side (Near Side). The geodynamic model requires that this distribution of thickness be arranged around the structural opening — the mechanical connection through which the lunar body was fed and on which it sat on the Earth. The crust must be thin around the old connection and thick on the side opposite it.

Why the model needs it

Two physical mechanisms act at the same time in the same geodynamic direction:

1. The thermal mechanism: The existence of a hot connection on one side (the near side) against more efficient radiative cooling to space on the opposite side (the far side) points to greater crustal thickening away from the connection. The mechanism is analogous to a pot standing on a flame: the wall farthest from the flame cools and hardens first.

2. The geometric mechanism (closure of the dome): The far side is the original surface of the dome and carried the terrestrial lithospheric shell with it all the way. This crust underwent a threefold linear thickening when this surface closed onto the far hemisphere. The geometric mechanism is analogous to a cloth spread over a dome and then gathered and drawn in around a ball: the same cloth now covers a much smaller surface area, and so its thickness grows. A shell of about 18 km that covered the dome closed onto one hemisphere and thickened threefold, into the 50–60 km far-side crust defined in Paper 3.

- **History of the near side:** The near side is the base of the structure. Its outer ring kept the sial of the Earth's surface on which the body rested and carried it along when the body peeled off, while the hot central contact lost this sial entirely to the heat and was smoothed under the action of gravity.

Mass-balance constraints

- **The law of conservation of mass in the crust:** The crustal thickness follows solely from a combination of closure, elastic/plastic deformation and thermal freezing, under a strict law of conservation of mass. Geometric compression must not be counted as an addition of new mass. When the shell is gathered it becomes thicker, but no material is added; the same mass is spread over a smaller area. Therefore the mass of the thick far-side crust must be explained in full by the mass of the shell that existed beforehand. Section B.3 shows that the shell of the dome and the far-side crust match each other in the mass balance to within about 10% (7% in earlier versions of the calculation).

What else it assumes (constraints on the survival of the structure)

- **Durability of the early record:** The derivation of the thick crust on the far side assumes that processes of mixing, internal transfer, changes in chemical composition, and later cratering by returning material or large impact basins did not erase the original asymmetry. What the closing dome created had to survive the subsequent geological events.
- **The global "helmet" approximation:** A smooth helmet shape is used as a mathematical approximation for the long-wavelength component (the global trend): the crust is described as thickest at one pole of the body (the centre of the far side) and thinning gradually and linearly toward the opposite pole (the centre of the near side), like a helmet thicker at the crown than at the rim. The model compares only the large-scale trend, not every local topographic detail (represented by Equation 3 in section B.3). No derivation requires that each local point be thicker than the one before it; basins, maria and local thickenings are expected to break the smooth curve, and the requirement concerns the overall trend only.

How it is tested

- The empirical test rests on global maps of crustal thickness derived from combined gravity and topography data, mainly from the GRAIL mission (section B.3 and Figure 5). The crust must show a gradual thickening as a function of the angular distance from the old contact on the near side, and the far-side crust must contain, in the mass balance, the mass of the terrestrial shell of the dome.

A.4 R4 — Orientation at the time of connection

What it requires

- **Firm locking of orientation:** The Moon always turns the same face toward the Earth (synchronous rotation — one rotation about its axis for each orbital revolution). The model requires that this face be tied directly to the side of the geodynamic connection: the side on which the Moon was attached to the pear is the side that faces the Earth today.

Why the model needs it

- **The relative-rotation constraint in a connected body:** A body whose mechanical connection prevents free relative rotation keeps its side of connection permanently turned toward the parent body. It still rotates in inertial space together with the rotation of the whole system.

- **The dancers analogy:** The mechanism is analogous to two dancers holding hands and turning together: each always faces the other (there is no relative rotation), and yet both turn once in space relative to the walls of the room. The connected Moon is in the same dynamical state: relative to the Earth it does not rotate, but relative to the distant stars it rotates once with each rotation of the system. In a suitable circular state, this orientation provides an exact synchronous initial condition for release. At the moment of detachment the Moon already shows the Earth a fixed face and rotates at a rate of one revolution per rotation.

What else must be solved (post-release dynamics)

- **Orbital evolution:** After the physical detachment, the model must solve the system of tidal torques, the libration and the dynamical changes of shape:
  - **Tidal torques (Tidal Torques):** Twisting forces that arise because the Earth exerts a stronger gravitational pull on the near part of the Moon's mass than on its far part.
  - **Libration (Libration):** The slight geometric rocking of the Moon, which allows an observer on the Earth to see over time a little more than half (about 59%) of its surface.
  - **Changes of shape (Deformation):** Plastic and elastic changes in the shape of the Moon, which change the dynamical way in which it responds to the tidal field.
- **The dynamical-orbit constraint:** The computed dynamical orbit must tie the former side of connection to the side that faces the Earth today. The mechanical wording “a connected body cannot rotate” is replaced by the exact definition of relative rotation: the connected body does rotate in space, but at the same angular rate as the system ($\omega_{\mathrm{rot}} = \omega_{\mathrm{orb}}$), and therefore it does not rotate relative to the parent body. This chapter is developed dynamically in Paper 4.

How it is tested

- The empirical test carries the values of shape, rotation rate and angular momentum at the moment of release from Paper 2 into the orbital calculations of Paper 4, and checks mathematically that the side of connection remains the side facing the Earth, or returns to being that side (section B.4). The existence of synchronicity today does not in itself constitute a confirming test of the model, since prolonged tidal evolution can produce it also from the initial conditions of alternative formation paths.

A.5 R5 — A defined order of cooling and preservation

What it requires

- **An asymmetric chronology of stabilisation:** The model requires that the far side, lying opposite the connection, stabilised mechanically and began to preserve its geological impact record before the fed, hot region (adjacent to the connection on the near side) did so.

Why the model needs it

- **The mechanism of the closing envelope:** The closure of the crustal envelope opposite the connection allows earlier morphological preservation on that hemisphere, while the feed region on the near side continues to receive a supply of heat and mass, and therefore continues to be heated and reworked (Recycling). Once the dome had closed structurally onto the far hemisphere, that surface stopped being renewed: it cooled radiatively, stiffened mechanically and began to preserve the impact craters that landed on it. By contrast, the near side, located directly above the hot feed flow, underwent continued tectonic and thermal reworking, and therefore its residual impact record began at a later point in time.
- **Definition of a geological unit:** The requirement concerns solely the time at which defined crustal units stabilised and preserved a record; it does not imply that every physical sample from the far side is chronologically older than every sample from the near side. A unit is defined as a region of crust with a single genetic origin and a uniform history, such as a highland plain (Highlands) or the floor of an impact basin.

The four geochronological clocks (avoiding conflict)

To preserve accuracy, the model separates firmly between four different isotopic clocks, and states that comparing a clock of one kind on the far side with a clock of another kind on the near side gives no indication of the order of closure:

1. Age of the ancient shell material: The age of the material itself before its rise, which in the model formed in the shell of the Earth before the rise of the dome.

2. Crustal solidification age (Solidification Age): The exact point in time at which the new crust solidified below the solidus temperature.

3. Impact age (Impact Age): The moment at which a crater-forming impact event melted part of an existing crust or reset (Reset) its isotopic clocks.

4. Age of the basaltic fill (Mare Basalt Age): The age of the dark mafic magma that filled the impact basins at a late stage.

- **Constraint on the scientific prediction:** The sweeping prediction that "all of the near-side crust formed chronologically later" does not follow from the scenario. The geodynamic prediction concerns only the order of mechanical closure and preservation between crustal units of similar origin and formation processes on the two sides of the body.

How it is tested

- The empirical test defines units of closure and preservation on both hemispheres, dates processes of similar genetic character, and checks whether the geodynamic order of events is preserved after a complete separation of the crater-forming events and the events of basaltic lava fill (section B.5 places the four clocks side by side for control).

A.6 R6 — Volatiles in the amount and state the mechanism needs

What it requires

- **Polymorphic constraint and phase rigour:** Volatiles are defined as components that pass easily into the gas phase — mainly water ($\mathrm{H_2O}$), carbon dioxide ($\mathrm{CO_2}$) and similar compounds. In Paper 3 their presence is essential to the dynamical operation of the mass-feed mechanism.
- **Mechanical roles in the system:** If volatiles serve to reduce density, to change rheological properties or to build the hydrostatic pressure of the feed, the mechanism must operate at concentrations and thermodynamic phase states that are physically possible all along the path of rise:
  - **Reduction of density:** Lowering the specific weight of the deep mantle material, which allows it to float and rise vertically.
  - **Viscous rheology:** Water changes the viscosity of molten or partially molten material, and allows it to flow and deform mechanically more easily (much as plasticisers do).
  - **Building ejection pressure:** As a free gas phase, the volatile components produce an expansion pressure that pushes the feed flow upward along a definite direction.
- **Depth constraint:** Each of these roles requires certain amounts of volatiles in a defined phase state (dissolved in the melt, as a dense supercritical fluid, or as free gas) at every given depth along the geodynamic conduit.

What does not satisfy it (ruling out partial confirmation)

- **Independence from local presence:** The mere finding of water at some geographic location on the surface of the modern Moon is not sufficient confirmation of requirement R6. The identification of water in lunar samples proves only its general presence in the body; it does not prove that it was present in the required mass, or in the correct phase state, at the geological locations where the feed flow needed it.
- **Ruling out dilute bubbles at depth:** Assuming the existence of an early vapour envelope does not mathematically permit placing dilute gas bubbles at every lithostatic depth. At great depth the high static pressure physically requires the volatiles to remain dissolved in the melt, or compresses them into a very dense volume. The phase state at every depth must be derived directly from the real values of pressure (P), temperature (T) and chemical composition (X), and not set arbitrarily to suit the needs of the mechanism.

The chemical–thermodynamic integration protocol

- **Firm overlap of ranges:** The pressure and concentration conditions of the volatiles needed to run the mechanism of Paper 3 must overlap entirely with the ranges that an independent geochemical and thermodynamic model allows.
- **Balance of loss and transfer:** The calculation must account for all the stages of mass loss, mass in the melt, mechanical ejection and dynamical transfer between the different reservoirs: part of the water escapes radiatively to space, part remains dissolved in the magmatic phase, part is expelled together with ejected material (Ejecta), and part is moved between the geological reservoirs. The water identified in laboratories today represents only the physical remainder that survived all these stages.

How it is tested

- The test derives from the equations of Paper 3 the minimum concentration and the thermodynamic phase state along the vertical feed path. These values are passed through a functional model of loss and geochemical differentiation, and from it the prediction is derived against the empirical concentrations measured in Apollo samples, such as the value of 615–1,410 ppm water recorded in closed melt inclusions (Melt Inclusions) (as set out in section B.6).

A.7 R7 — Crustal distribution and the centre of mass

What it requires

- **The geometric centre-of-mass offset:** The physical centre of mass of the Moon does not coincide with the geometric centre of its outer morphological shape. It shows a fixed linear deviation of the order of about 2 km toward the near-side hemisphere (Near Side; as set out in section B.7). The model requires that the distribution of crustal thickness — thick on the far side and thin on the near side — produce independently a geometric offset of this kind and in exactly this direction.

Why it follows (gravitational mechanism)

- **A crustal mass deficit:** In a body with a defined outer outline, replacing dense mantle material with light crustal material on the far hemisphere produces a mass deficit (a shortage of mass) on that side. Since the density of the crust is lower than the density of the mantle beneath it ($\rho_{\text{crust}} < \rho_{\text{mantle}}$), wherever the crust shows increased thickness, material of low density occupies a volume that would otherwise be filled by dense mantle material.
- **Shift of the balance point:** As a consequence, the far side holds less total mass than the near side within the same geometric outline.
- **The cork analogy (gravitational constraint):** The mechanism is analogous to a ball with a thick patch of light material such as cork set into one side: its gravitational balance point (the centre of mass) is necessarily shifted toward the opposite side. The structural contribution of the asymmetric crust pushes the centre of mass toward the thin side (the near side). This is a direct gravitational consequence derived from defined density conditions.

Components of the combined balance

The total centre of mass of the planetary body is a vector sum of several structural contributions, and the crustal term alone must not be relied upon:

1. Topography: The height of the outer surface.

2. The asymmetry of the mantle and the core.

3. Magmatic intrusions (Magmatic Intrusions): Bodies of dense silicate melt that solidified within the volume of the crust.

4. Basin fills (Mare Basalts): The dense, iron-rich basaltic lavas that filled the large impact basins on the near side.

- **The overall-sign constraint:** Each of these geological components moves the gravitational balance point by its own amount and in its own direction. The crustal contribution by itself pushes the centre of mass toward the near side, but requirement R7 will be defined as geodynamically confirmed only if the combined vector sum of all these structural contributions points toward the near side.

How it is tested

- **The single-density-model constraint:** The test protocol requires the use of a single, fixed mathematical model of material density for the simultaneous calculation of both the crustal thickness and the centre-of-mass offset. It is strictly forbidden to fit or change the density values independently merely in order to obtain the desired offset value.
- **Limitation of the thickness calculation:** The crustal thickness cannot be measured directly; it is computed from the observed gravity field under the assumption of fixed density values for the crust and the mantle. Were one density used to determine the thickness and another outline to obtain the offset, the agreement would be defined as artificial and fabricated.
- **A firm mathematical result:** Under a single, uniform density model, section B.7 yields a theoretical offset of 1.24–3.11 km for density contrasts varying in the range 300–750 kg/m³. This range of values contains exactly the empirical order of magnitude observed on the Moon. This value is a computed sensitivity range, and not an absolute independent prediction of the offset.

Summary of the set of seven core requirements (Part A)

With the formulation of requirement R7, the theoretical structure that links each physical trace on the Moon unambiguously to a defined stage in the formation path of the LEF model is complete:

- **R1 (affinity to terrestrial material):** Links the chemical and isotopic composition to the terrestrial source of the reservoirs.
- **R2 (a small metallic core):** Defines the mass balance and the hydrodynamics of the feed flow.
- **R3 (crustal-thickness asymmetry):** Derived directly from the mechanism of the dome's closure and thickening.
- **R4 (orientation at the time of connection):** Defined as a dynamical initial condition owing to the rotational locking of the connected body.

- **R5 (order of cooling and preservation):** Dictates the asymmetric chronology of crustal stabilisation.
- **R6 (volatiles):** Sets the thermodynamic and rheological constraints of the ejection conduit.
- **R7 (distribution of the centre of mass):** Aligns the geometric structure of the crust with the global gravity field.

Part B will now set each of these requirements, independently and in isolation, against the empirical data and measurements.

PART B

The seven requirements against the measurements

Part A established what a Moon built by elevation from the Earth must look like: seven requirements, and beside each the physical assumption that gives rise to it. Part B now sets each requirement beside the measurement that bears on it. For each requirement four firm steps are applied: what was measured, how the model explains it, the test that will settle the question, and the result that would not fit the requirement. Sections B.1–B.6 deal with the isotopes, the core, the crust and the maria, the synchronous rotation, the clocks of the lunar record and water; B.7 and B.8 continue with the centre of mass and the exposure zones.

The problem and the methodology of discrimination

- **The limited uniqueness of single observations:** A geological or gravitational property identified on the Moon can, in theory, fit several different formation mechanisms. A small metallic core, for example, or a thick crust on the far side, can be explained in more than one way, so their empirical discovery in itself is not unambiguous proof supporting the LEF model alone.
- **Definition of the scientific terminology in the model:** To preserve accuracy, three words are defined here in the following exact scientific sense:
  - **"Fits":** A definition denoting full agreement — the empirical observation matches what the model requires and predicts.
  - **"Open":** A definition denoting that the quantitative calculation or comparison needed to turn the morphological agreement into a decisive test has not yet been carried out; the

study has been defined by name and aim, but the data have not yet been fully processed.
    - **"Not met":** A definition reserved solely for a state of clear physical or numerical mismatch within the geodynamic conditions defined in advance.

B.1 R1 — What the isotopic similarity shows

1. What was measured

- **The triple oxygen system:** The ratios of the stable isotopes of oxygen were measured: $^{16}\mathrm{O}$, $^{17}\mathrm{O}$ and $^{18}\mathrm{O}$. Ordinary thermodynamic processes, such as magmatic melting, evaporation and fractional crystallisation, sort the isotopes by their atomic mass, and change the ratio $^{17}\mathrm{O}/^{16}\mathrm{O}$ by an amount approximately equal to half the change produced in the ratio $^{18}\mathrm{O}/^{16}\mathrm{O}$ (since the mass difference is half as large).
- **The $\Delta^{17}\mathrm{O}$ measure:** The quantity $\Delta^{17}\mathrm{O}$ measures exactly the distance by which the sample deviates from the global mass-dependent fractionation line. Processing and internal differentiation within a closed planetary body cannot shift this value, so $\Delta^{17}\mathrm{O}$ is an absolute fingerprint of the unchanging source reservoir. Separate bodies in the Solar System (such as the planet Mars and the various meteorite groups) have different and clearly distinct isotopic fingerprints.
- **The measured empirical value:** The research group of Young and colleagues (2016) measured and reported the difference in oxygen values between the Earth and the Moon:

$$\Delta^{17}\mathrm{O}_{\text{L-E}} = -1 \pm 5\ \mathrm{ppm}, \quad 2\mathrm{SE}$$

- **Interpretation of the numerical figure:** The isotopic fingerprint of the Moon minus that of the Earth equals -1 parts per million (ppm — a unit representing one part per million of the total isotope ratio), with an uncertainty range of $\pm 5\ \mathrm{ppm}$ at two standard errors ($2\mathrm{SE}$). This difference is a statistical zero within the limits of laboratory resolution, and shows that at the present level of precision the Moon and the Earth carry an identical oxygen composition [S1].

2. How the model explains it

- **Direct derivation from the scenario:** For the LEF model this isotopic identity is the direct and necessary prediction of the geodynamic scenario, since all the material of the Moon's mass was moved directly from the reservoirs of the Earth.

- **Comparison with alternative models:** The value of $5\ \mathrm{ppm}$ represents a reported laboratory uncertainty, not a theoretical physical threshold derived from the LEF equations; future measurements at higher resolution may reveal a tiny difference. In the alternative giant-impact model (Impact Hypothesis), in which most of the Moon's mass is derived from the material of the impactor (Theia), such an extreme isotopic agreement requires a full and massive hydrodynamic mixing between the two bodies. Since the scientific literature has presented such mixing mechanisms at the time of impact, the empirical figure in itself does not provide absolute discriminating power between a formation path of a connected body and a path of a mixed impact.

3. The test that will settle it

- **The multi-system balance protocol:** The test defines a mathematical derivation of the composition of the final product from defined terrestrial reservoirs (the deep reservoir and the sialic shell defined in Paper 3), their exact mixing according to the mass ratios set by the model, and a mathematical accounting of the mass losses and fractionation that take place during melting and evaporation. The predicted result must pass a simultaneous test (all at once) against several different isotopic systems (oxygen, titanium, chromium, etc.) across several separate lunar samples.

4. The result that would not fit the requirement

- The requirement would be considered not met if it is shown mathematically that no quantitative combination of terrestrial reservoirs and allowed thermodynamic processes is able to explain all the isotopic data and the chemical composition together in a single unified system.
- **Current status in the model:** The agreement of the oxygen isotopes is assigned the status fits; the development of the overall quantitative multi-system balance is assigned the status open.

B.2 R2 — The core is a quantitative target for transport

1. What was measured

- **Global seismic measurements:** The network of seismometers installed on the lunar surface by the Apollo missions recorded seismic waves from moonquakes (Moonquakes) continuously over years. The propagation speed of the waves and the way they are

reflected (Reflection) from the depth of the mantle reveal the thickness of the internal layers and the density boundaries.

- **Dimensions of the core layers:** The combined seismic model of Weber and colleagues defines a firm two-layer core structure [S2]:
  - **Solid inner core:** With a physical radius of about 240 km from the centre of mass.
  - **Liquid outer core:** Its outermost edge reaches a radius of about 330 km from the centre of mass (for a global lunar radius of 1,737.4 km).
- **The residual core mass:** The mass fraction of the core is estimated at about 1.4% of the total mass of the lunar body. This value is kept in the model as a comparison parameter from the scientific literature, not as an exact independent measurement. The data emphasise that the Moon is an extremely metal-poor body relative to the Earth, whose metallic core is proportionally much larger.

2. How the model explains it

- **A mechanism of entrainment and silicate filtering:** The LEF model explains this figure through a dynamical preference for silicate phases: the vertical feed flow that rose along the pear carried mainly light silicate material, containing a low concentration of droplets of free iron metal. The empirical figure fits this mechanism, but does not by itself prove that the source of the feed material lay beneath the layer of the terrestrial core before its final organisation.
- **The constraint of the balance equation:** Substituting the deep-mantle-material parameter from Paper 3 ($f_{\mathrm{int}} = 0.95$), the quantitative balance equation (Equation 1 in Part A) locks the required product value of the effective free-metal fraction in the fed material (X_{met}) and the hydrodynamic efficiency of its settling (ϵ_{core}) at the level of:

$$X_{\mathrm{met}}\epsilon_{\mathrm{core}} \simeq 1.47\%$$

- This firm numerical figure (shown graphically in Figure 4) becomes a defined and binding quantitative target, which the mass-transfer mechanism and the droplet hydrodynamics of Paper 3 must produce.

3. The test that will settle it

- **The volume-prediction protocol:** The test is defined as running a hydrodynamic model that computes quantitatively the range of core values (f_{core}) that the mechanism is physically able to produce. The calculation will rest on the function of the amount of metallic mass that the magmatic flow is able to sweep upward with it as a function of the speed and

viscosity of the flow, against the function of the gravitational settling of the droplets toward the centre of the lunar body afterwards.

4. The result that would not fit the requirement

- The requirement would be considered not met if it is shown that all the physically supported ranges of metal entrainment in the hydrodynamic equations yield a core mass that departs significantly from the allowed empirical seismic range defined in the scientific literature (1.0%–2.0%).
- **Current status in the model:** The existence of the small core is assigned the status fits as a fixed target parameter; the development of the hydrodynamic mass-transfer mechanism to the level of quantitative closure is assigned the status open.

B.3 R3 — The crust, the maria and the mass that came from outside

1. What was measured

- **High-resolution satellite gravity measurements:** The thickness of the lunar crust is not measured directly; it is inferred mathematically from the gravity field. The two spacecraft of the GRAIL mission (NASA) flew in a trailing orbit (one behind the other) around the Moon and measured with high precision the fluctuations of the varying linear distance between them. Every excess or deficit of mass in the subsurface exerted a differential gravitational pull: first on the leading spacecraft, and then on the trailing one [S3].
- **Isolating Bouguer anomalies (Bouguer Anomaly):** Combining the topographic height data measured by the LOLA laser altimeter (on board the LRO probe), the gravity component derived from the visible morphological topography was computed and subtracted. The gravitational remainder (the Bouguer anomaly) is attributed directly to the relief and the depth variations of the Moho — the density discontinuity between the light crustal material and the dense mantle material.
- **Seismic anchor and global values:** Converting the gravity relief into a linear crustal thickness (in kilometres) requires fixing a crustal density value and a calibration (anchoring) at the depth of one geographic point, based on the data of the seismic stations of the Apollo missions. The GRAIL data confirm the existence of a broad global asymmetry, and yield models with a global mean crustal thickness of 34–43 km [S3], a value that varies with the chosen density and with the seismic anchor used for calibration.

A numerical reading constraint: This figure must not be identified with or interpreted as a flat measurement of “a uniform 30 km on the near side and 60 km on the far side”. These values are only a geometric example for calibrating the model's equations.

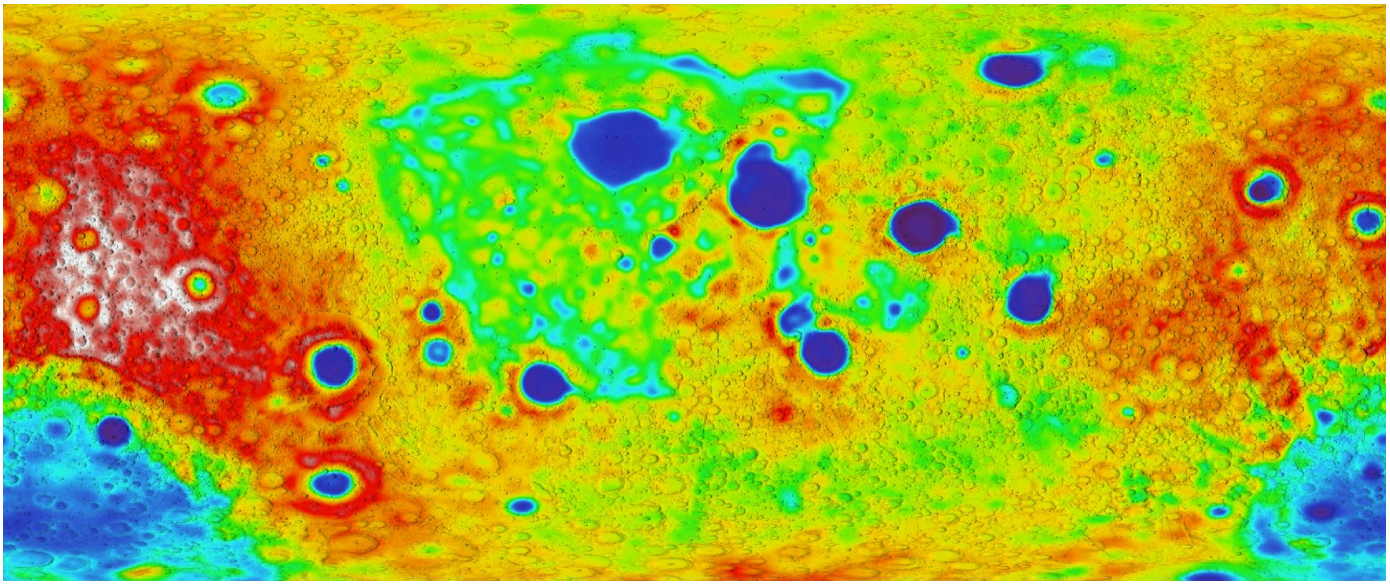


Figure 5. The crustal thickness of the Moon from GRAIL gravity and LOLA topography, cylindrical projection centred on 0° E, latitudes ±75° (the near side in the centre, the far side at both edges). Blue: thin crust beneath the near-side basins; red and white: the thick highlands of the far side. The far side is about 10 km thicker between the centre of the near side and the far side, and about 8 km thicker in the hemispheric means (Paper 12, §B.8): in the model, the shell of the dome thickened threefold; the near side is the peeled sial of the base, thinnest beneath the maria, the openings. Credit: LROC QuickMap (NASA/GSFC/Arizona State University); GRAIL crustal thickness (NASA/JPL-Caltech/MIT; Zuber et al. 2013).

Details of Figure 5 · The crustal-thickness map of the Moon (GRAIL / LOLA)

- **Projection parameters:** A cylindrical projection (Plate Carrée) with its geographic centre at 0° E, latitudes of $\pm 75^{\circ}$. The near side (Near Side) lies in the centre of the map, and the far side (Far Side) is spread over its two outermost edges (left and right).
- **Colour morphological legend:**
  - **Blue (thin crust):** Identified in the centre of the map beneath the large impact basins of the near side.
  - **Warm colours (red and white - thick crust):** Mark the high highlands of the far side at the edges of the map.
- **The measured asymmetry values:** The far-side crust is about 10 km thicker in a direct measurement between the centre of the near side and the centre of the far side, and about 8 km thicker in the calculation of the weighted means of the whole hemispheres (Paper 12, §B.8).

- **Data credit:** LROC QuickMap (NASA/GSFC/Arizona State University); GRAIL gravity processing (NASA/JPL-Caltech/MIT; Zuber et al. 2013).

2. How the model explains it

- **The geometric approximation equation (the helmet model):** To compare the empirical gravity map with the LEF mechanism, the long-wavelength global trend is represented by the following formula:

$$h\left(\psi\right) = h_0 - A\cos\psi, \quad A > 0$$

where the parameters are mathematically and firmly defined:

- ψ**:** The angular distance (in degrees) from the original geodynamic connection axis (0° at the centre of the contact on the near side, 90° on the middle belt, and 180° at the opposite pole on the far side).
- h**:** The predicted local crustal thickness.
- h_0**:** The base crustal thickness on the middle belt ($\psi = 90^\circ$).
- A**:** The amplitude of the global asymmetry (Amplitude).
- **The mechanism of angular variation:** Since the function $\cos\psi$ ranges between $+1$ (at the centre of the contact) and -1 (on the far side), the equation dictates a crust with a minimum thickness of $h_0 - A$ at the point of connection, and a maximum thickness of $h_0 + A$ on the far side.

Numerical derivation of the amplitude: Substituting the theoretical end values (30 km and 60 km) into the system of equations yields a base value $h_0 = 45$ ק"מ and an asymmetry amplitude $A = 15$ ק"מ. On the 90° belt a thickness of 45 km is obtained. These are point end values, not hemispheric means: the means integrated over the surface of a hemisphere for this example are 37.5 km and 52.5 km respectively (the reason is geometric: the integral of $\cos\psi$ over a hemisphere yields a mean value of exactly 0.5, so the hemispheric means are defined as $45 \mp A/2$). Wrongly substituting the hemispheric means as end values changes the intended amplitude and reduces it by half ($A = 7.5$ ק"מ).

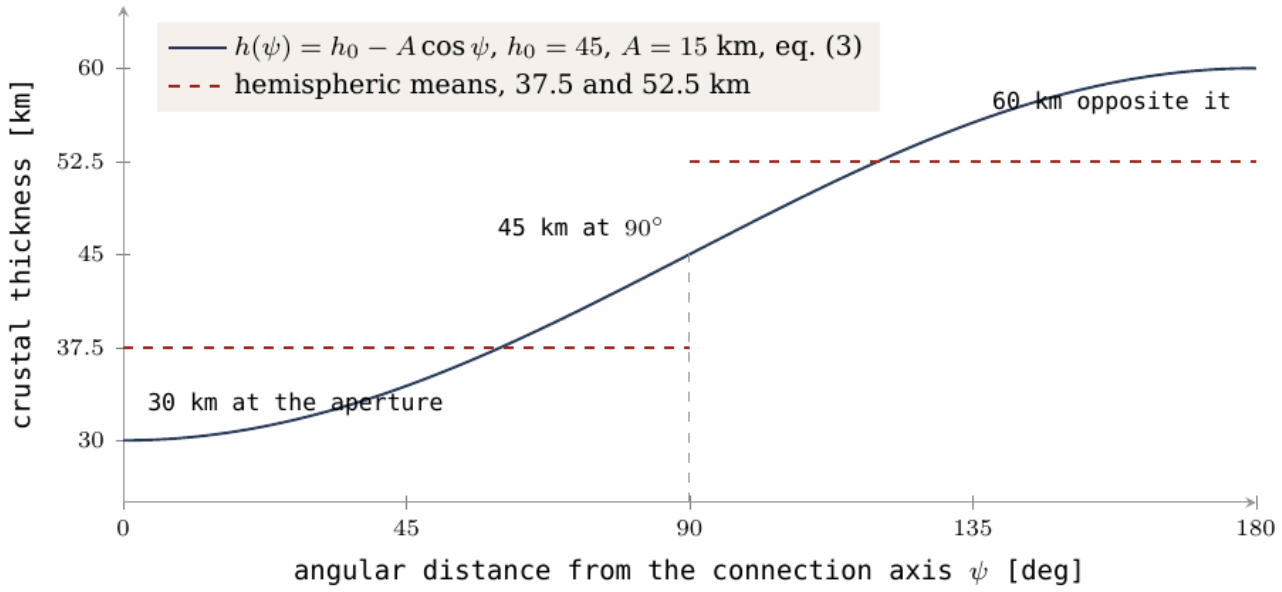


Figure 1. The long-wavelength crustal model of §B.3. End values of 30 and 60 km give $h_0 = 45$ and $A = 15$ km; the hemispheric means are 37.5 and 52.5 km, not the end values. The GRAIL global mean,43–34 km, is a constraint on h_0, not on A.

Details of Figure 1 · The long-wavelength crustal model

- **Behaviour of the curve:** The dark-blue curve (Equation 3) rises continuously from 30 km at the point of the opening ($\psi = 0^\circ$) to 60 km at the antipodal point ($\psi = 180^\circ$).
- **The hemispheric means:** Represented by the dashed red steps (37.5 km and 52.5 km), they lie well within the range of the extreme end values.
- **The GRAIL-mean constraint:** The empirical global mean set by the GRAIL mission (34–43 km) is a firm mathematical constraint on the height of the baseline of the system (h_0), but it does not define or limit the steepness of the rise represented by the amplitude (A). The steepness parameter must be read directly from the map.
- **Explanation of the structural LEF mechanism:** The asymmetry is explained in the model as a direct result of the gathering and closure of the envelope of the terrestrial shell of the dome, combined with non-uniform heating conditions. The far side carried the ancient 18 km lithospheric shell, which closed onto one hemisphere and thickened threefold into a crust of 50–60 km, while the side of connection (the near side) was kept at a high temperature. The near side is the peeled sial of the base, and it is thinnest beneath the mare basins, which are the openings of the original ejection.
- **Comparison with the Earthshine mechanism (Earthshine Hypothesis):** An alternative model in the scientific literature proposes a thermal link between a young, hot and glowing Earth and an asymmetric freezing of the Moon [S5]. According to this thesis, the heat radiation from the Earth kept the near side molten, while the far side cooled rapidly, and

crust-forming material thickened preferentially on the cold wall. These two different physical mechanisms predict the same global direction. Therefore the general asymmetry is defined in the LEF model as an agreement, while the patterns of internal mechanical deformation and the exact mass balances are what will provide the decisive discriminating power between them (Roy et al. 2014).

- **Analysis of the mare and rift structures:** The maria are concentrated on the near side, but their magmatic cover is influenced by deep magma production, local heat sources and fracture systems. The boundary structures of Oceanus Procellarum (Oceanus Procellarum) were interpreted on the basis of GRAIL data as a rectangle-like geometric system of rifts and magmatic intrusions (Dikes) formed as a result of thermal stresses [S4]. That study (Andrews-Hanna et al. 2014) ruled out the existence of a narrow connecting neck (an ancient bottleneck) to the Earth. The LEF model does not contradict this finding, since the model does not require a narrow neck; the lunar body peeled broadly from the wide base surface on which it sat (Paper 3, §C.4a), and the mare basins are the morphological record of the distributed openings that this surface left on the near side.

3. The test that will settle it

- **The belt-test protocol (Grid Moat Test):** The test requires that the geometric connection axis, the level of spatial resolution, and the application of a fixed basin mask (Basin Mask) be defined and fixed in advance, before the mathematical fit is carried out. The calculation will integrate area-weighted mean thicknesses within locked concentric angular belts, for an isolated estimate of the parameter A and an assessment of the residuals (Residuals) across different crustal density models.
- **Constraints of statistical control:** The basin mask deliberately removes the areas of the large basins from the statistical calculation, so that the local topographic thinning caused by the later impacts is not mistakenly interpreted as the long-wavelength global trend. Area weighting is essential because the angular belts near the axis are considerably smaller in geometric area than the middle belts. The test will also be run without the mask, as a sensitivity check. The restoration of crustal material excavated in the basins (Ejecta Restoration) may not be set in advance on the basis of the geometric shape one wishes to find. A geometric axis chosen directly from the map is a fitted parameter (Fitted parameter), and therefore requires verification against separate, independent data sets; otherwise the test will artificially show the pattern to which it was tuned in advance. This firm belt test is carried out in section B.8 on the geometry of the exposure zones.

4. The result that would not fit the requirement

- The requirement would be considered not met if the gravity analysis shows that the far-side crust cannot contain in its mass balance the mass of the shell that already existed in the terrestrial dome, or if the weighted belt map rules out the trend of gradual thickening as a function of the angular distance from the contact axis (ψ).
- **Current status in the model:** The general asymmetry is assigned the status fits; running the quantitative belt-test protocol and the full basin mask is assigned the status open.

The crustal mass-balance check protocol (R3 continued)

1. What was measured (mass-balance constraints)

- **The law of conservation of mass in the crust:** The thickening of the crust on the far hemisphere must rest on a defined source of mass, since geometric compression or plastic deformation cannot produce new mass. The calculation checks whether the terrestrial lithospheric shell, which rose as the skin of the lunar dome in the mechanism of Paper 3, contains a volume and mass that fully supply the construction of the existing far-side crust.
- **Parameters of the original shell of the dome (Paper 3):**
  - **Initial terrestrial shell thickness (h_s):** 18 km.
  - **Assumed material density (ρ_s):** 2,800 kg/m³.
  - **Radius of the base of the hemispherical dome (R_h):** 3,000 km.
- **The equation for the initial shell mass:** The curved surface area of a hemispherical dome is defined geometrically as $A_s = 2\pi R_h^2$ (half the surface area of a full sphere of the same radius). Hence the mass of the original shell is computed as follows:

$$M_{\text{shell}} = \rho_s h_s A_s = 2.85 \times 10^{21} \text{ ק"ג}$$

- **Mass fraction of the finished body:** This initial mass is equal to exactly 3.9% of the total mass of the modern Moon (M_L).

2. How the model explains it (comparison with modern measures)

- **The mass equation of a modern spherical shell:** For comparison of orders of magnitude only, the mass of a uniform modern spherical crust, with a firm outer radius of $R_L = 1,737.4$ ק"מ and an adopted density of $2,550$ ק"ג/מ"ק, is computed as a function of the linear thickness (h) from the volume of the shell (the volume of the whole Moon minus the volume of the inner sphere beneath the Moho):

$$M_{\text{crust}} = \frac{4}{3}\pi\rho_c \left[R_L^3 - \left(R_L - h \right)^3 \right]$$

- **Balance values derived from GRAIL data:** The table below shows the total crustal mass computed for the two ends of the GRAIL mean thickness, against the confirmed thickness of about 55 km on the far side. The last column isolates exactly half of this mass — representing the mass of the far hemisphere alone at the given thickness.

Crustal thickness in the example (h)	Computed global crustal mass	Fraction of the total lunar mass (M_L)	Mass of the far hemisphere alone
34 km (end of the GRAIL mean)	3.225×10^{21} ק"ג	4.39%	1.6×10^{21} ק"ג
43 km (end of the GRAIL mean)	4.057×10^{21} ק"ג	5.52%	2.0×10^{21} ק"ג
55 km (predicted far-side thickness)	5.15×10^{21} ק"ג	7.01%	2.6×10^{21} ק"ג

- **The no-tuning constraint on parameters:** This table is meant to illustrate the rigour of the mass-balance requirement, and is not a direct global reconstruction of the GRAIL data. The original shell of the dome closes and contracts over the area of the far hemisphere alone: bringing the data together sets 2.85×10^{21} ק"ג of ancient terrestrial shell mass against 2.6×10^{21} ק"ג of existing crustal mass on the far side at a thickness of 55 km.
- **Precision of the geometric agreement:** These two values, derived from entirely separate sources, match each other in the mass balance to within an extreme precision of about 10% (recalculations of Equation 5 with a density of 2,550 kg/m³ updated this value from earlier versions that showed an agreement of 7% and masses of 5.18×10^{21} kg and 2.7×10^{21} kg), without any arbitrary tuning of parameters between them.
- The shell thickness and the dome radius were set independently in Paper 3; the crustal thickness and its density were taken directly from the empirical lunar data of the gravity maps. The threefold linear thickening (from 18 km to 55 km) is a purely geometric product: a thin shell spread over the area of a broad dome, which is gathered and contracted into the small surface area of a single lunar hemisphere.
- **The near-side crust as a separate member:** The crust of the near side (with a mean thickness of 30–35 km) is not derived from the material of the closing shell of the dome. It is made of the sialic material (SiAl) of the terrestrial base surface, which peeled off and rose together with the lunar body (Paper 3, §C.4a). Its mass is computed separately (from Equation 5 for the near hemisphere at a thickness of 30–35 km) in the range 1.4×10^{21} to 1.7×10^{21} ק"ג (updated from earlier calculations that showed a value of about 2×10^{21} kg),

and it is an entirely independent and separate member within the global mass-balance equations.

- **The mineralogical-difference constraint:** Assigning the source material jointly to the term "sialic" does not by itself resolve the marked mineralogical differences that exist between the composition of the original terrestrial material and the lunar highland crust, which is characterised by a particularly high concentration of the mineral plagioclase (Plagioclase — a feldspar rich in calcium and aluminium). Defining the exact path of geochemical change from one composition to the other is an integral part of the overall balance, which is assigned the status open for future testing.

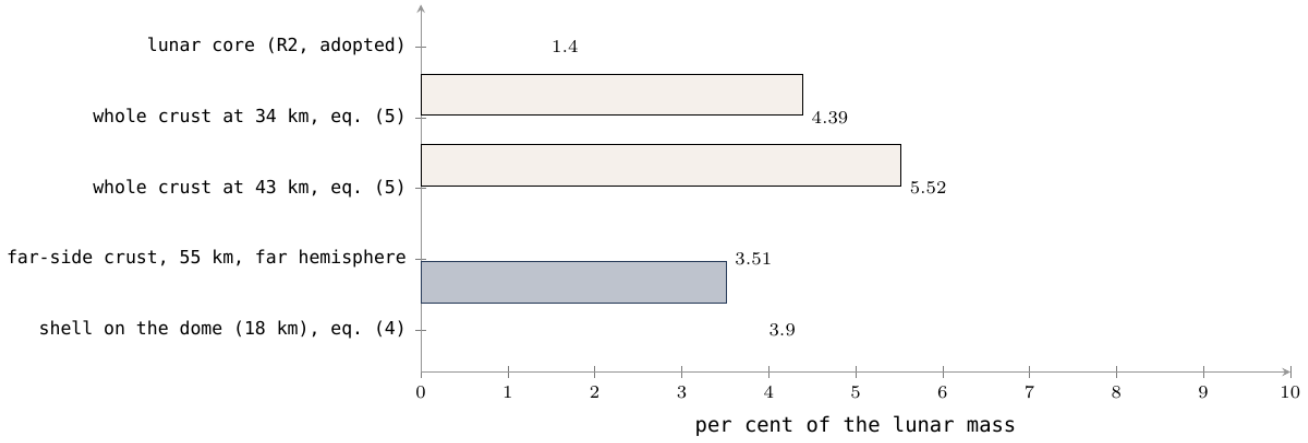


Figure 3. The mass balance of §B.3. The shell on the dome of Paper 3 (3.9% of the lunar mass) is the mass of the far-side crust alone (3.5% at a thickness of 55 km and a density of 2,550 kg/m³; an earlier version drew 3.67%); the near-side crust, the peeled sial of the base, is a separate member. The core fraction (R2) is the target that Paper 3 must supply.

Details of Figure 3 · The combined mass balance in the model

- **Morphological structure of the columns:** The figure sets the masses of the different components side by side as relative mass fractions of the total lunar mass (M_L).
  - **First column (supply):** Represents the mass of the shell that the dome carried with it according to Paper 3 (equal to 3.9% of the lunar mass).
  - **Second column (predicted demand):** Represents the mass of the far-side crust alone, which it must physically supply (equal to 3.5% of the mass, for a thickness of 55 km and a density of 2,550 kg/m³; updated from earlier drawn versions that showed 3.67%). These two columns show near-complete equality in the model.
  - **Two light columns (global control):** Show the total global crustal mass at the two ends of the thickness range derived from the GRAIL mission. These values are larger because they also include the mass of the near-side crust (the separate sialic member).

- **Red column (M_{core}):** Represents the mass of the metallic core derived from requirement R2 (1.4% of the mass), which is the second quantity that the feed mechanism of Paper 3 must supply quantitatively.

3. The test that will settle it

- The test protocol was defined in section B.3 (the weighted belt test and the global basin mask).

4. The result that would not fit the requirement

- The requirement would be considered not met if it is shown geochemically or mathematically that no combined balance of material volumes and mineralogical compositions is able to supply at the same time the crustal mass and the geometric spatial prediction using a single uniform and consistent set of parameters.
- **Current status in the model:** The existence of the broad asymmetry and the basic mass balance of the far side are assigned the status fits; locking the exact spatial profile, defining the sialic mass source of the near side and verifying it against the locations of the openings are assigned the status open.
- **Summary of the present findings:** The crust is thicker on the far side, as required; the terrestrial shell of the dome has exactly the right mass to build the far-side crust; and the geometric test that will fix the profile and the connection axis is fully defined.

B.4 R4 — Today's synchronicity and the connected past

1. What was measured

- **Modern rotational locking and orientation:** The Moon today shows a dynamical state of clear synchronous rotation, together with geometric and physical libration phenomena. Synchronicity means that the duration of the Moon's rotation about its spin axis is equal, to astronomical precision, to the time it needs to complete one orbital revolution around the Earth ($\omega_{\text{rot}} = \omega_{\text{orb}}$), a phenomenon that forces it to turn the same hemisphere (the near side) permanently toward the Earth.
- **The libration measure (Librations):** Tiny angular rockings of the surface as seen from the Earth. The librations are caused mainly by the eccentricity (the ellipticity) of the Moon's orbit (the orbital speed varies along the orbit while the rotation rate is constant) and by the tilt of its spin axis relative to the plane of the ecliptic. These geometric oscillations allow a

terrestrial observer to see over time a little beyond the hemisphere boundary (a total area of about 59% of the lunar surface).

- **The hammer-thrower analogy:** The dynamical mechanism is analogous to the hammer of a hammer thrower in athletics: as long as the thrower holds the hammer's cable and swings it round, the head of the hammer always turns exactly the same side toward the athlete while both turn together in space. A planetary body mechanically attached to its parent body must behave dynamically in this way. Requirement R4 examines whether the Moon kept that same early facing orientation also after its physical detachment from the pear.

2. How the model explains it

- **Kinematic inheritance of orientation:** The LEF model explains the existence of synchronicity as a direct result of the inheritance of the dynamical orientation vector from the moment of release and the connected past of the body.
- **An alternative explanation in the literature (tidal evolution):** Rotational locking in a ratio of $1:1$ is a common property in the Solar System, and is explained in science through a mechanism of prolonged tidal evolution (Tidal Evolution) [S6]. This mechanism is defined as a slow angular braking of a satellite's rotation, caused by the tidal friction and dissipation forces (Tidal Dissipation) that the planet induces within the volume of the satellite, until the rotation rate is fully synchronised with the orbital rate. Most of the massive moons in the Solar System reached geometric locking in this way. Since the two different physical mechanisms (early inheritance of orientation versus later tidal locking) lead to the same orbital ratio of $1:1$, the present empirical figure does not in itself hold absolute discriminating power. To make a scientific separation, the test must focus on the exact characterisation of the hemisphere that was locked and on the dynamical path that led to it.

3. The test that will settle it

- **The dynamical-integration protocol:** The test requires a firm mathematical transfer of the parameters of the morphological shape, the rotation rate and the exact angular-momentum vector at the moment of physical release from the equations of Paper 2 into the complex orbital and tidal calculation model of Paper 4. The dynamical calculation must show in a controlled way that the former side of geodynamic connection (the wall that peeled off the pear) is indeed the side that is mechanically bound to remain facing the Earth throughout the orbital evolution, or to return and be locked in that state under the action of the tidal torques.

4. The result that would not fit the requirement

- The requirement would be considered not met if it is shown through the orbital-mechanics equations of Paper 4 that no supported dynamical path is able to preserve or restore the required geometric relation and orientation, or if the calculation shows that the tidal torques would have locked a different hemisphere of the body toward the Earth.
- **Current status in the model:** The existence of modern synchronicity is assigned the status fits as a condition of general dynamical compatibility; proof of direct mechanical continuity from the connected past is assigned the status open.

B.5 R5 — Four different clocks

1. What was measured

- **Ages of the lunar record:** Characterising the chronology and the stratigraphic order of the Moon rests on reading isotopic and geomorphological values. The data show several different kinds of geological age, which measure separate physical events.
- **The four lunar clocks distinguished in the model:**
  1. **Crystallisation age of the crustal material:** Measures the point in time at which a mineral grew from the magmatic melt (melting) and began to trap the daughter products of radioactive decay in its crystal structure (the isotopic clock begins to tick). Strong, late thermal heating can reset this clock (Resetting).
  2. **Impact age or impact-melt age (Impact Melt Age):** Dates a meteoritic impact event that melted, altered or reset an existing crustal component already fixed in place. In the LEF model these craters were formed by the landing of returning terrestrial material (Paper 6), and their ages date these events and not the formation of the hemisphere.
  3. **Age of the basaltic fill (Mare Basalt Age):** Dates the crystallisation of the dark, dense mafic lava that flooded the impact basins at later stages.
  4. **Exposure or preservation age (Exposure/Preservation Age):** A measure not obtained directly from laboratory analysis of a physical sample. It is estimated statistically from the density of impact craters on the surface using a predictive chronological model, and sets how long the surface remained exposed and preserved its geological record. The measure depends on burial under deposits, erosion, or reaching a state of crater saturation (Saturation — a state in which new impacts erase old ones).

A firm numerical reading constraint: The methodological rule in the table states that ages may be compared only within the same kind of clock ($1 \to 1$, $3 \to 3$). Comparing a clock of one kind on the far side with a clock of another kind on the near side is defined as a serious error and says nothing about the order of closure.

- **Empirical age data of mare samples:**
  - Samples of the Chang'e-5 mission (China) record a young basalt of about 2.0 billion years (Ga) on the near side [S7].
  - Samples of the Chang'e-6 mission record volcanic activity on the far side (within the South Pole-Aitken basin) about 2.8 billion years ago (Ga) [S8].

2. How the model explains it

- **Characterising the late fill ages:** The two values measured in the Chang'e samples (2.0 Ga and 2.8 Ga) are fill ages belonging solely to the third clock. They are examples of late, secondary volcanic activity on the two sides of the body; no indication should be derived from them that the whole far side underwent thermal closure at this single point in time, or that the near side remained physically attached to the Earth until the time this basalt formed.
- **The constraint of the moment of release:** In the LEF model the Moon detached and was released at the corotation line a long time (billions of years) before these late lavas erupted.
- **Analysis of the age of the SPA basin (South Pole–Aitken):** The chronological value attributed to the SPA basin dates a specific event within the basin, and depends firmly on whether the material sampled in the analysis represents the impact melt of the basin itself or of a later basin that developed inside it. A chronological gap such as $4.5 - 3.9 = 0.6\ \mathrm{Ga}$ (600 million years) is not a physical measurement of the duration of the body's mechanical construction; it subtracts a reading of the second clock (impact age) from an assumed start date (the formation of the system). This window remains in the scenario as the series' timeline scheme, subject to a separate requirement of an internal chronological consistency check.

3. The test that will settle it

- **The uniform clock-comparison protocol:** The test identifies and maps defined units of closure and preservation on both hemispheres, and isolates fully the events of impact and lava fill. The calculation will compare the first clock (primary crystallisation age) with the first clock, and the fourth clock (preservation age) with the fourth clock, on crustal units formed

by the same genetic mechanism on both sides, and only from this uniform matching will it determine which hemisphere is originally older.

4. The result that would not fit the requirement

- The requirement would be considered not met if the data show that representative crustal units on both hemispheres require, in direct comparison (clock 1 against clock 1), a chronological order opposite to the one predicted — that is, that the primary crust of the near side froze and stabilised before the crust of the far side, beyond the ranges of laboratory uncertainty and the allowed later changes.
- **Current status in the model:** Fixing the timeline and matching the gradings are assigned the status fits; applying the uniform clock-comparison protocol to the primary crustal data is assigned the status open for calculation (the modern basalt data do not provide the confirmation needed for the order of crustal formation).

B.6 R6 — The water measured against the water the transport requires

1. What was measured

- **Analysis of melt inclusions (Melt Inclusions):** The research group of Hauri and colleagues (2011) measured quantitatively concentrations of 615–1,410 ppm water in lunar melt inclusions trapped hermetically within crystals of the mineral olivine ($\left(\mathrm{Mg,Fe}\right)_2\mathrm{SiO_4}$) that formed before the geological eruption [S9].
- **Physical explanation of the measurement (the sealed-bottle mechanism):** A melt inclusion is a tiny droplet of ancient magma that was sealed within the volume of an olivine crystal as it grew beneath the surface. The crystal acts in the system as a completely sealed bottle: when the lava erupted onto the surface and underwent massive degassing (the loss of gaseous components to space), the droplet buried within the olivine kept its original volatile inventory.
- **The empirical mass value:** The parameter ppm is defined in parts per million by mass (Parts per million by mass), so the upper concentration of 1,410 ppm is equal to exactly 0.14% water by mass ($0.0014 M_{\mathrm{magma}}$). Hauri and colleagues emphasised that these values are identical to the orders of magnitude recorded in magmas derived from the upper mantle of the Earth; the lunar interior is not dry.

2. How the model explains it

- **A local constraint of the magma samples:** These empirical measurements are valid for the specific magma preserved within the inclusions and for the lunar source regions (Mantling sources) that gave rise to them, and do not represent a homogeneous or uniform concentration throughout the volume of the Moon at the time of its formation.
- **Separation between the inclusion concentration and the mantle concentration:** There is a firm physical difference between the concentration of volatiles in the magma and their concentration in the source mantle (Source rock): during a process of partial melting (Partial melting) of the source material, the volatiles show a low partition coefficient (Incompatible elements) and pass preferentially into the liquid phase (the melt). As a consequence, the magma formed is considerably richer in volatile components than the parent mantle that gave rise to it.
- **The constraint of the global flux-balance equation:** The change in the overall lunar volatile inventory is represented in the model as a function of time by the following dynamical equation:

$$\frac{dM_{\mathrm{v,L}}}{dt} = \dot{M}_{\mathrm{v,in}} - \dot{M}_{\mathrm{v,loss}}$$

where the parameters are defined as follows:

- $M_{\mathrm{v,L}}$**:** The total mass inventory of volatiles in the Moon.
- $\dot{M}_{\mathrm{v,in}}$**:** The positive supply rate of volatiles arriving from the terrestrial mantle together with the feed flow.
- $\dot{M}_{\mathrm{v,loss}}$**:** The negative rate of volatile mass loss (radiative escape to space, degassing, or ejection back to the parent body).
- **The analogy of the bathtub with the pulled plug:** This equation is equivalent to a bathtub whose tap is open while at the same time the plug at its bottom is pulled. The final water level (the residual inventory measured today) gives no direct indication of the total amount of mass that flowed through the system, unless both flux rates ($\dot{M}_{\mathrm{in}}, \dot{M}_{\mathrm{loss}}$) are computed quantitatively. The equation requires the construction of a dynamical flux-and-composition model, and does not by itself reconstruct the initial inventory. The existence of an early vapour atmosphere is not confirmation of the density chosen for the deep material under conditions of high pressure; the volatiles that lighten the mass must be dissolved within the structure of the material itself at depth, in the thermodynamic phase state that the lithostatic pressure dictates.

3. The test that will settle it

- **The geochemical–thermodynamic model protocol:** The test derives from the equations of Paper 3 the minimum concentration value and the phase state along the vertical feed path. These values are passed through the flux-balance equation (Equation 6), which subtracts the total mass lost to space, and the computed theoretical remainder is compared directly with the empirical values of 615–1,410 ppm recorded in the inclusions.

4. The result that would not fit the requirement

- The requirement would be considered not met if the thermodynamic model shows that the chemically possible range of volatile values does not overlap with, or is lower than, the minimum range needed to drive the feed flow and reduce the density in the geodynamic mechanism of Paper 3.
- **The statistical-control constraint:** Finding a single dry or water-free sample is not evidence against the requirement, since it cannot show that all the geological formation reservoirs were originally dry.
- **Current status in the model:** The presence of internal volatiles is assigned the status fits on the basis of the empirical inclusion data; closing the quantitative model of their dynamical role and the fluxes is assigned the status open.

Summary of the model components assessed in sections B.1–B.6

At this point of the analysis, bringing the data together shows:

1. An oxygen isotopic identity shared by the two bodies (a difference of $0\ \mathrm{ppm}$ within the limits of measurement).

2. The existence of a small metallic core (1.4% of the mass).

3. A global asymmetry of crustal thickness (a thick far side, with a mass balance matching to within 10% the 18 km terrestrial shell of the dome).

4. Preservation of a fixed facing orientation (synchronous rotational locking).

5. Evidence of late volcanic activity on both hemispheres (fill lavas of 2.0 Ga and 2.8 Ga).

6. The presence of internal water within protected magma inclusions.

- All these findings actually exist on the Moon with the status fits, as the scenario of the LEF model structurally requires. For each property the dedicated calculation protocol that will turn the morphological agreement into a decisive control test has been defined exactly.

B.7 R7 — A joint calculation of crustal thickness and the centre-of-mass offset

1. What was measured

- **Structural separation of two centres:** In lunar geodesy and geophysics two separate planetary centres are defined [S10]:
  - **The centre of figure (Center of Figure):** The purely geometric centre of the outline of the Moon's outer surface, computed by a mathematical fit of an optimal sphere to the global topographic data.
  - **The centre of mass (Center of Mass):** The mechanical and gravitational equilibrium point of all the mass and material making up the body, which is the dynamical focus around which the Moon moves in its orbit.
- **The empirical offset vector:** In the real Moon these two centres do not coincide. They lie a linear distance of about 2 km apart, with the centre of mass permanently offset toward the near-side hemisphere (Near Side), which faces the Earth directly [S10].

2. How the model explains it

- **A mechanism of asymmetric mass deficit:** The anorthositic crustal material is lighter and less dense than the mafic mantle material lying beneath it ($\rho_c < \rho_m$). Where the crust is particularly thick (the far side), light material occupies a geometric volume that would otherwise be filled by dense mantle material, and so this hemisphere holds less total mass. The mechanical system is analogous to a hollow ball whose inner wall is lined on one side with a thick layer of cork (a light material) and on the opposite side with a thin layer: the equilibrium point (the centre of mass) is necessarily shifted away from the side thickened with cork and moves toward the side with the thin lining, where the heavy core material lies closer to the surface. The broad asymmetry of the lunar crust (thick crust on the far side and thin on the near side) physically forces the shift of the centre of mass toward the near side.

Firm mathematical formulation of the offset

For the calculation, a firm coordinate axis x is chosen, running from the geometric centre of figure and pointing directly toward the centre of the connection region (the near side), where the crust shows its minimum thickness. The centre of mass is defined mathematically by a total volume integration:

$$\delta x = \frac{1}{M_L} \int_V \rho\left(\mathbf{r}\right) x \, dV$$

where dV is an infinitesimal volume element, $\rho\left(\mathbf{r}\right)$ is the local density function at position $\mathbf{r}$, and M_L is the total lunar mass. For a body with perfect spherical symmetry the integral vanishes, since every mass contribution at coordinate $+x$ is exactly offset by a symmetric contribution at coordinate $-x$; only the asymmetric component of the mass distribution affects the value of the offset (δx).

Scenario A: the thin-shell approximation

Assuming a crustal model with the thickness represented in Equation 3 ($h\left(\psi\right) = h_0 - A\cos\psi$), and defining the firm density-contrast parameter $\Delta\rho = \rho_m - \rho_c$ (representing the mass missing for each cubic metre in which crustal material stands in place of mantle), a central symmetric core does not affect the mass moment along this axis. In the thin-shell approximation ($h \ll R_L$, where all variations of crustal thickness are approximately concentrated at the lunar radius R_L), the solution of the integral yields the following formula:

$$\delta x \simeq \frac{4\pi\Delta\rho R_L^3 A}{3M_L} = \frac{\Delta\rho}{\bar{\rho}_L} A$$

This equation (Equation 8) shows that the linear offset value is exactly equal to the global asymmetry amplitude (A), multiplied by the ratio between the density contrast ($\Delta\rho$) and the mean density of the Moon ($\bar{\rho}_L = 3,346$ ק"ג/מ"ק). The constant, symmetric thickness component (h_0) drops out of the mechanical calculation entirely; only the part that varies angularly from side to side, $A\cos\psi$, moves the equilibrium point. The order of magnitude of the change is clear: for a density contrast of a few hundred kg/m³ (equal to about one tenth of the mean density), an asymmetry amplitude $A = 15$ מ"ק dictates an offset equal to about one tenth of that value — that is, in the predicted range of 1–2 km. The positive sign ($+$) confirms mathematically that the offset necessarily points toward the thin (near) side.

Scenario B: integration over a finite thickness

In geological reality the outer surface keeps an approximately spherical outline, and most of the change in thickness is concentrated at the base of the crust (the Moho boundary, which varies in depth). A full integration over the finite thickness of the layer, where $b = R_L - h_0$ defines the mean radius of the crust–mantle boundary, yields the following equation:

$$\delta x = \frac{4\pi\Delta\rho}{3M_L}\left(b^3 A + \frac{bA^3}{5}\right)$$

This equation (Equation 9) keeps the basic structure of Equation 8, but its first term places the deep radius b in place of R_L: since the varying boundary lies deeper within the volume of the body, its mechanical lever arm about the centre of mass is shorter, and so the computed offset value is somewhat smaller. The second term, containing the component A^3, is a tiny correction arising from the finite dimensions of the thickness variation itself.

Firm numerical values of the offset calculation (δx)

The values are computed under locked example conditions: base thickness $h_0 = 45$ ק"מ and asymmetry amplitude $A = 15$ ק"מ (derived from end values of 30 km and 60 km, as set out in Figure 1).

Density contrast ($\Delta\rho$, kg/m³)	Offset in the thin-shell approximation (km)	Offset with finite-thickness integration (km)
300	1.345	1.243
500	2.242	2.072
750	3.362	3.108

- **Sensitivity analysis and range of fit:** Reading along the rows of the table shows that the effect of the finite thickness reduces the offset value by only a few per cent, and reading down the columns confirms mathematically that the offset value is directly proportional to the density contrast ($\Delta\rho$). The data show that the empirical order of magnitude observed on the Moon (about 2 km toward the near side) is reproduced exactly within the parameter range of the model: the middle row, for a standard density contrast of 500 ק"ג/מ"ק, yields a computed offset of 2.072 km.
- **The inverse-solution and moment-of-inertia constraint:** This agreement is not an independent prediction, but a fit to the data (an inverse solution — Inverse Problem), since the choice of the exact density value is based on using the empirically known answer to lock the input parameter. Changing the value $\Delta\rho$ while keeping the total mass requires a calibration of the internal density values in the deep layers. A full global model must meet at the same time the mass constraints, the tidal response (Tidal Response) and the measures of the normalised moment of inertia (I/MR^2), which measure the way mass is distributed between the centre and the surface and limit the density structures allowed in the mantle and the core.

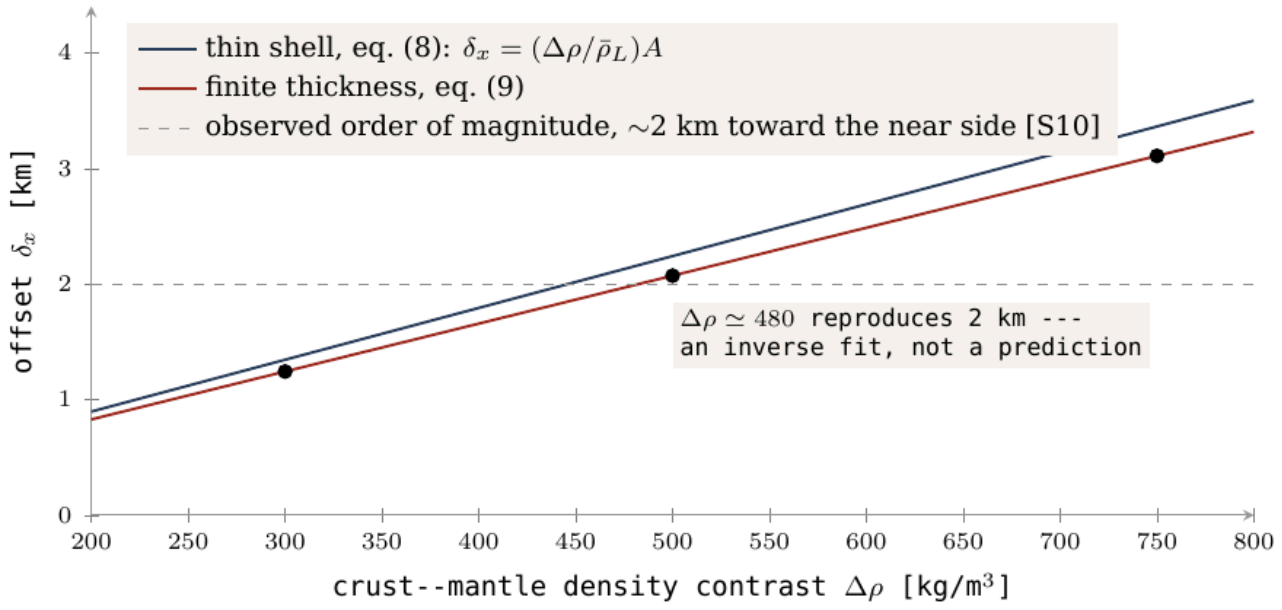


Figure 2. The joint test of §B.7: the centre-of-mass offset produced by the crust of Figure 1, over a range of crust–mantle density contrasts. The sign points toward the thin (near) side, and the magnitude covers the observed order of magnitude; the density that reproduces exactly 2 km is a fit to the data.

Details of Figure 2 · The combined test of crustal amplitude and offset

- **Horizontal axis (X):** The crust–mantle density contrast ($\Delta\rho$), in the range 200 to 800 kg/m³.
- **Vertical axis (Y):** The computed offset value (δx) in kilometres.
- **Graphs and calibration points:**
  - **Solid dark-blue line:** Represents the thin-shell approximation equation (Equation 8).
  - **Solid red line:** Represents the finite-thickness equation (Equation 9). Both lines are entirely linear and pass through the origin $(0, 0)$.
  - **Black points:** Mark the three computed reference points of the table.
  - **Horizontal dashed grey line:** Marks the reference line of the observed empirical order of magnitude of 2 km.
- **Explanation of the crossing point:** The red line (finite thickness) crosses the empirical 2 km line exactly at the point where $\Delta\rho \simeq 480$ ק"ג/מ"ק. The note on the graph stresses firmly that this crossing is a calibrated fit and not an independent prediction.

The spatial-direction constraint

The one-dimensional analysis assumes that the direction of the offset, the location of the connection and the thin side are all aligned along a single geometric line. In the real Moon,

however, there is an angular gap: the centre of the contact as drawn in the model lies at 24.1° N and 335.2° E (Paper 3, §B.4), a location that places it at an angular distance of about 34.0° from the mean sub-Earth reference point ($0^\circ, 0^\circ$ — the mean Earth axis from the centre of the Moon). An axial model centred on the centre of the contact does not predict an offset straight along the mean Earth line, but predicts a tilted offset vector pointing directly toward the contact point. Therefore the geophysical test must compare the spatial vector components of the direction and not only the absolute linear length.

3. The test that will settle it

- **The combined-fit protocol (Global Joint Inversion):** The test requires running a global mathematical inversion that performs a simultaneous joint fit of the crustal-thickness values, the density model, the topography, and the vector components of the centre-of-mass offset within a single, uniform and fixed coordinate system.

4. The result that would not fit the requirement

- The requirement would be considered not met if it is shown mathematically that there is no realistic, physically supported density model in the literature able to explain the size of the offset and its tilted direction together with the other internal constraints of the lunar interior (the normalised moment of inertia and the tidal response).
- **Current status in the model:** The direction of the positive sign and the sensitivity range of the order of magnitude have been derived exactly, with the status fits; carrying out the global combined-fit protocol is assigned the status open.

B.8 R8 — The exposure zones

1. What was measured

- **Continuous development of the surface:** The lunar surface shows morphological, gravitational and chronological properties that vary with their spatial position. The main measures include the linear crustal thickness (in kilometres), the density and population of the oldest impact craters (Crater Density), and the geological characterisation of the large impact basins.
- **Definition of the three concentric zones:** The data show that these properties are arranged in a structure of three concentric (circular) belts around the centre of the near side of the Moon. The geometric boundaries of these belts were defined in the model in two calibration stages:

1. **Earlier exposure rule (the September tests):**
   - **The central contact zone:** $0^\circ - 41^\circ$ (exactly 12.5%, or one eighth, of the lunar surface area).
   - **The base ring (the limb belt):** $41^\circ - 90^\circ$ (37.5% of the area).
   - **The far side:** $90^\circ - 180^\circ$ (50.0% of the area).
2. The current translation rule of Paper 3: places the centre of the contact at the coordinate 24.1° N, 335.2° E, and defines the geometric boundaries at the following values:
   - **The contact cap:** $0^\circ - 31^\circ$.
   - **The middle belt:** $31^\circ - 70^\circ$.
   - **The far part:** $70^\circ - 180^\circ$.

2. How the model explains it

- **The model exposure mechanism:** The LEF model explains the concentric order through the way in which the lunar surface was exposed to space at different stages during its formation and its detachment from the pear. The curved surface of the dome (the far side) was open to space from the beginning. By contrast, the flat base sat on the terrestrial parent body; as the hemisphere rounded into a sphere, the base lifted off it from its outer edges inward.
- **The suction-cup analogy:** The physical mechanism is analogous to a suction cup being peeled off a glass surface: it comes free first at its outer edges, and last at its geometric centre. The last part to be released to space was the contact itself, which was exposed only at the moment of physical release.
- **Derivation of the three main properties:** From this chronological order of exposure three complete physical phenomena are derived:
  1. **Maximum crustal thickness on the far side:** The former shell underwent mechanical thickening during the closure of the dome.
  2. **An old crater density on the far side:** Since this part was exposed to space for the longest time, it must carry the earliest geological record of the crater population.
  3. **Zonal characterisation of basins:** The basins formed in each band record firmly the dynamical and thermal conditions that prevailed in that very zone at the time of their formation.
- **The symmetric-axis constraint:** The model predicts that crustal thickness, preservation age and the character of the basins must each be arranged independently as a function of the angular distance from the centre of the near side, and not according to any other planetary axis.

The firm geometric test protocol

To avoid an artificial fit to the data, the test is defined with quantitative thresholds and firm control axes, all set before the calculation is carried out:

Characterisation of the geometric axes

- **The main axis (the sub-Earth point):** Defined at 0° N, 0° E — the point on the lunar surface lying directly beneath the Earth, since the contact zone faced it directly and the Moon kept this orientation.
- **Control axes and sensitivity protection:** The calculation is run in parallel on three arbitrary control axes (the opposite antipodal point, the lunar north pole, and the centre of Oceanus Procellarum at 23° N, 320° E; the updated contact in Paper 3 is centred at 24.1° N, 335.2° E). The geometric arrangement must hold with the highest significance for the main axis, and must not show a mathematical improvement for the arbitrary axes. This is a statistical safeguard: had an identical arrangement been obtained around the north pole, the pattern would be defined as a random product that says nothing about the validity of the mechanism.

The belt methodology and statistical integration

- **Area weighting (Area Weighting):** To compute the combined means, a firm area weighting is applied to each cell of the grid. The weighting is necessary because in map projections the grid cells shrink in their true area as one approaches the poles of the grid; without this correction, tiny cells would receive the same statistical weight as massive cells in the equatorial band.
- **Trend analysis using the Spearman measure:** The lunar surface is divided into twelve concentric angular belts, each of a fixed width of 15°. The statistical measure for determining the trend (denoted as parameter T5) is Spearman's rank correlation coefficient (r_s), computed between the ordinal number of the belt and its mean thickness. Beside it are reported the values of the differences between the means of the three broad belts (T1).
- **Interpretation of the Spearman correlation in the model:** The measure ranks the different belts mathematically on a scale of values, and examines how closely the ranking follows the order of angular distance outward from the axis:
  - A value $+1$ represents a perfect monotonic rise (each belt consistently thicker than the one before it).
  - A value 0 represents a complete absence of geometric order in the system.
  - A value -1 represents a consistent monotonic decline.

- The measure is intended to examine the very existence of a directional order and not the mathematical function of the rising curve, in full accord with the model's prediction.

The constraint of including the maria (no masking)

- **The maria as a core element of the prediction:** The dark mare basins are the thinnest crust on the near side. In the LEF model they are defined physically as the original ejection openings of the contact disc (Paper 3, §B.5), representing its youngest surface. Therefore the prediction itself dictates an extremely thin crust and a scarcity of old impact craters at the centre of the axis.
- **An absolute ban on masking (No Masking):** Subtracting or masking the mare areas from the main statistical test is strictly forbidden; removing them would take the prediction itself out of the system and void the validity of the test. The standard mask used in the scientific literature for crater counting (which treats the later basalt as a surface cover hiding an older crater population lying beneath it) is run and reported in parallel only as a side control and comparison term, so that the reader can examine the results under the two different reading methodologies. These two geometric boundaries are represented and drawn on the gradient map in Figure 6.

4. The result that would not fit the requirement

- The requirement would be considered not met if the value of Spearman's correlation coefficient (r_s) for the main axis does not show a positive statistical significance considerably higher than that of the arbitrary control axes, or if the weighted belt means show that the crust of the contact cap ($0^\circ - 41^\circ$ or $0^\circ - 31^\circ$) is thicker or older than the outer belts of the far side.

Gravity terms and the mascon mechanism

- **Bouguer anomalies (Bouguer Anomaly):** Represent the record of the gravity field measured by the spacecraft of the GRAIL mission, after subtracting the gravitational pull due to the visible outer morphological topography. The gravitational remainder directly reveals the distribution of mass and density hidden in the subsurface beneath the crust.
- **Mascon (Mascon — mass concentration):** Defined physically as a particularly strong positive Bouguer anomaly located beneath impact basins. The presence of a mascon shows that dense mafic mantle material ($\mathrm{Fe\text{-}Mg\ rich}$) rose vertically and occupies a shallow position close beneath the thin crust of the basin.

- **Ringed mascon basin (Ringed Mascon Basin):** A gravity structure that shows, in addition to the central mass concentration, a distinct circumferential ring of opposite gravitational sign (a mass deficit) surrounding the centre of the basin.
- Constraint on reading Figure 6: the map is divided geometrically by two concentric circles around the sub-Earth point $(0^\circ, 0^\circ)$:
  1. **Inside the solid circle ($0^\circ - 41^\circ$):** The domain of the contact disc.
  2. **Between the solid and the dashed circle ($41^\circ - 90^\circ$):** The domain of the limb belt.
  3. **Outside the dashed circle ($90^\circ - 180^\circ$):** The domain of the far-side hemisphere.
- **The empirical distribution of mascons:** The most massive non-ringed mascons are fixed deep within the domain of the contact disc; the ringed mascon basins are concentrated in the limb belt; while the ordinary impact basins lying on thick crust are located beyond it, on the far side.

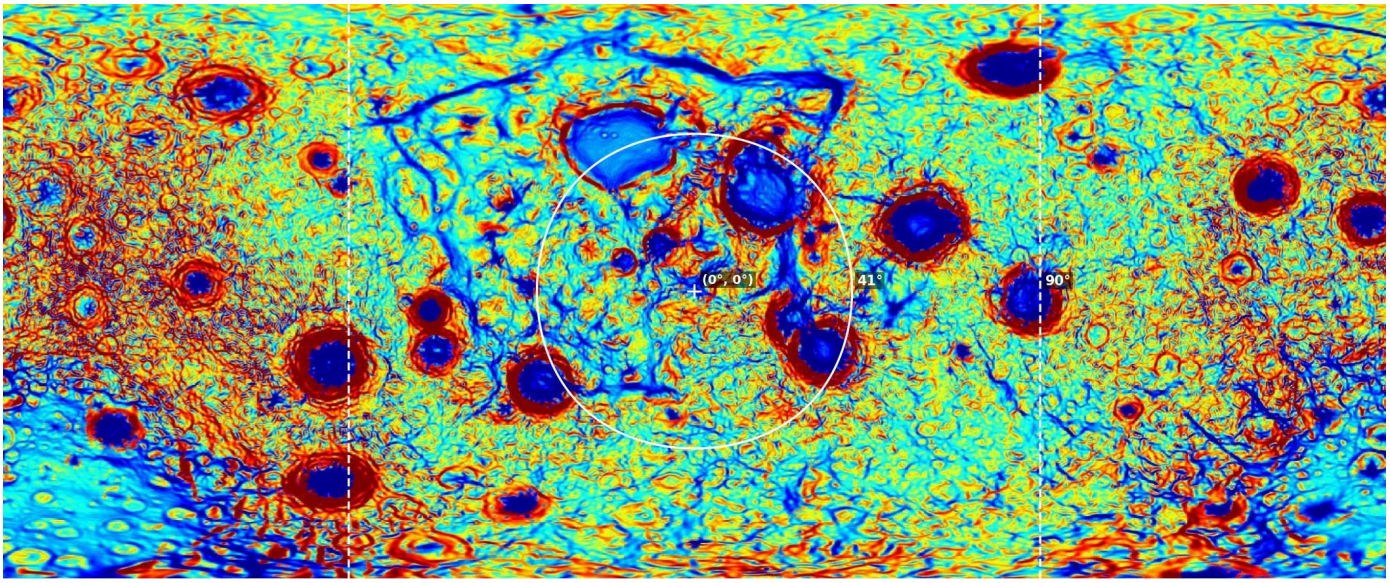


Figure 6. The three exposure zones on the GRAIL Bouguer gradient map. Circles at 41° (solid) and 90° (dashed) from the sub-Earth point (0°, 0°) mark the contact disc, the limb belt and the far side of the translation rule as used for the September tests (Paper 3, §B.4; the current boundaries in §B.8). Blue: mass concentrations beneath thin crust; red: their margins. The strongest non-ringed mascons lie within the contact disc, the ringed basins in the belt, and the basins on thick crust beyond it (tests T1–T5). Cylindrical projection centred on 0°E, latitudes ±75°; the far side is split between the two edges of the map. Credit: LROC QuickMap (NASA/GSFC/Arizona State University); GRAIL Bouguer gravity gradients (NASA/JPL-Caltech/MIT).

Figure 6 · The three exposure zones on the GRAIL Bouguer gradient map

- **Projection parameters:** A cylindrical projection (Plate Carrée) centred on the coordinate 0° E, latitudes $\pm 75^\circ$. The far side is split between the two outermost edges of the map.
- **Gravity legend:**

    - **Blue:** Marks strong positive Bouguer anomalies (mass concentrations/mascons beneath thin crust).
    - **Red:** Marks the raised margins of the anomalies.
- **Geometric boundaries (the September tests):** A solid circle at an angular radius of 41° and a dashed circle at an angular radius of 90° mark the boundaries of the contact disc, the limb belt and the far side (these boundaries were updated numerically to 31° and 70° respectively, as set out in section B.8).

Protocol table of the five statistical tests (T1–T5)

The table defines firmly the passing constraints and the mismatch thresholds. The numerical range lying between the passing threshold and the lower threshold line is defined as a range of indecision (the data do not confirm the prediction with the required strength, but do not stand in physical contradiction to it).

Test	Empirical data product	Model prediction	Numerical threshold for passing	Would not fit if
T1 — Crustal thickness	GRAIL global crustal-thickness model calibrated to the Apollo seismic anchor [S3], grid at a resolution of 1°.	The mean thickness must rise monotonically from the contact disc to the limb belt and to the far side.	- The mean of each belt is greater than the previous one by ≥ 5 ק"מ. - Difference far side minus contact disc: ≥ 15 ק"מ. - In sub-belts of 15°: no more than a single reversal of trend.	- Difference far side minus contact disc < 10 ק"מ. - The ordering of the three belts is not monotonic.
T2 — Old craters	Craters of diameter ≥ 20 ק"מ assigned stratigraphically to the pre-Nectarian or Nectarian period [S12]. Areal density per 10^6	The far side must be the densest, the limb belt intermediate, and the contact disc the sparsest. A theoretical upper bound on the	- Far/limb ratio ≥ 1.3. - Far/disc ratio ≥ 1.5. - Limb/disc ratio ≥ 1.1.	- Far/limb ratio < 1.1. - The contact disc shows a density equal to that of the far side.

Test	Empirical data product	Model prediction	Numerical threshold for passing	Would not fit if
	km² (the mare areas are included in the test).	far/limb ratio $= 3$ (for a population that rose whole on the dome).		
T3 — Ringed mascon basins	Impact basins with a mass concentration and a distinct outer ring in the GRAIL Bouguer gradient [S3, S4].	The geometric centres of the basins are concentrated in the limb belt.	$\geq 60\%$ of the ringed mascon basins lie within an angular range of $41^\circ - 105^\circ$ from the axis.	$< 40\%$ of the basins, or an entirely uniform distribution in the function $\cos\theta$ (randomly smoothed).
T4 — Non-ringed mascons	Impact basins with extreme positive Bouguer anomalies without the presence of a distinct outer ring.	The centres lie exactly within the domain of the contact disc and its geometric margins.	$\geq 70\%$ of the ten strongest mascons lie within an angular radius of 50°.	$< 50\%$ of the strong mascons.
T5 — Uniqueness of the axis	All the data sets of the tests above, rerun simultaneously on the sub-Earth axis, the Procellarum axis, and two arbitrary control axes.	The predicted arrangement holds significantly on the first two axes, and is entirely absent on the control axes.	The statistical trend measures of each of the control axes are lower than half ($< 50\%$) of the value of the measure for the main axis.	A single arbitrary control axis shows an arrangement and organisation of the data at a level of correlation equal to or higher than that of the main axis.

Empirical results and quantitative runs (September 2026)

The results below represent the first empirical run of the statistical tests T1, T2 and T5 on the data sets published in the scientific literature.

T1 — Crustal thickness (quantitative run)

The test was run simultaneously on four different mathematical models of crustal thickness derived from the GRAIL archive [S3] (grids at a resolution of 0.25°, with fixed global means of

34 km, 35 km, 43 km and 43 km respectively). The four models rest on different geophysical assumptions that shift linearly the height of the baseline of the whole crust.

The area-weighted belt means are given in the table below (the exact values and differences were derived from the grid data before numerical rounding):

Crustal model from the GRAIL archive	Contact disc (km)	Limb belt (km)	Far side (km)	Difference: far minus disc (km)
Model 1 (global mean 34 km)	28.3	30.6	38.0	9.65
Model 2 (global mean 35 km)	29.1	31.4	39.2	10.0
Model 3 (global mean 43 km - a)	37.0	39.3	47.3	10.3
Model 4 (global mean 43 km - b)	36.8	39.1	47.4	10.6

- Analysis of the T1 result: the ordering trend is fully monotonic across all four geophysical models: in every single model the crust shows its minimum thickness in the contact disc, thickens in the limb belt, and reaches its maximum thickness on the far side ($h_{\mathrm{contact}} < h_{\mathrm{rim}} < h_{\mathrm{far}}$).
- **The mismatch-threshold constraint:** The numerical differences between the far side and the contact disc are 9.65 km, 10.0 km, 10.3 km and 10.6 km. Since the lower threshold line was set in advance at a threshold value of < 10 ק"מ, in Model 1 the empirical difference (9.65 km) enters the mismatch range by 0.35 km. Models 2–4 lie within the range of indecision (10.0 km lies exactly on the threshold line, 10.3 km and 10.6 km above it).
- **Firm methodological determination:** Test T1 did not pass the required passing threshold (≥ 15 ק"מ), and in Model 1 it lies below its lower threshold line by a gap of 0.35 km. The largest morphological step occurs in the transition from the limb belt to the far side; by contrast, the linear step from the contact disc to the limb belt is considerably smaller than the threshold value of 5 km per step set in the model.

T5 — Uniqueness of the axis (quantitative run)

- **Calculation of the global correlation coefficient:** Dividing the surface area of the body into twelve concentric angular belts of width 15° around the sub-Earth axis yields a constant and particularly high Spearman rank correlation coefficient of $+0.87$ across all four GRAIL crustal models.

- **Statistical significance:** The probability value (p-value) associated with this correlation is:

$$p = 2 \times 10^{-4}$$

This value shows that the statistical chance that such a strong monotonic ordering across twelve belts would arise at random, without a real geodynamic trend in the system, is equal to only two cases out of ten thousand trials (0.02%).

- Behaviour of the gradient in Model 1: the belt means rise continuously and monotonically along ten consecutive belts — out of twelve — from a minimum value of 26.8 km in the belt of $30^\circ - 45^\circ$ to a maximum value of 45.6 km in the belt of the opposite pole, $165^\circ - 180^\circ$.
- **Run of the arbitrary control axes:**
  - The lunar north-pole axis yields a negative correlation value of -0.43.
  - The south-pole axis yields a weak positive correlation value of $+0.43$.
  - The axis of the extreme eastern edge (the eastern limb) yields a value of $+0.76$.
  - The Oceanus Procellarum axis (Procellarum) yields a value of $+0.74$.
- **Firm methodological determination:** Test T5 passed fully. The geodynamic axis predicted by the LEF model organises and orders mathematically the distribution of crustal thickness at a correlation level significantly higher than any other arbitrary control axis. The Procellarum axis comes close in value to the system because it lies geographically only a few tens of degrees from the sub-Earth axis and shares with it an integral part of the same gravity pattern; by contrast, the rotation poles of the Moon — to which the LEF mechanism has no dynamical relation — order the crustal data at a weak level of correlation and in directions opposite to each other.

T2 — The oldest craters (quantitative run)

The test was run in full on the Robbins global crater catalogue [S13] for all impact craters with a geometric diameter equal to or greater than 20 km (a total set of 6,973 craters of all geological ages, since a catalogue sorted by stratigraphic age alone is not available in the archive data).

The quantitative values are computed in units of crater density per 10^6 קמ^2 of surface area:

Methodology of the crater-count run	Contact disc	Limb belt	Far side
Maria included in the set (the firm main test)	99	164	219
Applying the conventional mare mask (the control term)	160	193	218

- **Analysis of the T2 result (maria included):** The graded density ratios between the zones yield firm values of:
  - **Far side / limb belt ratio:** 1.34 (passes the passing threshold set at ≥ 1.3).
  - **Far side / contact disc ratio:** 2.21 (passes the passing threshold set at ≥ 1.5).
  - **Limb belt / contact disc ratio:** 1.65 (passes the passing threshold set at ≥ 1.1).
- **Firm methodological determination:** Test T2 passed fully in all three statistical ratios. The density of old craters on the far-side crust is more than twice the density recorded within the contact disc at the centre. The scarcity of young craters that characterises the mare basins is explained in the model as later landings that took place after the moment of release — a mass of terrestrial material thrown into space and captured gravitationally by the Moon (Paper 6, §M.1), together with a later settling of space-dust components — on the surface of the near side, which was exposed last from the suction cup of the pear.
- **Analysis of the control term (applying the mare mask):** When the calculation is run with the basalt areas masked (counting per 10^6 קמ^2 of unmasked highland area only), density ratios of 1.13, 1.37 and 1.21 respectively are obtained. The predicted monotonic ordering is fully preserved, standing above the lower threshold lines that were set. This finding shows that the concentric organisation of the surface is an inherent global property that does not depend on the way the maria are treated statistically; applying the mask narrows the differential contrast but does not reverse or cancel the trend.
- **The limitation of the overall count:** Since the present empirical count was carried out on craters of all ages (owing to the limitation of the Robbins catalogue), it also includes young, later crater populations that landed in a distributed way over all regions of the body after the moment of release. The presence of these later impacts dilutes uniformly the differential statistical contrast between the belts and cannot produce it artificially. The count sorted by stratigraphic age alone (pre-Nectarian and Nectarian), as the test protocol originally defines, has not yet been carried out in the model.

Protocol of the geometric update and spatial integration (15 October 2026)

1. What was measured (update of the orientation and boundary conditions)

- **Shift of the geodynamic contact axis:** While the early calibration tests (September 2026) rested on the theoretical sub-Earth axis located at the standard coordinate ($0^\circ\mathrm{N}, 0^\circ\mathrm{E}$), the updated geometry of the model places the centre of the physical contact at a firm coordinate of 24.1° N, 335.2° E [Appendix A]. This location shows an angular deviation of 34.0° relative to the mean sub-Earth axis.

- **Update of the boundaries of the concentric belts:** The definition of the three exposure zones on the surface of the body was recalibrated on the basis of the current translation rule of Paper 3, which replaced the earlier threshold radii (41° and 90°) with the following structural threshold values:
  - **The contact cap:** $0^\circ - 31^\circ$ (replaces the contact disc).
  - **The belt (the middle belt):** $31^\circ - 70^\circ$ (replaces the limb belt).
  - **The far part:** $70^\circ - 180^\circ$ (replaces the far side).
- **Methodological constraint:** The impact-crater density data and the crustal thickness of Model 1 were recalculated (rerun) on this updated geometry. This is not to be regarded as running a new test, but as a recalculation of the current zones against the same thresholds and mismatch thresholds set in advance in the table, with the contact cap, the belt and the far part placed in the stead of the contact disc, the limb belt and the far side respectively.
- **Status of the basin tests:** The basin-distribution tests T3 and T4 have not yet been run on the system; all the raw data files, the mathematical processing scripts and the weighted belt means are fixed and stored in the reproduction package of the paper.

Firm numerical values of the recalculation (the current geometry)

The values and differences were derived exactly from the data grids before numerical rounding. The ratio measures are given in geometric order: far part/belt , far part/contact cap , and belt/contact cap.

Data product (current geometry)	Contact cap	The belt	The far part	Ratios or linear difference
Craters ≥ 20 ק"מ per 10^6 km² (mare areas and the SPA basin removed from the calculation)	53	187	220	Ratios: 1.17 / 4.15 / 3.55
Craters ≥ 20 ק"מ per 10^6 km² (mare areas counted in full)	26	148	215	Ratios: 1.45 / 8.42 / 5.81
Crustal thickness (Model 1), km (mare areas remain in the system)	26.3	30.3	36.2	Difference: far minus cap = 9.9 km
Crustal thickness (Model 1), km (mare areas and the SPA basin removed)	31.3	33.0	38.9	Difference: far minus cap = 7.5 km

2. How the model explains the results of the recalculation

Analysis of the crater population (Robbins ≥ 20 ק"מ [S13])

- **Analysis with the maria and the SPA basin removed:** When the basalt areas of the maria and the later morphological structure of the South Pole–Aitken basin (South Pole–Aitken) are removed both from the count numerators and from the area integrals, a uniform "like-for-like" comparison is obtained (primary highland crust against highland crust). The densities yield values of 53, 187 and 220 craters per 10^6 km².
- Verification against the thresholds of test T2: the ratio values obtained (4.15 and 3.55) pass the passing thresholds set in advance (≥ 1.5 for the far/disc ratio and ≥ 1.1 for the limb/disc ratio). The far part/belt ratio (1.17) lies within the range of indecision, since it stands above the lower threshold line (< 1.1) but below the theoretical passing threshold (≥ 1.3). The monotonic ordering trend is fully preserved.
- **Analysis with the maria included (the September convention):** Yields extreme density values of 26, 148 and 215 craters per 10^6 km², which dictate marked division ratios of 1.45, 8.42 and 5.81.

Analysis of crustal thickness (GRAIL Model 1 at a resolution of 1°)

- **Analysis with the maria included (the firm T1 convention):** Since the maria are defined in the model as the physical record of the structural openings, including them in the calculation is essential. The weighted means yield values of 26.3 km, 30.3 km and 36.2 km. The required monotonic ordering holds fully, with the crust of the far part exceeding the crust of the contact cap by 9.9 km. This difference lies a tiny distance of only 0.1 km below the original lower threshold line (< 10 ק"מ).
- **Analysis with the maria and the SPA basin removed:** Yields thickness values of 31.3 km, 33.0 km and 38.9 km, which narrow the global linear difference to 7.5 km.
- The grid constraint of Models 2–4: Models 2–4 require extrapolation and processing of the full grids, and are therefore left for future runs in the model. On the early September geometry these models yielded difference values (far minus disc) higher by 0.37 to 0.92 km than Model 1; this trend is expected to be reflected in the recalculation as well.

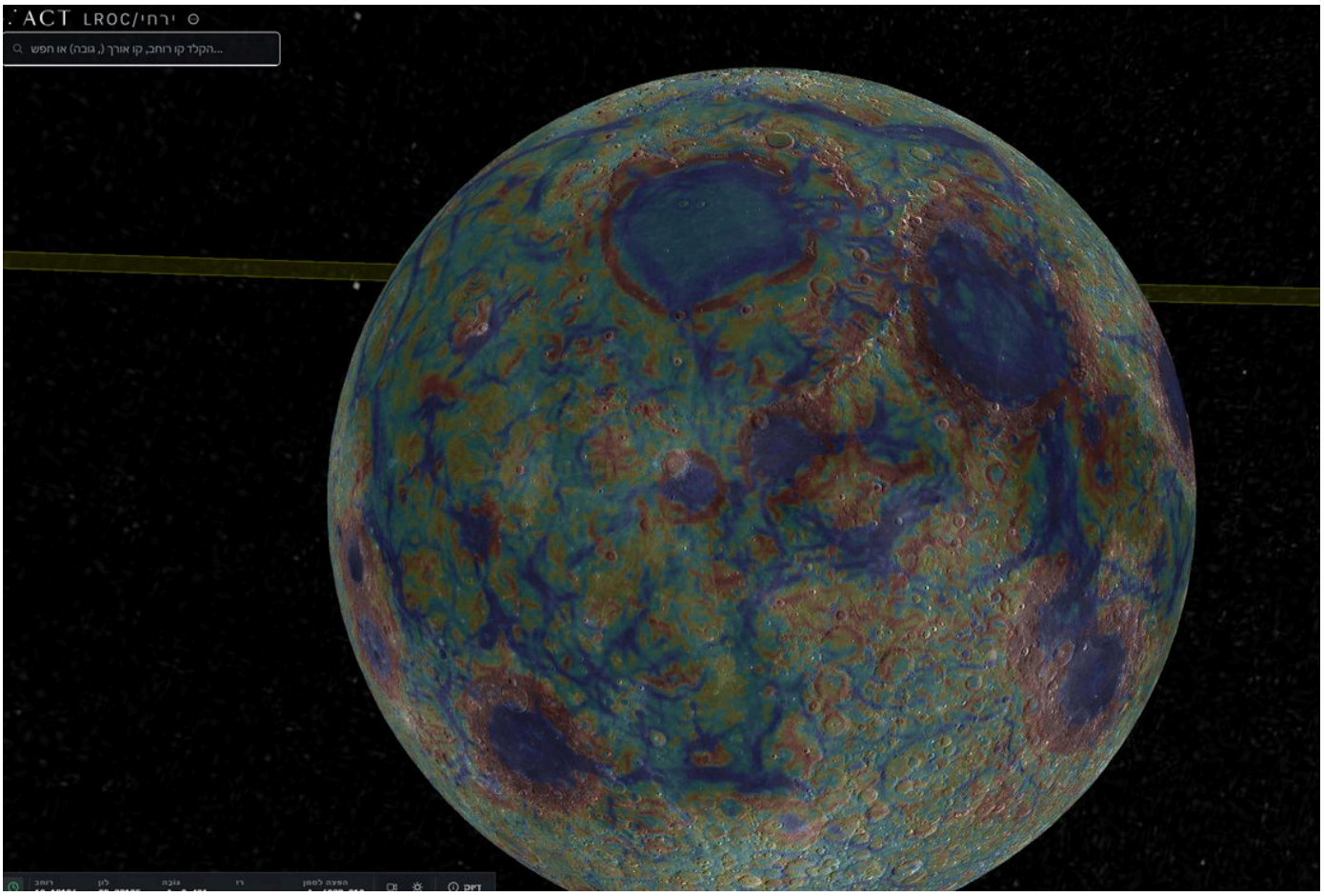


Figure 7. The contact zone — the centre of the near side. The region that was the residual contact disc of the base, exposed only at release: the strongest mass concentrations, the thinnest crust, the youngest and smoothest maria, without rings. Credit: LROC QuickMap (NASA/GSFC/Arizona State University); GRAIL Bouguer gradient layer (NASA/JPL-Caltech/GSFC/MIT).

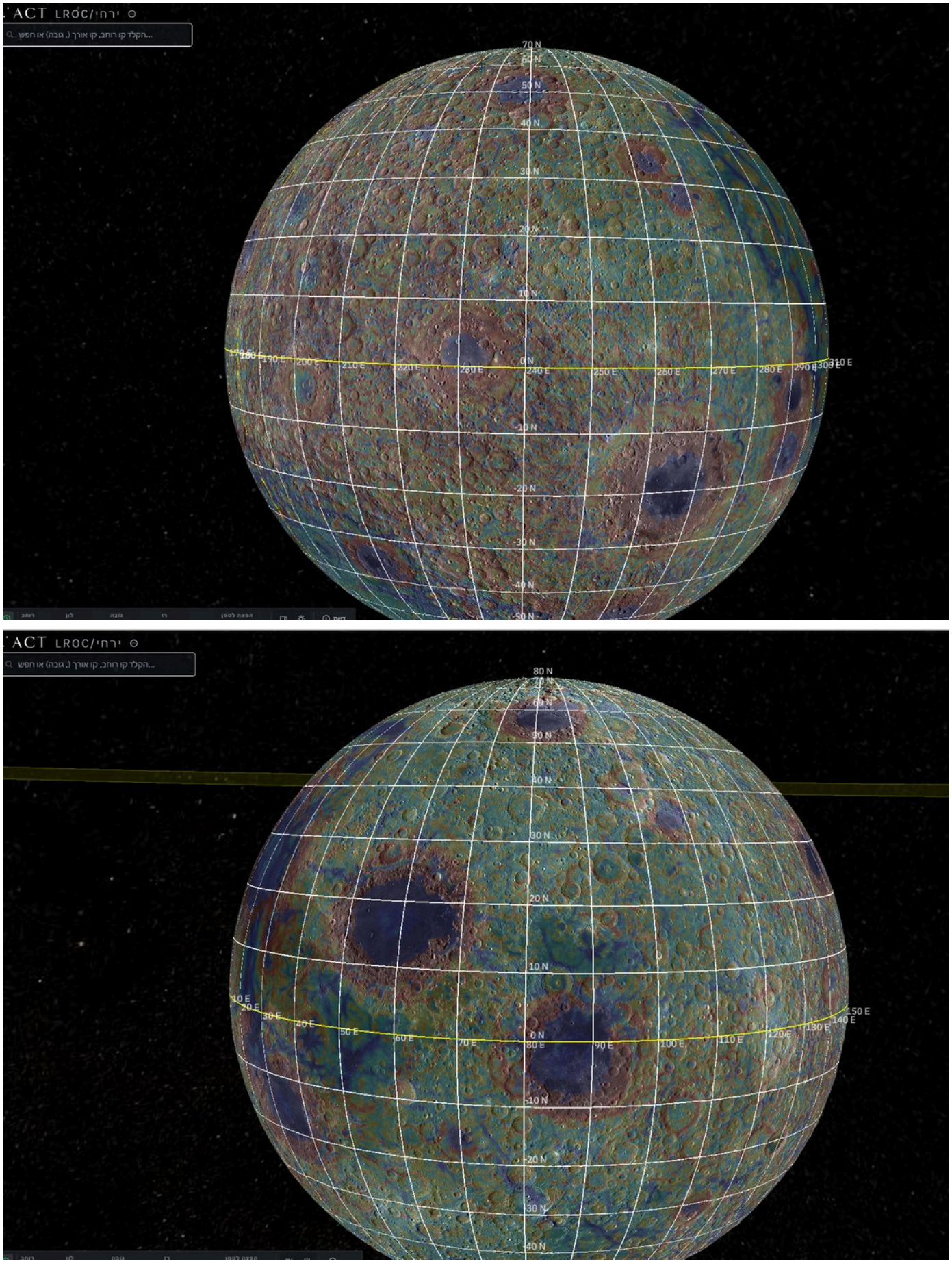


Figure 8. The limb belt — the base ring. Left, the edge of the disc around 270°E; right, around 90°E. Basins with a mass concentration and a distinct ring, on both limbs, the eastern and the western: the region exposed as the contact shrank from the margins onto the disc during the rounding. Its appearance on both limbs is what a

closing hemisphere gives: the base came free along all its margins, and not on one side only. Credit: as in Figure 7.

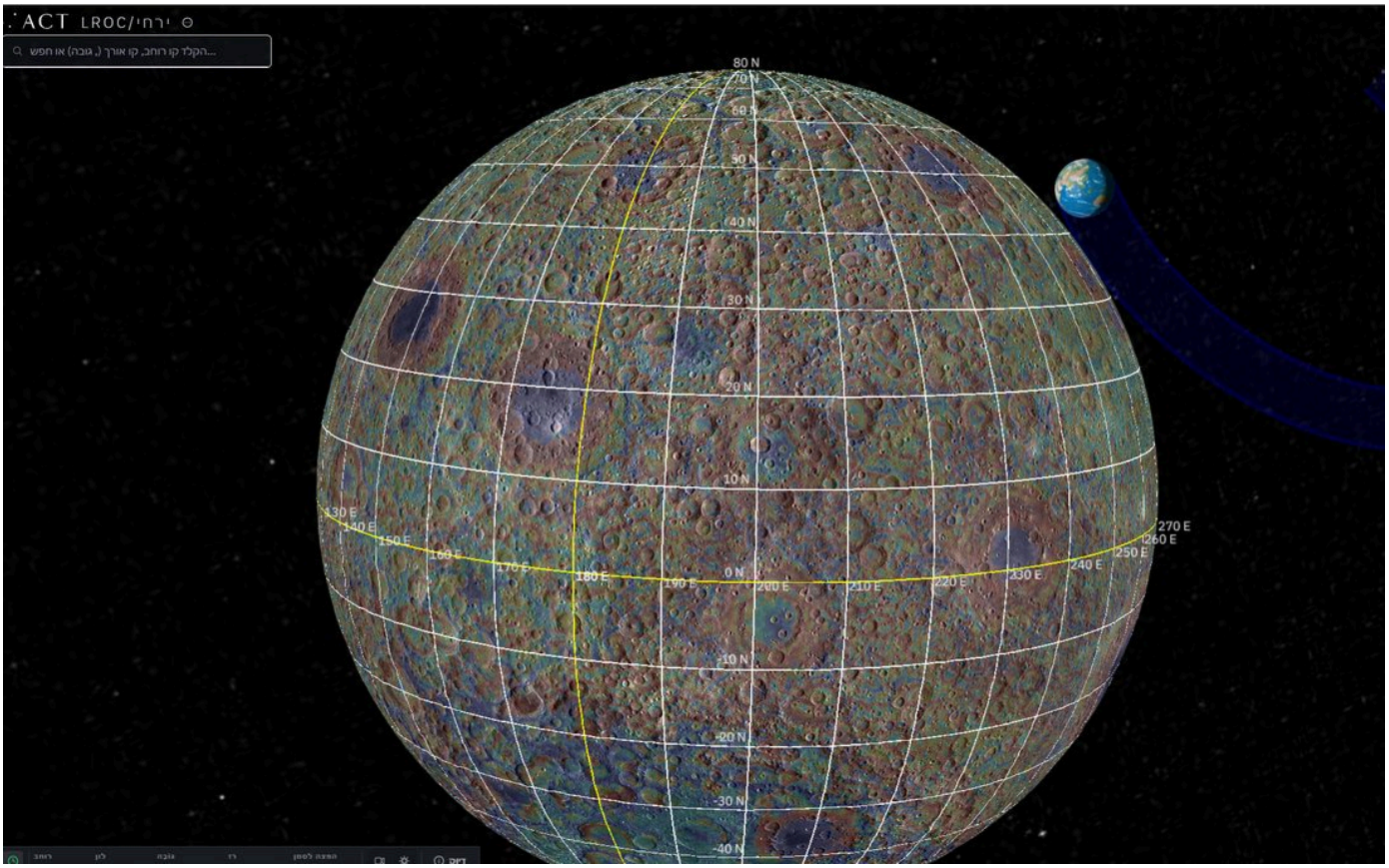


Figure 9. The former surface of the dome — the far side, centred on 180°E, with the Earth above the edge of the disc around 250–270°E. Basins on thick crust, without a mass concentration and without a ring: the crust that was the shell, thickened threefold in the closure, exposed since the stage of the flat dome and carrying the oldest record. Credit: as in Figure 7.

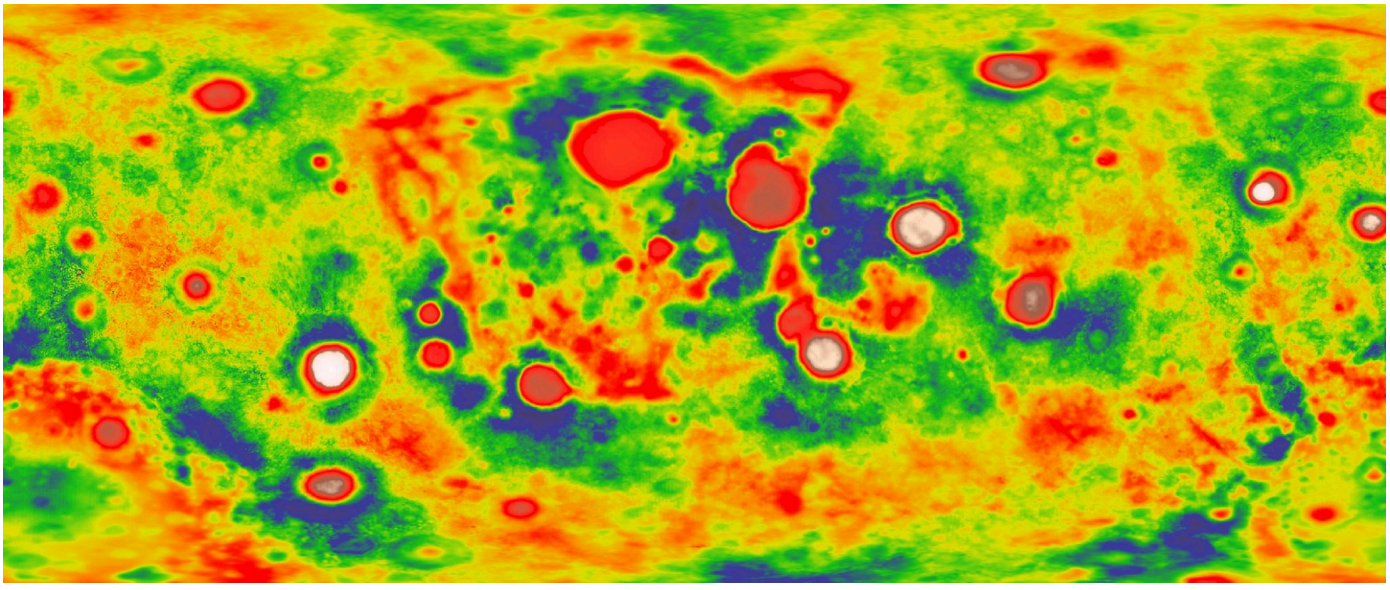


Figure 10. The GRAIL Bouguer gravity anomaly, spherical-harmonic degrees 6–660, cylindrical projection centred on 0°E, latitudes ±75°. The lowest degrees, the broadest variations over the whole Moon, were removed so that features on the scale of a basin stand out. Red and white: positive anomalies, the mass concentrations;

blue: negative. Orientale (left, 95°W) shows the classic target pattern — a positive centre inside a negative ring — the signature of a ringed basin of the belt (T3); Imbrium and Serenitatis in the centre are positive without a distinct negative ring (T4). Credit: LROC QuickMap (NASA/GSFC/Arizona State University); GRAIL Bouguer gravity (NASA/JPL-Caltech/MIT; Zuber et al. 2013).

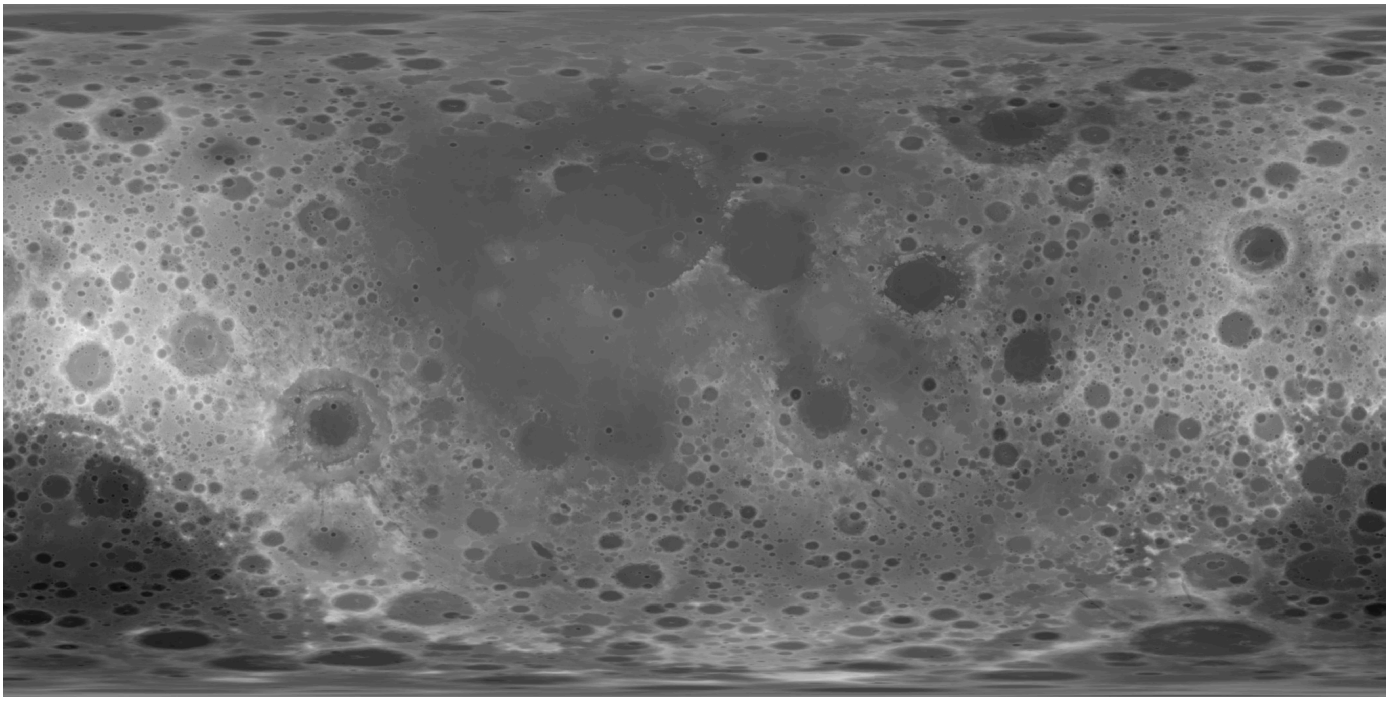


Figure 11. The global elevation of the Moon (SLDEM2015, Kaguya TC + LOLA), cylindrical projection centred on 0°E: dark is low, light is high. The maria and the basin floors on the near side are the depressions; the highlands of the far side at both edges of the map are the heights, and they carry the oldest crater population of the Moon — carried upward on the shell and compressed with it, denser than that of the near side because of the closure of the dome (Paper 3, §B.6; test T2), its craters carried whole and at their size (Paper 7). Credit: LROC QuickMap (NASA/GSFC/Arizona State University); SLDEM2015 (Barker et al. 2016, NASA/GSFC/JAXA).

Details of Figures 7–11: the order of structures on the lunar surface

The figures show how the tectonic and gravity structures vary along the spatial sequence, as a function of the distance outward from the axis at the centre of the near side.

- **Figure 7 · The contact zone — the centre of the near side:** Depicts the area that was the residual contact disc of the geodynamic base, which was exposed to space last, at the moment of physical release. The region is characterised by the strongest non-ringed mass concentrations (mascons), the thinnest crust, and the youngest and smoothest maria, without outer ring structures.
- **Figure 8 · The limb belt — the base ring:** Shows a left view (the edge of the disc around 270°E) and a right view (around 90°E). The region shows impact basins with an internal mass concentration and a distinct circumferential ring in the Bouguer gradient on both

geographic limbs (the eastern and the western alike). This pattern represents the area exposed as the margins of the contact contracted inward during the mechanical rounding. The appearance of the structures on both limbs is a direct geometric product of the closure of a hemisphere: the base came free and tore away vertically along all its circumferential margins, and not on a single geographic side.

- **Figure 9 · The former surface of the dome — the far side:** Centred on the coordinate 180°E (with the Earth located above the edge of the disc in the range 250°E–270°E). Shows impact basins standing on thick, high crust, without mass concentrations and without gravity rings. The region is defined as the crust that was the original shell of the dome, which thickened linearly threefold during the closure; it was exposed to space first, and therefore carries the oldest chronological record of the body.
- **Figure 10 · The GRAIL Bouguer gravity anomaly (spherical-harmonic degrees 6–660):** Cylindrical projection centred on 0°E, latitudes $\pm 75^{\circ}$. The lowest degrees (the broad global variations) were removed mathematically in order to bring out structures on the scale of individual basins:
  - **Warm colours (red and white):** Mark positive anomalies (mass concentrations/mascons).
  - **Blue:** Marks negative anomalies (mass deficit).
  - **The Orientale basin (Orientale - 95°W, at the left edge):** Shows the classic target pattern — a positive gravity centre enclosed within a negative ring — which is the physical signature of the ringed basins of the belt (test T3).
  - **The Imbrium and Serenitatis basins:** Located in the centre and show a strong positive anomaly without a distinct outer negative ring (test T4).
- **Figure 11 · The global elevation of the Moon (SLDEM2015, Kaguya TC + LOLA):** Cylindrical projection centred on 0°E. Dark colours represent low topographic areas (depressions), and light colours represent high areas. The mare basins and the basin floors of the near side are the main topographic depressions. The high highlands of the far side at both edges of the map represent the global heights, and carry the densest and oldest crater population (test T2); this crust was carried vertically upward on the shell of the dome and compressed with it (Paper 3, §B.6), while its ancient craters were carried whole and kept their original diameter (Paper 7).

4. The result that would not fit the requirement

- The requirement would be considered not met if any of the first three statistical tests (T1, T2 or T3) falls below the lower numerical threshold line set for it, or in the case of a

complete mismatch in the axis-uniqueness test T5.

Model status and separation of alternatives (Competitor Hypothesis)

- **Current state in the model:** Defined as open to full testing; the concentric arrangement is the decisive observation for the geometry of the dome (Paper 1); its absence would not fit this geometry.
- **The competing hypothesis (the geochemical thermal anomaly):** The scientific alternative accepted in the literature explains the lunar asymmetry through a single ancient thermal and compositional anomaly centred on the Procellarum region (Procellarum KREEP Terrane — PKT) [S4] — a massive near-side region that was significantly hotter and enriched in heat-producing radioactive elements (K, REE, P, Th, U) relative to the rest of the body.
- **The discriminating power of the limb belt:** The thermal-anomaly model (PKT) successfully predicts and explains a pronounced two-sided contrast (Dichotomy) between the near side and the far side, but it cannot predict or explain the geometric existence of the limb belt (the middle belt) between them, nor the selective concentration of ringed mascon basins within it. The middle belt is therefore the geophysical property that distinguishes and fully separates the two explanations: a simple two-sided contrast is expected in both models, whereas a third zone, forming an intermediate belt with a basin type of its own, follows exclusively from the chronology and order of exposure of the LEF mechanism.
- **Summary of the residual data state:** The predicted axis organises the crustal data at a significance level higher than the control axes (T5 passed); the densities of ancient craters follow the predicted concentric order and all the ratios stand above the pass threshold (T2 passed); the global thickness ordering is confirmed in all four GRAIL models (the ordering holds; the model 1 differences are 9.65 km and 9.9 km in the recalculation). The basin tests T3 and T4 await a full run in the reproduction package.

B.9 Test results instead of a cumulative score

The methodological problem and the statistical separation

- **Rejection of the cumulative score:** Part B has so far set each of the first eight requirements against the empirical measurements. The natural next step would have been to add the results mathematically into a single number — indicating the total of supported rows against those that remain open — and to read the sum as the global probability of success of the model. The paper rejects this approach, and instead presents the isolated

state of each requirement in a unified table that specifies what the data already confirm and which calculations remain to be completed.

- **Mutual dependence of the balance items:** The evidential weight of the different rows must not be combined as though they were independent experiments, and no “probability of success” may be computed from the number of matching rows alone. The physical reason is simple: a thick far-side crust (R3) and a centre of mass offset toward the near side (R7) are not two separate geophysical facts about the Moon. The second property follows directly and deterministically from the first, because light crustal material replacing dense mantle material on one side necessarily shifts the gravitational equilibrium point toward the opposite side (§A.7). The mare basins, which fill the thin-crust domains of the near side, and the young surfaces they carry, are a third morphological view of the same single mass distribution. Counting them as three separate successes is equivalent to weighing one package on three separate scales and reporting three independent weights.
- **Concealment of the classification of results:** A weighted numerical score would conceal and blur the methodological difference between the three types of result that the paper is careful to keep apart: a result derived directly from physical equations, an a posteriori fit of parameters, and an independent prediction not yet tested against data. The table reports each requirement in its own rheological and geometric terms, and the model is assessed exclusively on the basis of the tests capable of examining it.

Status table of requirements R1–R18: confirmation and open tasks

The table defines the exact position of each requirement on the scientific path from the status ”required” to the status ”tested”. It consists of the 8 original requirements (R1–R8) and the 10 later requirements (R9–R18).

Mechanical requirement	What is empirically supported at present	The calculation or measurement needed to complete the test
R1 Terrestrial origin	Significant isotopic similarity in the systems measured in laboratories.	Development of a unified multi-system balance of the sources of the reservoirs and processes.
R2 Little metal	Seismic interior data consistent with a small metallic core.	Hydrodynamic calculation of the transfer of metal in the flow and of its separation.

Mechanical requirement	What is empirically supported at present	The calculation or measurement needed to complete the test
R3 Asymmetry	Large-scale asymmetric crust and non-uniform basaltic activity.	Derivation of a quantitative mathematical profile and a source–crust mass balance.
R4 Orientation	Synchronous rotation locking in a ratio of $1:1$ at present.	Demonstration of the continuity of the connection's direction vector throughout the orbital evolution.
R5 Order of preservation	Separate histories of crystallisation ages, impacts and late activity.	A uniform chronological comparison of units measuring the order of closure.
R6 Volatiles	Presence of internal water in deep magmas sampled in inclusions.	Calculation of a concentration and thermodynamic phase state dynamically sufficient for construction.
R7 Centre of mass	Geometric offset toward the near side; matching vector sign in the model.	A joint mathematical solution for the size of the offset, its direction and the internal structure.
R8 Exposure zones	The test was run statistically: - T5 passed ($r_s = +0.87$ around the predicted axis, in all 4 models). - T2 passed (empirical ratios: 1.34 / 2.21 / 1.65). - T1 did not pass (far side/disc difference: 9.65, 10.0, 10.3, 10.6 km. Model 1 fell 0.35 km below the existing lower threshold line at 10 km, models 2–4 in the indecision range; the monotonic ordering is preserved in all of them).	Running test T2 on a crater population sorted by stratigraphic age; running the basin tests T3 and T4.
R9–R15 Paper 3 pear	R15 passed successfully: documentation of an increased crater density at the poles ($1.09 \rightarrow 1.53$) that depends on crater diameter (according to Paper 7).	Requirements R9–R14 are stated together with their test criteria in section C.6.
R16–R18 Post-pear	R16 supported: the crust of the mare floors is young (a density of 0.8 sharp craters of diameter ≥ 20 ק"מ	Requirement R18 is stated together with its test protocol in

Mechanical requirement	What is empirically supported at present	The calculation or measurement needed to complete the test
	per 10^6 km², against 18.3 on the plains and 207 in the highlands, Paper 6). R17 supported: structural documentation of two material exits (Paper 8, Figure C.1).	section C.7.

Quantitative analysis of the extended test row R8

Row R8 represents the only structural requirement of Part B that has undergone a full run as a controlled statistical test with quantitative thresholds set in advance in section B.8:

- **Test T5 (axis uniqueness):** Examines whether the distribution of crustal thickness is ordered around the predicted geodynamic axis (facing the Earth) at a significance level higher than the arbitrary control axes. A Spearman coefficient value of $+0.87$ confirms mathematically that from belt to belt, outward from the centre of the axis, the crust thickens monotonically almost without exception, and no control axis or pole reaches this level of correlation.
- **Test T2 (crater density):** Compares the density of large craters (≥ 20 ק"מ) between the three exposure zones. The data show that on the far side the density is 1.34 times that of the limb belt and 2.21 times that of the contact disc, and in the limb belt 1.65 times that of the contact disc — full agreement with the chronological order of exposure of the suction cup.
- **Test T1 (global crustal thickness):** Identifies the same monotonic gradient in crustal thickness across the four GRAIL models, with difference values (far side minus contact disc) of 9.65 km, 10.0 km, 10.3 km and 10.6 km. Since the pass threshold value was set at ≥ 15 ק"מ and the lower threshold line at 10 ק"מ, model 1 falls 0.35 km below the lower threshold line and defines T1 as not passed (these values were updated from earlier versions that placed the contrast above the threshold line and called T1 open). The gradient itself is preserved in all of them.

The last two rows summarise the findings of Part C: Of the ten geodynamic requirements added there, R15, R16 and R17 already have measured and confirmed quantitative support from the equations of Papers 6, 7 and 8; the remaining residual requirements are stated together with the tests intended to decide them.

PART C

Further signatures of the rise and of the connection

Parts A and B dealt with the seven classical requirements and with the exposure zones of R8 — properties that every theory of the Moon's origin must explain. Part C turns to signatures that only the LEF route predicts: the crater record carried upward on the shell (C.1), the place on the Earth from which the Moon rose and the place on the Moon where it remained attached (C.2), the openings through which the material passed (C.3), the manner of the release (C.4), the deep structure left behind (C.5), and the requirements added following the pear of Paper 3 and the papers that came after it (C.6–C.7). Several of these rest on the reading of maps, and the part states for each the test that would separate the model's reading from other readings.

The problem and the methodological approach

- **The uniqueness limitation of morphological appearances:** A matching structural appearance identified in geological or gravity mapping can admit several different geophysical interpretations and theories.
- **Test-based separation of interpretations:** The scientific approach of the paper is strict retention of the author's morphological observations as the basis for working hypotheses (Hypotheses), while defining dedicated quantitative tests whose purpose is to separate completely between the different possible interpretations.

C.1 The craters as a record that rose with the shell

1. What was measured (geometric and rheological characterisation of the record)

- **Evolution of the original crustal population:** The properties of the ancient population of impact craters that formed on the terrestrial lithospheric shell before the Moon acquired its final spherical structure were measured. The shell is defined in the model as the light outer crust (SiAl) of the parent body above the source region, which rose vertically as the outer skin of the growing body and became the existing lunar crust (§A.3).
- **The crater transport mechanism:** Since the shell was in a viscous-rigid state, the impact craters formed on it moved together with it as a solid, much as marks drawn on a flexible

sheet move with it when the sheet is lifted. The geometric closing of the dome forced the gathering and concentration of the existing record into a much smaller surface area.

- **Crater compaction parameters (Paper 7):** The quantitative measurements of Paper 7 establish that the rigid shell carried the ancient craters structurally intact and preserved their original linear diameter, whereas the mechanical rounding of the attached body pushed and compacted the craters toward one another in geographic space:
  - **The median diameter at the poles:** 1.06 times larger than the median diameter of craters in the equatorial belt, with the same size threshold retained. Four of five statistical size tests confirm that crater size was preserved independently, and no test shows signs of linear shrinkage of the craters themselves.
  - **Variation of density with size (R15):** Crater density rises significantly as a function of crater diameter: the ratio stands at 1.09 in the diameter range 20–30 km, and climbs to a value of 1.53 in the diameter range 60–100 km (detailed in §C.6). This shows that the more massive and older craters existed during a longer part of the mechanical rounding, and therefore underwent greater spatial compaction. The morphological record underwent geographic compaction (mutual approach), but the craters themselves did not shrink in diameter.

Rule-cancellation constraint: The early calibration rule of Paper 3 — which assumed that a crater on the dome can keep its geometric circularity while physically shrinking to a value of 0.58 of its initial diameter, and that a population can therefore crowd by up to a factor of 3 — was cancelled entirely following the precise size tests of Paper 7.

2. How the model explains this

- **Asymmetry of exposure of the record:** The far-side hemisphere exclusively carried with it the original shell of the dome and its ancient craters; by contrast, the middle belt (the limb belt) and the contact cap (the contact disc) lay beneath the protected base of the pear, and were exposed to space and received craters only at later stages, after the physical peeling and release.
- **The chronological gradient vector:** The direct geodynamic prediction following from this mechanism dictates that the oldest craters must show their maximum density on the far side, a lower density in the middle belt, and a minimum (sparsest) density within the contact cap (this is the order of the statistical gradient measured in test T2, as detailed in §B.8). This ancient population is the defined target of identification, and the model does not claim that all craters visible on the surface today formed at this initial stage.

Mechanical and rheological constraints of the crust

- **The circularity-preservation mechanism (Pure Compaction):** Preservation of the circular morphology of the ancient far-side craters requires a physical bound on non-uniform deformation (shear); a circular shape does not indicate an absence of mechanical stresses, but their character. Geometry establishes that uniform hydrostatic compaction (Pure compaction) reduces the area of a circle without making it elliptical, whereas shear stresses (Shear) or vector stretching in a preferred direction change its geometry to an elliptical shape. The closing of the dome is defined physically as a mechanism of the first type (uniform compaction). A circular crater standing on an ancient floor unit (Highlands) on the far side is a direct product of the model, whereas identification of craters systematically elongated along a preferred vector would not fit the thesis.
- **The rheological viscosity-threshold constraint:** For the terrestrial sial shell to preserve the crater structures on it over time, the material must not undergo plastic or visco-elastic flow on the time scale of the rise of the pear. The Paper 3 model defines the rigid viscosity bound of the system:

$$\eta_{\text{shell}} \gtrsim 10^{23}\ \text{Pa}\cdot\text{s}$$

This extreme viscosity value (in pascal-seconds) ensures a crust rigid enough to hold the crater geometry in a solid and stable state while the body of the dome closed and deformed beneath it. The existence of annular compaction (Annular compaction) carried at the crater rims — the mechanical squeezing felt by the crater ring as the lithospheric surface around it is drawn centripetally inward — remains an independent mechanical prediction of the model, and must be separated stratigraphically from systems of cracks and rifts formed by later meteorite landings or by thermal subsidence processes (Subsidence).

3. The test that would decide

- **The sorted morphometric analysis protocol:** The test requires strict selection and sorting of crater populations on the basis of criteria of relative age and degree of structural preservation (according to harmonised catalogues). The measurement will compute the areal density values, the diameter distribution function and the geometric ellipticity parameter, and will analyse them as a function of the exact angular distance from the centre of the updated contact axis ($24.1^\circ\text{N}, 335.2^\circ\text{E}$), while applying full masking control over the basaltic burial areas of the maria.

4. The result that would not fit the requirement

- The requirement would be considered not met if the statistical analysis shows that the ancient crater population in the far-side hemisphere is not denser than that of the middle belt by a factor of at least 1.3 ($1.3\times$), or if it is shown that the median diameter of these craters is significantly smaller on the far side, or if they show a systematic ellipticity deviation along a preferred vector.
- **Ejection constraint:** The absence of craters on young basaltic fill surfaces is not in itself a trace of the old mechanical connection. The characterisation of the source of the returning terrestrial material (Ejecta) and the kinematic energy of its landing are developed separately in Paper 6.

C.2 Terrestrial source region and lunar connection point

1. What was measured and predicted (location of the rise and of the connection)

- **The two-sided identification constraint:** If the Moon's mass rose and separated from the Earth, the model must determine and state precisely two separate geographic locations: the geodetic location on the Earth from which the structure rose, and the lunar location at which the body remained attached until the moment of final release.
- **The source coordinates on the Earth:**
  - **Base of the pear:** Located near the island of Kiritimati (Christmas Island in the Line Islands of Kiribati), at about 1.9° north and 157.5° west, directly above the Pacific LLVP anomaly (the large low-velocity province in the lower mantle, lying just above the core–mantle boundary).
  - **Back of the pear (the westward lean):** Leaned back westward and northward against the direction of the rotation vector. The author's early reading marks this back in the north-central Pacific, at about 25° north and 165° east (Paper 3, §B.1).
  - **Tip of the pear (where the Moon was formed):** According to the findings of Paper 9 (The Stripes), the centre of the tip is fixed near the coordinate 9.3° north and 169.3° east, at a linear distance of about 3,760 km west of the base at Kiritimati (3,770 km in earlier versions). This region is bounded between the two morphological bends of the stripes, which lie 3,360 km apart. The stripes in the North Pacific are the folded record of the leading wall (the front) of that pear, which folded as it collapsed back.
- **The contact coordinates on the Moon:**
  - **Centre of the construction mound:** Located near 16.6° north and 13.1° east.
  - **Centre of the last contact (the connection axis):** Fixed near 24.1° north and 335.2° east, and within it lies the Mare Imbrium basin, which is the main feed opening (Paper 3,

§B.4), alongside the Mare Serenitatis basin, defined as the secondary ejection opening. The author's early reading (about 23° north and 320° east) serves as a reference and comparison item only.

2. How the model explains this

- **Modern geomorphological characterisation:** These identifications are based on the reading of present-day landforms and gravity maps, and are not palaeogeographic coordinates reconstructed independently from models of tectonic plate motion. The existence of young basaltic oceanic crust in the Pacific today does not contradict the model, since the surface has been renewed; the reading refers to the deep hemispheric asymmetry of the lithosphere and of the mantle beneath it.
- **Agreement with the orbital release latitude:** The centre of the contact on the Moon is fixed near 24° north. On the other hand, the orbital-mechanics equations of Paper 4 derive that the Moon was physically released from an ancient latitude of exactly 22.5°. This convergence shows that the contact point read from the surface morphology, and the release latitude derived independently from the orbital dynamics, agree with each other within a small error of a few degrees.

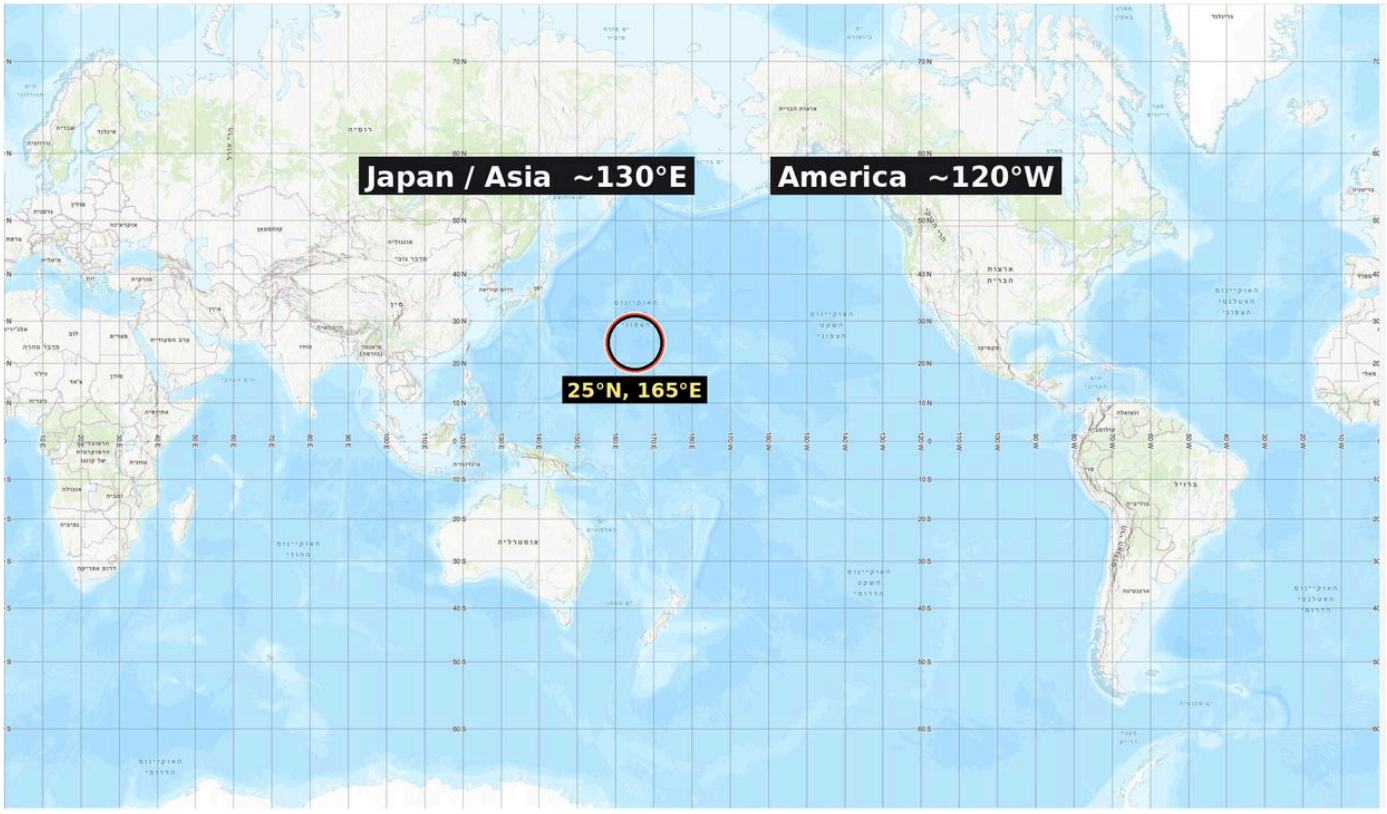


Figure 12. The source region. The marked region, centred near 25° north, 165° east, in the north-central Pacific, is the back of the pear, leaning westward; its base lay to the east, near Kiritimati (1.9° north, 157.5° west); the tip, according to Paper 9, lies near 9.3° north, 169.3° east. The region lies between the margin facing Asia

(about 130° east) and the margin facing America (about 120° west). The present sea floor is younger than 180 million years; the reading refers to the hemispheric asymmetry of the lithosphere beneath it (Paper 3, §B.1). Credit: base map, Google Maps, © Google; annotations by the author.

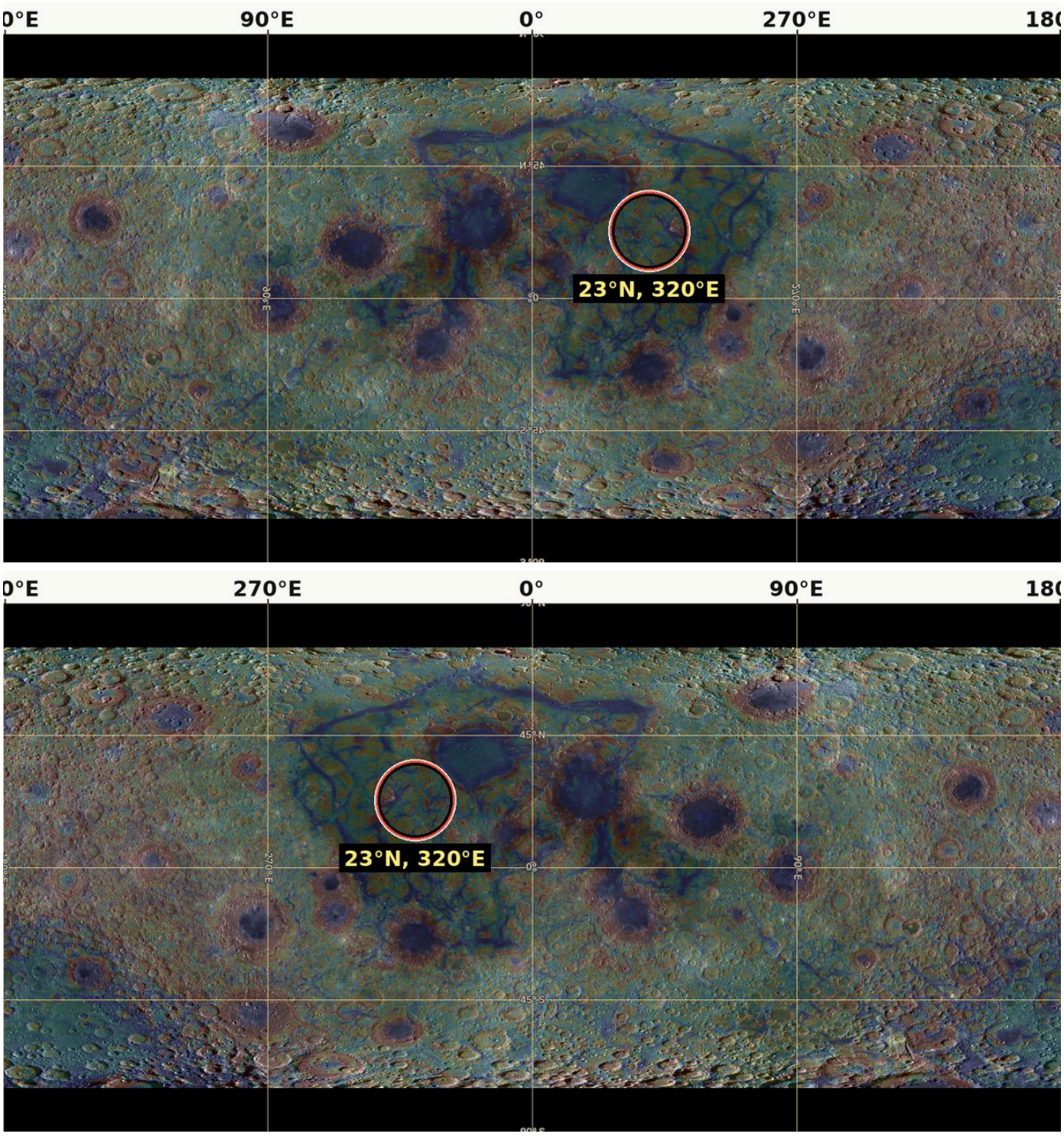


Figure 13. The contact on the Moon. The early reading of the contact, about 23° north, 320° east, on a cylindrical map centred at 180° east (left) and at 0° east (right); the contact as outlined later by the author is centred at 24.1° north, 335.2° east (Paper 3, §B.4), with Imbrium the main exit and Serenitatis the secondary, close to the latitude from which the Moon left, 22.5° (Paper 4). Credit: LROC QuickMap (NASA/GSFC/Arizona State University); GRAIL Bouguer gradient layer (NASA/JPL-Caltech/GSFC/MIT).

Details of Figures 12 and 13: the source and contact regions

- **Figure 12 · The terrestrial source region (credit:** base map Google Maps; annotations by the author): Shows the Earth's side of the model. The back of the pear is located in the north-central Pacific, bounded geographically between the margin facing the Asian continent (about 130° east) and the margin facing the American continent (about 120° west). The base of the structure lay to the east, near Kiritimati, and its tip at 9.3° north, 169.3° east. Since the present ocean floor is younger than 180 million years, the figure does not present the modern crust as a direct remnant, but attributes the record of the event to the deep hemispheric asymmetry beneath the lithosphere (Paper 3, §B.1).
- **Figure 13 · The contact on the Moon (credit:** LROC QuickMap / GRAIL Bouguer Gradient): Shows the Moon's side in two layouts: once with the far side at the centre of the map (left) and once with the near side at its centre (right). It shows the location of the contact at 24.1° north, 335.2° east (Imbrium the main, Serenitatis the secondary), close to the dynamic release latitude of 22.5° derived in Paper 4.
- **Application of the geometric translation rule:** Passing from the point-like imprint deficit to the mapping of the entire source requires applying the translation rule of Paper 3: the total surface area is preserved up to a uniform hydrostatic compaction factor. The rim of the base of the terrestrial dome translates to the edge of the visible disc (the limb at 90°), the top of the crown translates to the hidden far side ($90^\circ - 180^\circ$), and the contact surface becomes the near-side cap with a radius of 41° according to the September tests, or with a radius of 31° according to the updated rule (Paper 3, §B.8). What physically touched the Earth must remain facing it. A simple mirror simulation (as examined in Paper 9 at a ratio of $\times 1.37$) is limited exclusively to the surface of the narrow contact, and does not replace the mapping rule of the whole body.

3. The test that would decide

- **The combined mapping protocol:** The test requires defining mathematically in advance the boundaries of the terrestrial source region, the boundaries of the lunar connection region and the ranges of geodetic location error. The global mapping model will be examined on the basis of its ability to predict and reproduce separate geophysical properties and signs that were not part of the parameter set used to select and calibrate the regions in advance, in order to avoid the fault of circular inference (Circular Reasoning). The area calculation in Paper 3 sets the dimensions, but does not constitute independent confirmation of the ancient geographic location of the region.

4. The result that would not fit the requirement

- The requirement would be considered not met if it is shown mathematically that applying the translation rule and the geometric mapping to the predicted source regions produces structural distortions or contradictions that do not fit the crustal thickness map or the gravity field of the present-day Moon.

C.3 Three groups of openings and rings

1. What was measured and predicted (classification of the ejection structures)

- **Characterisation of the crustal openings:** The openings are defined as the geological points at which the lithospheric shell was pierced and deep mantle material flowed upward through them. The dynamic geometry of Paper 3 divides the openings into three distinct groups, on the basis of the length of time each group was open directly to space:
  1. **Openings on the crown (the far hemisphere):** Were exposed to space first, and ejected magmatic material during most stages of the construction of the body while exposed.
  2. **Openings on the flank:** Were exposed to space gradually during the rise and rounding of the dome.
  3. **Openings beneath the base (the near hemisphere):** Were covered and completely sealed against the Earth throughout all stages of construction, and became exposed only at the moment of physical detachment and release; the deep material passed through them without their being exposed to space before the detachment.
- **Late geological constraint:** This structural division does not claim that every modern basaltic basin (mare) is an ancient opening. Even after the mechanical connection to the Earth was detached, crustal structures and density anomalies formed during the early formation stages continue to exert a focused influence on late volcanic activity (secondary volcanism) originating in the deep lunar mantle.

2. How the model explains this

- **Distinction between a gravity ring and a topographic rim:** In a gravity map (Bouguer gradient), a ring is defined as a geometric circle of mass excess or mass deficit, often lying deep beneath the surface. By contrast, a rim is defined as an external topographic expression only, of elevated ground. Establishing the existence of a gravity ring requires processing gravity data (GRAIL) and applying a crustal density model, whereas mapping an external rim requires altimetry data (LOLA).

- **Rejection of a global viscosity law:** There is no general physical law dictating that only material of a certain intermediate viscosity value can produce a gravity ring. Therefore, identification of the structures requires a combination of geological classification, measurement of rim height, mapping of the subsurface structure and its chemical composition, and not only a visual colour estimate on a Bouguer gravity-gradient map.

3. The test that would decide

- **The basin classification protocol (Basin Classification Framework):** The test requires defining and locking in advance the geometric location groups before carrying out the analysis of the gravity rings. The protocol requires publication of a complete global basin list and a quantitative height index computed relative to the topographic surroundings. The empirical data will be compared simultaneously against competing models of meteorite impact, magmatic filling, thermal subsidence (Subsidence) and magmatic intrusion (Dikes).

4. The result that would not fit the requirement

- The requirement would be considered not met at the level of the openings if a mathematical mismatch is shown between the empirical basin data and the three-group classification hypothesis.
- **Multi-level model structure constraint:** A mismatch at the level of the openings is limited exclusively to this geological classification thesis, and does not affect the physical corotation equation of Paper 2, since the corotation equation that sets the dynamic release conditions rests on global mass and momentum balances and does not depend on the internal sorting of the crustal openings.

C.4 Release and rounding

1. What was measured and predicted (the separation mechanism)

- **The dynamic distance constraints:** The physical mechanism of the moment of detachment is governed by two rigid threshold distances from the centre of the Earth:
  - **The corotation radius (Corotation Radius):** The geodynamic distance at which a body rotating synchronously with the Earth's daily rate moves at exactly the speed required for a free circular Keplerian orbit; inside this radius a released body will collapse and fall back gravitationally, and beyond it, it will be accelerated and carried outward. At the moment of release (an Earth day of 5.77 hours), the corotation radius lay at a fixed distance of 16,399 km in the equatorial plane (about 2.6 Earth radii).

- **The Roche limit (Roche Limit):** The dynamic distance within which the differential tidal forces (the difference in the Earth's pull between the near and far parts of the satellite) exceed the self-gravity of the body that holds its mass together, leading to mechanical disruption.

2. How the model explains this

- **Separation of corotation from tidal survival:** Paper 2 separates mathematically between reaching the corotation radius and the structural survival of the body against the Roche limit. A simple comparison between the Moon's self-gravitational acceleration and the gradient of the terrestrial tidal field provides an engineering scale similar to the Roche limit of a dense, rigid body, but does not by itself prove the physical rate of rounding, the mechanical failure of the interface or the degree of structural stability of a hot, viscous body. A weak or high-temperature planetary body may undergo mechanical tearing even if it lies beyond the theoretical Roche limit; the comparison is a reference measure and not a continuous writing of a dynamic history.
- **The peeling chronology (Peeling Mechanism):** The order of stages is fixed in the model: the lunar body underwent mechanical rounding to the shape of a sphere while attached to the pear, and at the moment of release its outer shell was rigid and round (Paper 7). The detachment took place by a mechanism of gradual peeling:
  1. Along the broad 6,000-km base, the sial material (SiAl) on which the body sat peeled mechanically off the deep sima layer (SiMa) and was carried along with the Moon.
  2. Along the reduced 3,000-km contact disc, the deep feed heat melted the sial material completely and smoothed the surface under the influence of the gravity field.
  3. When the Moon's centre of mass crossed the physical corotation radius, the corotation forces pulled the remainder of the melted contact interface free and detached it completely (Paper 3, §C.4a, C.5). There was no bottleneck or narrow, thin stalk.

Interpretation of the lithospheric peeling mechanism

The term peeling defines a mechanical separation occurring along the geological continuity layer of lowest mechanical rigidity (the weak layer at the interface). The body sat on the terrestrial sial crust, beneath which lay the sima; the thermo-mechanical boundary between them, which was hot and mobile (low viscosity), was the weakest working surface in the connection (as detailed in the environment table).

The physical analogy is equivalent to a strip of adhesive tape (masking tape) lifted from a painted wall and taking with it the whole paint layer (the sial) while leaving the underlying plaster layer (the sima) exposed on the wall: the lunar body swept the sial along with it and left the sima behind. The separation process reduced the diameter of the attached area from 6,000 km to 3,000 km, where the heat flux of the feed melted the sial instead of lifting it, and the ground underwent gravitational smoothing. The final stage required no mechanical blow or severing of a stalk; when the centre of mass reached the corotation radius, the tangential rotational speed of the body became exactly equal to the free orbital speed, and the corotation forces pulled away the remainder of the molten contact and separated the bodies.

3. The test that would decide

- The model requires explicit identification, within the crustal structure and in the gravity field, of the geodynamic signatures derived from this chronology:
  - **The middle belt (the limb belt of R8):** Must carry within it the mass of the peeled and displaced sial.
  - **The contact cap (the contact disc):** Must show the lowest crustal thickness values together with the strongest non-ringed mascon anomalies, without negative gravity rings around them (as detailed in Figures 6 and 7).

C.5 The deep anomaly and the north–south asymmetry

- **Integration of the mantle anomalies (lobe A):** Lobe A is defined physically as the part of the deep reservoir that lay directly beneath the terrestrial source region and fed the rising lunar body magmatically. Its deep remnant today is interpreted in the model as the Pacific LLVP (Large Low-Shear-Velocity Province) ridge, while the African LLVP anomaly is defined as the symmetric opposite lobe, which was not disturbed (R11, §C.6).
- **Limitations of the tomographic interpretation:** Low seismic velocity is not a direct or quantitative measurement of a crustal or mantle mass deficit, and there is no mathematical model in this paper capable of carrying a specific structural anomaly through billions of years of plate-tectonic mixing and geochemical differentiation. Seismic waves slow both in hotter material and in material of different chemical composition (iron-rich, for example), and a slow region can be denser than its surroundings rather than lighter; a wave-velocity map alone does not define where mass is missing.
- **Constraint on the hemispheric (north–south) asymmetry analysis:** Examining the asymmetry between the northern and southern hemispheres requires prior mathematical separation of the principal asymmetry between the near side and the far side, and full

subtraction of the local morphological influence of the large impact basins. It is strictly forbidden to rely on data from the South Pole–Aitken (SPA) basin alone to confirm the asymmetry: the SPA basin is so massive that by itself it distorts the averages and makes the southern hemisphere look morphologically different from the north. The global north–south asymmetry signal must show statistical significance also after full subtraction of the SPA basin.

C.6 Requirements added following the pear

Paper 3 defines the geodynamic shape of the original connection: a single bulge of the Earth's own material, stretched into the shape of a pear that holds itself mechanically by its own gravity (an equilibrium figure), whose western tip (the dome) closed as the body of the Moon, and whose body collapsed and fell back into the mantle at the moment of release. This physical description adds a set of 7 further requirements (R9–R15) on the Moon and on the Earth, stated under defined confirmation conditions and mismatch thresholds.

Table of the structural requirements of the pear (R9–R15)

Mechanical requirement	What the pear physically requires in the model	Geophysical test protocol	Would not fit if
R9 — A pear under its own gravity	A connection about 9,000 km wide at the Earth's surface, narrowing in one continuous arc to a contact of 3,000 km, holding itself as an equilibrium figure and lying entirely within the radius of the corotation line.	Theoretical calculation of the equilibrium figure of a fast-rotating Earth under internal pressure, with an asymmetric body at its tip (Paper 3, §C.9).	No self-supporting geometric shape exists mathematically for any value of density and rotation rate (day length) in the admissible ranges.
R10 — The pear returns	The residual mass of the body of the pear, equivalent to a range of 7 to 12 lunar masses (about 11% of the mass of the whole Earth), collapses and falls back in its entirety at the moment of release; no material component of the connection remains in orbit.	Identification of a dense, iron-rich pile of material at the base of the mantle (CMB) directly beneath the base of the pear; complete absence of material remnants in near orbit.	A remnant or debris of the connection is identified in orbit around the Earth, or in the case of an absence of a sunken mass anomaly beneath the source region.

Mechanical requirement	What the pear physically requires in the model	Geophysical test protocol	Would not fit if
R11 — The Pacific LLVP	The base of the pear (near Kiritimati at 1.9°N, 157.5°W) lies directly above the Pacific LLVP (the returned material); the African LLVP is the opposite lobe, which was not disturbed (Paper 3, Part E).	Comparison of the base coordinates with the outlines of the LLVP in published tomographic models; the Pacific LLVP must be richer in iron, lower and broader than the African one.	The base lies outside the outlines of the LLVP, or in the case in which the two LLVP regions (Pacific and African) show complete identity in composition and in morphological shape.
R12 — The KREEP complex	The torn rim of the terrestrial sial crust, which underwent mechanical folding inward at the rim of the base during the peeling, carries and fixes the KREEP elements on the near side around the bulge basin (Procellarum).	Mapping of the KREEP distribution against the theoretical base-rim lines; analysis of the most primitive anorthosites, which must show depletion of potassium (K) and thorium (Th).	The KREEP concentration shows no geometric relation to the rim of the base, or if the primitive anorthosites retain the terrestrial sialic geochemical signature.
R13 — The core as a gauge	The small metallic core (1.0%–2.0% of the Moon's mass) traces and quantitatively measures the physical proximity of the feed flow to the sinking terrestrial iron.	Development of a hydrodynamic transport model of iron droplets within the rising silicate flow, defined as a function of the depth of the geological source.	The computed core mass fraction requires a source lying directly within the Earth's core, or yields a core with no metallic phase at all.
R14 — Absence of foreign material	The lunar body is built of the Earth's material only: no mass component of an external impactor (planetesimal) exists in its volume.	Combined analysis of the multi-system isotopic balance (R1), and examination of the basalts of the Pacific plumes rising from the LLVP, which must carry a purely terrestrial signature.	Identification of a distinct non-terrestrial chemical or isotopic component within the total volume of the Moon or within the materials of the Pacific magmatic plumes.
R15 — Crater density by size	The mechanical rounding of the attached body pushed and compacted the ancient craters at the poles; the degree of spatial compaction	Calculation of the net ratio of crater density between the poles and the equator as a function of crater diameter, with full removal	The graded density ratio is constant or decreases, and does not rise with increasing crater

Mechanical requirement	What the pear physically requires in the model	Geophysical test protocol	Would not fit if
	increases the larger and older the crater.	of the maria and the SPA basin: 1.09 in the range 20–30 km, climbing to 1.53 in the range 60–100 km.	diameter above the threshold of 20 km.

Details of the row mechanisms and the physical constraints

- **R9 (balance of the equilibrium figure):** The massive body of the pear does not hold up mechanically by virtue of the rigidity or structural strength of its material — since no solid material is strong enough on such a planetary scale — but, like the Earth's equatorial bulge, the bulge is stabilised as an equilibrium figure (Equilibrium figure) through a full dynamic balance between its own gravity field, the internal hydrostatic pressure, and the centrifugal forces of rotation. All parts of the body must lie within the radius of the corotation line, so that their tangential speed is lower than the free orbital speed and they remain mechanically bound to the Earth; only the material of the tip, on reaching the line, undergoes gravitational release.
- **R10 (the collapse mechanism of the connection):** With the detachment of the tip, the body of the pear (equivalent to about nine lunar masses) loses the external tension vector and collapses gravitationally back into the Earth's interior. Since all the mass lay within the corotation line, no material component can remain in a residual orbit. The swallowing left morphological folds mapped in Paper 9 (the stripes on the front, and the arcs on the back), whose tests are defined there and are not counted as an independent requirement in this chapter.
- **R12 (geochemistry of the KREEP complex):** The KREEP elements are defined as incompatible elements (Incompatible elements) that do not fit into the crystal structure of common minerals during magmatic crystallisation, and therefore concentrate systematically in the last residual liquid phase. On the Earth they are fixed within the sialic continental crust. In the LEF model, KREEP is the chemical record of the torn rim of the terrestrial sial, which underwent mechanical folding inward during the peeling of the body at the rim of the base. The lunar highlands are built of anorthosite (Anorthosite), composed almost entirely of the mineral plagioclase; the most primitive anorthosites are extremely poor in potassium and thorium. Since they originate in the terrestrial sial crust, the model requires that these elements were carried away and depleted from them into the structure of the folded rim (Procellarum PKT).

- **R15 (confirmation of crater compaction):** This measure is the only mechanical requirement of group C.6 that was run, computed and fully passed the pass thresholds in the quantitative measurements of Paper 7 (§D.1–D.2). The measured graded ratio values ($1.09 \to 1.53$) confirm that the massive craters underwent greater spatial compaction at the poles as a consequence of the length of their stay on the shell during the rounding of the attached body.

C.7 Requirements added after the pear

Three papers completed chronologically after the development of the pear model add three independent requirements (R16–R18): the definition of the maria as structural openings (Papers 5 and 6), the residual trace on the far side of the heat energy that passed through them (Paper 8), and the application of the same physical mechanism to the Mars system (Paper 10). Each requirement takes a defined component of the existing model — the ejection openings of section C.3, the heat flux of the feed, and a satellite composed of interior material of the parent planet — and defines its geophysical trace at the location where it can be measured with strict empirical rigour.

Table of the later structural requirements (R16–R18)

Mechanical requirement	What the model physically requires	Empirical test protocol	Would not fit if
R16 — The 34 maria are the openings	The mare basins are 34 openings above feed domes on the near side (Paper 5, Appendix A). The dark plains (Procellarum, Tranquillitatis) are not openings but crust melted in place (In-situ melting). The returning terrestrial material came out through the openings; craters that landed on a liquid, active mare floor underwent complete melting (Paper 6).	Comparison of the density of sharp craters of diameter ≥ 20 ק"מ on the mare floors against the plains and the highlands, over the 34 geometric footprints; analysis of the melted fraction by size and time of landing.	The mare floors turn out to be rich in sharp craters at a level equivalent to plains of the same geological age, or if the shallow craters within them were buried under late lava flows and did not undergo melting.
R17 — Two exits and a dome to their south	The heat flux that passed beneath the far side had two exits, located opposite the Imbrium basin (the main one)	Mapping of all the centres of the SPA basin on a single geometric map: exits in the north, a	The crater deficit and the thin-crust centre show no geometric relation to the points

Mechanical requirement	What the model physically requires	Empirical test protocol	Would not fit if
	and opposite the Serenitatis basin (the secondary one). The centre of the dome, the crater minimum and the thermal centre lie to the south of both (a push vector perpendicular to the Earth's rotation axis due to Coriolis forces, and the elongation of the body increased distances: 17°–21° in the flattened stage, 24°–29° today) (Paper 8, §C.3–C.4).	small dome of thin crust above each, the dome and the crater minimum at the centre, and the dense mass and the lowest topography in the south (Paper 8, Figure C.1).	opposite the near-side openings, or if they are located geographically to the north of them.
R18 — Martian moons from a single equatorial conduit	Phobos and Deimos were fired as lithospheric plugs from the Martian crust from a single ejection conduit located at Pavonis Mons ($1.48^\circ\mathrm{N}, 247.04^\circ\mathrm{E}$) on the equator, driven by the pressure of buried gas. The orbits circulated within a ring of condensed water vapour (Paper 10, §C–G).	Spectral and geochemical analysis of Phobos against the composition of the Martian plate; gravity mapping to locate a mass deficit beneath the vent of Pavonis Mons; measurement of the orbital inclination of the two moons within a range of $\leq 2^\circ$ from the equator.	Samples of Phobos (from the MMX mission) show a planetary origin that is not Martian or a mixed composition with an impactor component (Theia-like), or if the gravity field beneath the vent of Pavonis shows a positive mass excess.

Details of the row mechanisms and the data integration

- **R16 (the melting mechanism of the ancient craters):** The geological mare is defined in the model as an active crustal opening located directly above a magmatic feed dome. The broad dark plains are not deep openings, but areas of crust that underwent local thermal melting in place. Since the openings served as the main source for the ejection of the returning terrestrial material (the particles that created the impact craters), and since a crater that landed on a surface of liquid magma melted and was swallowed into it, the mare floors must show a deficit of sharp craters with fresh rims. The quantitative measurement data of Paper 6 confirm this property fully: exactly 0.8 sharp craters of diameter ≥ 20 ק"מ

per million km² were counted on the mare floors, against a value of 18.3 on the plains and 207 in the ancient anorthositic highlands. This gap shows that the density of sharp craters in the maria is lower by more than a factor of twenty ($20\times$) relative to the plains, and confirms the residual requirement as having confirmed status.

- **R17 (the signature of the thermal heat flux in the SPA basin):** The thermal energy that passed through the near-side ejection conduits left a direct geophysical record on the opposite side of the Moon, within the South Pole–Aitken (SPA) basin. These two heat entry points are located antipodally opposite the Imbrium and Serenitatis basins. The dynamic equations dictate that the centre of the dome, the thermal centre and the crater-minimum zone must in principle remain to the south of these two points: the fast rotation vector of the Earth forced a hydrodynamic deflection of the flow perpendicular to the rotation axis (Coriolis forces), and the mechanical transition of the body from a flattened shape to a spherical shape increased the linear distances on the surface from a range of 17°–21° in the early flattened stage to a range of 24°–29° today. This is a purely geometric change of the outline of the body that alters the distances, and not a displacement of the SPA basin in space. Figure C.1 of Paper 8 aligns all these centres on a unified map in the predicted geodynamic order — from the northern exits to the dense mass and the minimum topography in the south — and fixes this requirement as having confirmed status.
- **R18 (the ballistic mechanism in the Mars system):** An independent application of the ejection mechanism to the early planet Mars (whose physical radius was 1.5 times its modern dimensions and whose rotation period was 5 hours) predicts that Phobos and Deimos were fired ballistically from a single crustal conduit located at the volcano Pavonis Mons. The released body begins its orbit at an inclination equal to the geographic latitude of its launch site. The coordinate of Pavonis Mons, located at 1.48° north (according to the IAU files), is the location closest to the equator among the four massive volcanoes of the Tharsis province; the two moons today orbit the planet within an angular uncertainty range of less than 2° from this axis. Final confirmation of the composition is defined as open and is expected to come from the data of the MMX mission (Japan). Paper 10 is an isolated and fully independent test, and its statistical results do not affect the parameter set of the terrestrial case.

Geodynamic axis-orientation constraint: The Earth's rotation axis may have undergone a change or reorientation (True Polar Wander) after the moment of physical detachment of the body, an event that would today place the SPA basin off a straight engineering line opposite the terrestrial heat source (the burner). Since the rotation axes were deliberately left outside

the system of equations for future work in the model, determining their exact position is not a binding requirement in this paper.

Summary of all eighteen requirements in the model (end of Part C)

With the statement of requirement R18, the full theoretical structure of the paper is complete, bringing together a set of 18 strict geophysical and geometric requirements:

- R1–R7: the seven classical requirements (isotopic composition, core dimensions, crustal asymmetry, synchronous orientation, chronology clocks, volatile inventory, and centre-of-mass offset).
- R8: the concentric exposure zones and the global Spearman correlation test (T5).
- R9–R15: the seven geodynamic requirements derived directly from the equilibrium figure of the body of the pear (Paper 3).
- R16–R18: the three later structural requirements added following the completion of Papers 5, 6, 8 and 10.

Part D will now bring together the test protocol and the final mismatch thresholds for all the requirements, and will define precisely which values remain open for future calculation.

PART D

Consistency across the papers, tests and conclusions

Methodological characterisation and the system approach

- **The physical closure constraint:** A morphological or gravitational match on the surface of the present-day Moon does not by itself close the physical gap in the geodynamic construction mechanism. A Moon that looks as the model states it should look is a necessary but not sufficient condition (Necessary but not sufficient): the dynamic mechanical stages that built it — the deep feed flow from below, the vertical rise of the lithospheric shell, and the separation at the corotation line — must all operate in full accordance with the laws of physics.
- **The unified system approach:** The model applies a fixed and uniform set of conditions across all the papers of the series, and makes explicit the logical and mathematical

dependencies between their claims. This structure allows the reader to follow which results rest on one another, and to identify to which specific requirement each empirical finding that does not fit pertains.

- **Integration of the 18 requirements:** The previous parts of the paper (A to C) stated the mechanical requirements individually: requirements R1–R8 for a Moon built in an attached state, requirements R9–R15 derived from the mass balances of the pear, and requirements R16–R18 added following the later papers. Part D steps one methodological step back and treats all these requirements as a single integrated system, through four main axes:
  1. Mapping of the inputs and open conditions that each paper in the series passes to the next (§D.1).
  2. Analysis of the geochronological and isotopic time-scale constraints (§D.2).
  3. Collection of all the tests and mismatch thresholds in a single matrix table (§D.3).
  4. Definition of the scientific novelty of the work and presentation of the explicit conclusions (§D.4, §D.5).

D.1 The conditions that Paper 12 receives from the series

The series divides the mechanical work among its papers in a structured way: Paper 2 defines the orbital and momentum conditions, Paper 3 builds the geodynamic mechanism of the pear, Papers 4–8 follow the evolution after the release and interpret the surface, and Paper 10 examines the principle on the Mars system. Each paper passes a confirmed result as an input (Input) to this part, and at the same time leaves an open condition that the result alone cannot decide, which is kept as an independent requirement with a test protocol of its own.

Table of the flow of results and open conditions in the series of papers

The rows are arranged by the chronology of the scenario: papers in dark blue build the body and release it (Papers 2–4); papers in red read the surface of the Moon (Papers 5–8); and white rows represent the external control of Mars and the present integration of Paper 12.

Paper in the series	The geodynamic result passed on as input	The mechanical condition that remains open (required as a test in R1–R18)
2 — LEF Mechanism in Principle: corotation	Definition of the orbital states and momentum vectors in explicit mathematical terms of mass and rotation rate.	The survival of the body against tidal forces and the physical separation do not follow automatically from the general equality of speeds alone.

Paper in the series	The geodynamic result passed on as input	The mechanical condition that remains open (required as a test in R1–R18)
3 — LEF Mechanism in Practice: Part A	Construction of an example geometric state showing a positive radial overpressure value ($\Delta P_{\mathrm{rad}} > 0$).	The transition-radius constraint: the radial balance holds from a radius of about 1,000 km (where the transition carries 0.46 lunar masses and a lateral load of $\approx 10^{22}$ ניוטון), which dictates a connection on the broad base of the dome and not a narrow neck.
3 — LEF Mechanism in Practice: Part B	Hemisphere parameters: a density of 1,300 kg/m³, a curved surface area three times that of the far hemisphere, and the geometric translation rule.	Derivation of the exposure zones and the crustal asymmetry from a uniform rule; the viscosity constraint of the shell ($\eta \gtrsim 10^{23}\ \mathrm{Pa \cdot s}$) is a theoretical requirement and not a measured value.
4 — The dynamic orbit	Release at an ancient latitude of 22.5°, at a distance of 17,287 km, into a circular orbit; calculation of the evolution of the orbit and of the orientation.	The orbital calculation rests entirely and depends strictly on accepting the early release conditions from Papers 2–3.
5 — The mare basins	Mapping of the 34 maria as structural openings above feed domes on the near side; definition of the dark plains as crust melted in place; classification of R8.	Definition of the chronological control requirement R16 (detailed in §C.7).
6 — Crater morphology	Characterisation of the returning terrestrial material and its ballistic landing on the shell; the 34 openings as the source of the material; melting of craters on active mare floors.	The kinematic speed of the returning material is not demonstrated in the paper; requirement R16 (detailed in §C.7).
7 — Geometry of the poles	Measurement of the rounding of the attached body on the craters; crater density toward the poles increases as a function of crater diameter.	Confirmation of the size-dependent crater-compaction requirement R15 (presented in §C.6).

Paper in the series	The geodynamic result passed on as input	The mechanical condition that remains open (required as a test in R1–R18)
8 — The South Pole–Aitken basin	Interpretation of the crater-poor centre in SPA as a sign of a heat source standing antipodally opposite the near-side openings (Imbrium and Serenitatis); a dome to their south.	Definition of the thermal and geometric asymmetry requirement R17 (presented in §C.7).
10 — The Mars system	Running of an independent external control test under the physical parameters of Mars.	The Martian moons as the product of two crustal plugs fired from a single ejection conduit on an expanded, fast-rotating Mars; orbital tuning by means of a vapour ring.
11 — The present paper (the requirements)	Full integration of the observational matches, setting of the statistical pass thresholds and mismatch thresholds.	The statistical tests examine the products, but do not replace the solution of the physical equations of the feed, the rise and the release.

Analysis and explanation of the row mechanisms

- **The corotation and momentum mechanism (Paper 2):** The corotation radius is the distance from the Earth's centre of mass at which a body completing one synchronous rotation in the early Earth day moves at exactly the tangential speed required for a free Keplerian orbit. In the day of 5.77 hours that prevailed at the time of detachment, this distance is fixed in the equatorial plane at 16,399 km. Paper 2 locks the values of the masses, the rotation rate, and the total angular momentum vector of the system — a conserved physical quantity ($L = \mathrm{const}$) that no internal mechanical process can create or destroy. The general equality of speeds at this point is solely a kinematic meeting point; it does not by itself prove that the hot lunar body survived the disruptive tidal forces of the planet, or that it succeeded in being released mechanically as a unified mass. For this reason, the release process is kept as an open requirement managed in this part.
- **The overpressure and lateral-load mechanism (Paper 3 - Part A):** The radial overpressure is defined physically as the outward pushing force exerted by the deep mantle material, remaining after subtraction of the lithostatic weight of the column of material lying above it. A positive value ($\Delta P_{\mathrm{rad}} > 2\ \mathrm{mGal\ equivalent}$) is a necessary condition that allows the feed material to rise vertically through the crust. The balance equations show mathematically why the mechanical connection to the pear cannot exist in the structure of a narrow stalk or bottleneck: the dynamic balance conditions converge only

where the geometric transition radius is equal to or greater than about 1,000 km. At this point, the transition cavity already contains a massive mass equivalent to 0.46 lunar masses and must carry an extreme lateral shear load (Lateral load) of the order of:

$$F_{\text{shear}} \approx 10^{22} \text{ ניוטון}$$

A mechanical stress of such dimensions cannot be carried by a narrow stalk; it requires distribution over the entire width of the base of the dome. For this reason, the model rules out the existence of a neck and is built on a broad base.

- **The geometry of the dome and the rheological constraint (Paper 3 - Part B):** The body is represented at this stage as a hemisphere with a mean density of 1,300 kg/m³, whose curved surface area is three times the area of the far hemisphere into which it closes and contracts. This geometric closing dictates a thickening of the shell by a factor of approximately three. The geometric translation rule used to map the dome onto the finished sphere yields simultaneously both the boundaries of the concentric exposure zones of requirement R8 and the profile of the thickened crust on the far side.

The crustal viscosity constraint: The rheological viscosity bound value ($\eta \gtrsim 10^{23}\ \mathrm{Pa \cdot s}$) is defined as a binding physical requirement and not as a material value measured in the laboratory. The lunar crust must hold this value throughout the rise of the dome so that its plastic flow rate is slow enough, which allows the ancient craters to keep their circular morphology during the deformation and closing of the dome beneath them.

- **Separation of the viscosity layers:** There is a strict methodological prohibition on projecting or transferring the low viscosity values of the separating surface (Interface) at the base of the shell to the outer, crater-preserving surface. The two components belong to different structural layers with opposite properties: the separating surface lying between the shell and the interior of the body is the layer of lowest mechanical rigidity (the softest) in the body, and its low viscosity is what allows the lithospheric shell to slide mechanically and ride upward on the elongating dome. By contrast, the outer surface carrying the craters must remain in a rigid, solid state.

Constraint on the test-case values: The numerical values specified in the Paper 3 rows (a density of 1,300 kg/m³, a transition radius of 1,000 km, and a lateral load of $10^{22}\ \mathrm{N}$) are defined test cases (Test cases) set for the purpose of verifying the balance equations at a specific point in time, and do not represent a single static state of the system over the whole duration of the scenario, since the dimensions of the body change continuously during the rise.

The statistical tests of Paper 9 (The Stripes) do not appear as an independent row in this table, because their findings on the Earth's side (the location of the tip and the folds on the front) are incorporated into this chapter in sections C.2 and C.6.

D.2 The chronology remains a central constraint

1. The scientific problem and the isotopic constraint

The central question is whether the long time scale of the scenario — a gradual mechanical construction of the pear lasting hundreds of millions of years — can be reconciled physically with the systemic isotopic clocks that date the separation of the metallic phase from the silicate phase in the early Earth and Moon.

2. How the geochemical clock works ($\mathrm{Hf–W}$)

The hafnium–tungsten isotope system ($\mathrm{Hf–W}$) is used in science to bound and constrain processes of metal–silicate separation [S11]. It is a short-lived decay clock that stops ticking entirely after only a few tens of millions of years:

- The short-lived radioactive isotope hafnium-182 ($^{182}\mathrm{Hf}$) decays by double beta decay to the stable isotope tungsten-182 ($^{182}\mathrm{W}$).
- Geochemically, the element hafnium is lithophile (Lithophile) — it remains exclusively in the silicate reservoir (the crust and mantle) when a metallic phase separates from it.
- By contrast, the element tungsten is siderophile (Siderophile) — it tends to concentrate preferentially in the metallic melt and descends with it gravitationally into the core.
- **The isotopic signature mechanism:** If metal separation occurred early, while the parent isotope $^{182}\mathrm{Hf}$ was still present and active in the system, the remaining silicate mantle keeps its hafnium inventory. As this short-lived hafnium decays over time, the silicate mantle accumulates an excess (Anomalous excess) of the daughter isotope $^{182}\mathrm{W}$ relative to the source material. The size of this excess records quantitatively and strictly how early in planetary history the separation occurred.
- Once the parent isotope $^{182}\mathrm{Hf}$ has been completely exhausted from the system by decay, the clock stops: new anomalies and excesses of tungsten-182 can no longer be produced, and any late metal-separation event will not change the isotopic ratio of the mantle.

The chemical equilibrium constraint: The interpretation of the chronometric system depends entirely on the accretion history (Accretion) and on the degree of chemical equilibration (Chemical equilibration) that prevailed between the metallic and silicate melts

during core sinking (how much tungsten was exchanged between the sinking iron droplets and the silicate mantle through which they plunged). It is methodologically prohibited to assume a "partial equilibrium" as a free parameter intended to extend the differentiation duration arbitrarily to fit any time scale; partial equilibrium does widen the range of interpretations, but it must be computed precisely for a declared and consistent historical model [S11].

3. The radioactive decay equation and the calculation of the reset bound

To clarify the rate at which the clock operates, under the assumption of a fixed half-life of 8.9 million years (Myr) for $^{182}\mathrm{Hf}$, the law of radioactive decay is represented mathematically as follows:

$$\frac{N\,(\,t\,)}{N_0} = 2^{-t/\,(\,8.9\,\mathrm{Myr}\,)}$$

where N_0 is the initial concentration of the radioactive parent isotope, $N\,(\,t\,)$ is its residual concentration remaining after time t, and the exponent counts exactly the number of half-life cycles that have elapsed. After a small number of cycles (a few tens of millions of years), the parent isotope is effectively exhausted and the clock stops.

- **Calculation of the parent residue at the end of the construction window:** At the end of a construction window of 600 million years (Ma) — which amounts to many tens of successive half-lives — the solution of the equation yields a tiny residual fraction of:

$$\frac{N\,(\,600\,\mathrm{Ma}\,)}{N_0} \approx 5.1 \times 10^{-21}$$

This repeated halving, tens of times over, leaves in practice not a single atom of the parent isotope $^{182}\mathrm{Hf}$ in the system; the clock stands completely still for most of the time window. This window represents the estimated timeline of the LEF model: about 600 million years from the start of the development of the bulge to the moment of physical release at the corotation line 3.873 billion years ago (internal division: about 400 million years for the growth of the bulge and the pear, about 100 million years for the dynamic flattening stage, and about 100 million further years until the detachment).

- **Constraint on the preservation of the initial signature:** An isotopic system that has decayed and become extinct can preserve ratio differences formed at the early stages, but it must under no circumstances be treated as a clock that can be reset (Resetting) at the end of the construction window. Hence, the W signatures identified in the Moon today were set and locked geochemically at the beginning of the window, while the parent isotope was

still present in the mass. The geodynamic scenario must carry these signatures throughout the 600-million-year window without resetting them.

4. Status of the requirement

The model must present a unified account and a single system of equations that computes simultaneously the isotopic signatures of tungsten, the fraction of metallic mass transferred to the core (f_{core}), and the evolution of angular momentum within a strict and consistent chronology (the same time values must be used for the dynamic position of the reservoirs, the time of separation of their metallic phase, and the total mass that reached the lunar body). This stringent chronology requirement is currently defined as open for calculation, and it is listed with the other open tasks and is not taken as being met by itself in the model.

D.3 Map of tests

1. The scientific problem and the mapping strategy

The central question addressed in this chapter is: what are the precise findings that would not fit a given requirement, and what is the extent of the part of the model that each such finding concerns? All the mechanical and geometric requirements stated in Parts A to C are collected here in a single table.

The scope-of-conclusion column is defined as being of the highest methodological importance, since it sets the limits of the influence of each result: every empirical mismatch is isolated and concerns a specific physical or morphological branch of the explanation, leaving the other parts of the system standing independently; the row of the exposure zones (R8), which is emphasised and highlighted in the model, represents the central observation for the geometry of the dome (Paper 1).

Matrix of tests and scope of the model's conclusions

The claim tested in the system	The explicit physical requirement	Test protocol and mismatch condition	Scope of the conclusion
Terrestrial origin (R1)	Simultaneous chemical and isotopic agreement in several systems after subtraction of permitted separation processes.	There is no unified mathematical solution within the source domains and the laboratory measurement errors.	Would not fit the claim of an exclusively terrestrial origin in the version tested.
Metal supply (R2)	The range of core-mass values (f_{core}) produced by the model must be consistent with the seismic interior data.	The hydrodynamic transport and differentiation equations predict a mass fraction exceeding the permitted empirical range.	Would not fit the mass-transfer mechanism of Paper 3.
Volatiles and feed (R6)	The conditions of pressure, density, flow speed and thermodynamic energy hold together in the same state of matter.	There is no physical overlap between the states of matter supported by the data and the dynamic transport requirements.	Would not fit the given geodynamic construction route and mechanism.
Rise and preservation (§C.1)	The lithospheric shell rises with the dome and structurally preserves the record of the ancient craters on it.	There exists no common rheological domain capable of satisfying the two opposite requirements simultaneously.	A mismatch in the version of the dome mechanism as developed in Paper 3.
Crust and mass balance (R3)	The same original shell mass yields the possible empirical values of thickness and geological composition.	Identification of a crustal mass excess or deficit that cannot be resolved within the material fluxes permitted in the model.	A mismatch in the general balance equation between the external and the internal material.
Chronological order of closure (R5)	Existence of a strict and directional order of preservation and freezing between identified genetic crustal units on both sides.	Parallel dating of comparable units dictates significantly the reverse order of times.	A mismatch in the specific chronological prediction of the model.

The claim tested in the system	The explicit physical requirement	Test protocol and mismatch condition	Scope of the conclusion
Crust and centre of mass (R7)	The density model and the asymmetric crustal thickness yield a matching geometric and gravitational offset vector.	Absence of a joint mathematical solution for the mass integral within the supported measurement-error frame.	A mismatch in the specific structural and gravitational explanation proposed.
Exposure zones (R8)	A consistent concentric ordering of thickness, basin type, and the density of old craters as a function of angular distance from the contact axis.	Absence of such a concentric order, or identification of its ordering around another arbitrary planetary axis.	The observation that would bear against the geometry of the dome (Paper 1).
Rings and openings (§C.3)	Existence of a clear measurable and quantitative difference between the three groups of openings set in advance.	The statistical difference is absent, or is fully explained by the control axes and the interpolations.	A mismatch in the morphological branch only, without affecting the corotation equation.
Openings and craters (R16)	The 34 mare basins are the structural openings and the exclusive source of the returning terrestrial material.	The mare floors turn out to be rich in sharp craters at a rate equivalent to plains of the same age.	A mismatch in the model's thesis of the source of the returning material.
The sign on the far side (R17)	The morphological centres of the SPA basin are arranged geometrically to the south of the two exit points on the near side.	Absence of an ordering or spatial relation around the antipodal points opposite the openings.	A mismatch in the interpretation of the SPA basin as the residual sign of the thermal burner.
The Mars system (R18)	The two moons were fired ballistically from a single equatorial ejection conduit located at Pavonis Mons.	The composition of Phobos is not Martian, or a positive mass excess is shown beneath the vent of Pavonis.	A mismatch in the control test on Mars, without affecting the parameters of the Earth–Moon case.

2. Chronological analysis and explanation of the requirements and thresholds

- **Terrestrial origin (R1):** The isotope ratios of the lunar body must match terrestrial material in several isotopic systems simultaneously (oxygen, titanium, chromium, tungsten, and volatile elements), after precise accounting for the processes of separation, mass-dependent fractionation and evaporation that the geodynamic scenario permits. The requirement would be considered not met if it is shown mathematically that no mixture of the source domains of the terrestrial mantle and crust can produce a solution matching all these systems within the given measurement-error limits.
- **Metal supply (R2):** The parameter f_{core}, which represents the mass fraction of the core in the whole body and is applied in Equation 1, is fixed on the basis of the empirical seismic data at about 1.4%. If the hydrodynamic equations of the process of droplet transport and differentiation within the feed flow predict a mass fraction exceeding the range permitted in the literature (1.0%–2.0%), this would not fit the mass-transfer mechanism of Paper 3, but the global model would remain standing.
- **Volatiles and feed (R6):** The mechanical mechanism relies on the presence of volatile components to reduce the lithostatic density of the feed material, alter its viscous rheology and produce the required ejection pressure (§A.6). The finding of water values in melt inclusions (615–1,410 ppm) within olivine crystals demonstrates their general presence in the lunar interior, but is not proof that they were present in the volume and in the precise thermodynamic phase state in which the deep feed flow needed them (§B.6). The equations of pressure, density, flow hydrodynamics and energy must all converge under the same single physical state of matter. If the thermodynamic model rules out an overlap between the phase states supported by the data and the transport requirements, this would not fit the geodynamic construction route.
- **Rise and preservation (§C.1):** The same lithospheric shell unit must ride mechanically upward together with the elongation of the dome, and at the same time keep the circular morphology of the ancient craters on it. This condition dictates the existence of a layer with unique rheological properties (the behaviour of the material under stress and load) capable of meeting these two opposite needs at once: a separating working surface of low rigidity (soft) at the base of the interface, and above it a rigid outer shell with a viscosity value of $\eta \gtrsim 10^{23}\ \mathrm{Pa \cdot s}$. The absence of such a common parameter domain in the equations of planetary mechanics would not fit the dome mechanism of Paper 3.
- **Crust and mass balance (R3):** The ancient terrestrial dome shell (with a computed mass of 2.85×10^{21} ק"ג) must fully supply the mass of the modern far-side crust (2.6×10^{21} ק"ג, as detailed in §B.3), while the near-side crust is defined as an independent and separate item in the system (the sialic mass of the base). Identification of a mass excess or deficit that cannot be resolved or interpolated within the permitted material fluxes would not fit the

balance equations between the material that came from outside the body and the material that developed within it.

- **Chronological order of closure (R5):** The crustal units underwent thermal closure and stabilised mechanically in a fixed chronological order. The model predicts that the anorthositic highland crust of the far side is significantly older than highland crust with similar genetic characteristics on the near side. If a parallel isotopic analysis of comparable units dictates a reverse chronological order, the specific chronological prediction of the system will not be met.
- **Crust and centre of mass (R7):** The lunar centre of mass is offset by 2 km from the geometric centre of figure toward the near side. The mineralogical density values and the asymmetric crustal thickness must yield, in the solution of the integral, an offset vector matching the measurements both in linear magnitude (length) and in spatial direction. The absence of a joint mathematical solution within the supported measurement-error frame would remove the validity of the structural explanation.
- **Exposure zones (R8 - the core row):** Since the closing of the dome took place in a fixed chronological and geometric sequence (the suction-cup mechanism), the crustal thickness, the typological gradient of the basins, and the density of the oldest craters must all be arranged monotonically as a direct function of the angular distance from the axis of the centre of the near side (sequence: the contact cap, the middle belt, and the far part), independently of any alternative planetary axis. This is the strongest and most rigorous spatial prediction of the model. The absence of this concentric ordering, or identification of its statistical organisation around another arbitrary control axis, would not fit the spatial geometry of the dome.

Control on the R8 run data: sections B.8 and B.9 detail the current status of the tests: tests T5 (axis uniqueness with a correlation of $+0.87$) and T2 (crater density in ratios of 1.34 / 2.21 / 1.65) passed fully. Test T1 (global crustal thickness) was defined as not passed; the thickness differences (far side minus contact disc) in the four GRAIL models stand at 9.65 km, 10.0 km, 10.3 km and 10.6 km, with model 1 falling 0.35 km below the lower threshold line set at 10 km, while the other models lie in the indecision range, and the monotonic gradient itself is preserved fully across all four. The recalculation on the updated geometry preserves the ordering in both measures (the crustal thickness yields a difference of 9.9 km in model 1).

- **Rings and openings (§C.3):** The three groups of openings (crown, flank, and beneath the base) are set and mapped independently before the gravity rings are examined, and must show a measurable difference in their geophysical character. The absence of a difference,

or its full resolution by the statistical control equations, would not fit the morphological branch only; as established in section C.3, this mismatch is isolated and does not affect the independent validity of the corotation equation of Paper 2.

- **Openings and craters (R16):** If the 34 mare basins are the structural openings and the source of the ejection of the returning terrestrial material, the mare floors must show a deficit of sharp craters, since a crater that returned onto a molten, active surface underwent melting and was swallowed into it. The measurements of Paper 6 record a low density of 0.8 sharp craters (≥ 20 מ"ק) per 10^6 km² in the maria, against values of 18.3 on the plains and 207 in the highlands (§B.9). A future identification of mare floors rich in sharp craters at a rate equivalent to plains of the same geological age would not fit the claim regarding the source of the returning material.
- **The sign on the far side (R17):** The morphological and thermal centres of the SPA basin must be offset and arranged geometrically to the south of the two antipodal entry points of the Imbrium and Serenitatis basins (mapped in Figure C.1 of Paper 8). The absence of angular organisation around these points would not fit the interpretation of the SPA basin as the physical residual sign of the terrestrial thermal burner.
- **The Mars system (R18):** The two moons predict Martian source material only, fired ballistically from a single equatorial ejection conduit at Pavonis Mons. A determination that Phobos samples consist of non-Martian material (or a mixed composition with an impactor component), or the identification of a positive mass-excess anomaly in the subsurface beneath the vent of Pavonis Mons, would not fit the control test on Mars, without changing or affecting the parameter set of the terrestrial case.

3. Constraints on a claim that a requirement is not met

- **Definition of "no solution" (a range and not a point):** A methodological determination that a requirement is not met under the formulation "no solution" requires a reasoned mathematical definition of the whole parameter domain and of the error ranges, and not an isolated check of a single numerical point. A scientific claim stating that the mechanism does not operate must show that no possible physical solution exists across the whole dynamic range that the empirical data permit.
- **Rejection of confirmation from a single point of success:** On the other side, the mathematical success of a single combination or choice of parameters (a point match) does not provide confirmation or grounding for a complete formation route. The presentation of all the statistical samples, the laboratory error ranges, and the full

calculation procedures is intended to allow an external reader to carry out an independent assessment of the model under the requirement of strict scientific control:

$$\text{ניתנת לבדיקה} \rightarrow \text{ניתנת להערכה} \rightarrow \text{חסינה} \rightarrow \text{מפורטת} \rightarrow \text{תחזית}$$

This chain dictates that every geodynamic claim in the model must be formulated as an independent prediction, specified at a level of mathematical detail that allows its quantitative calculation, robust (Robust) to reasonable changes and noise in its input data, open to assessment and critique by external researchers on the basis of the global data files provided, and holding inherently the capacity to be tested against the data. Every row in this map of tests is built to meet the final stage of this criterion.

D.4 The novelty of the work

1. Defining the limits of knowledge and of existing physics

The basic physical and geophysical tools used in the paper — including the laws of the gravity field, the principle of mass conservation, the mechanism of differentiation, models of thermal cooling of the crust, and planetary tidal mechanics — are all existing physics, known and verified in the scientific literature. In addition, the thesis that a hot early Earth can induce a structural and thermal asymmetry within the volume of the Moon is not new in science [S5]: the research group of Roy and colleagues (2014) proposed earlier that the heat radiation and Earthshine (Earthshine) of a young, molten Earth preferentially heated the near side, and thereby provided an explanation for the formation of the thicker anorthositic highlands on the cold far side.

2. The novel element in the LEF model

The unique scientific contribution of the LEF model, supported and tested within the present series of papers, is the definition of the structured dynamic relation between the deep mantle feed flow (while the physical interface existed) and the vertical rise of the outer lithospheric shell as the outer surface of that same body, together with the mechanical requirement that these two mechanisms together produce matching mass balances, internal structures and geological records.

The novel element does not lie in any isolated physical process, but in their combination and integration within a single planetary body: the interior of the Moon was filled and grew from below by a feed flow, while its outer skin (the crust) was composed directly of the shell of the Earth itself, which was lifted and rose on the structure of the dome.

3. The contribution of Paper 12

The main contribution of the present paper is the formulation and organisation of all the geophysical tests under a uniform mathematical and coordinated framework, and in particular:

- Establishing the direct quantitative link between the spatial prediction of crustal thickness and the vector components of the centre-of-mass offset (Equations 8 and 9).
- Construction of the strict mass balance linking the original material inventory of the terrestrial shell with the crustal mass of the far side (Equation 4 against Equation 5).

The derived equations are based exclusively on the principles of standard mechanics, and their application to the LEF scenario is not a claim to the discovery of a new physical law. The paper does not claim historical priority for any component in isolation, before a comprehensive literature review has been carried out; Appendix B specifies individually, component by component, what is defined as standard physics, what is a previous observation or prior knowledge in science, and what is the component that constitutes a working hypothesis unique to the model itself.

D.5 Conclusions and their geodynamic interpretation

The following sections bring together the final scientific conclusions of Paper 12, with each statement followed by its explicit mechanical interpretation with respect to the model:

Conclusion A

Terrestrial origin (R1), the metallic-phase limitation (R2), the volatile inventory (R6), and the crustal asymmetry (R3) are quantitative observational targets that the geodynamic scenario must explain in a single unified system. Their general existence at present is not in itself an independent confirmation of the transport route.

- **Interpretation of the conclusion:** These four properties are global threshold criteria that every theory dealing with the origin of the Moon must reproduce successfully. The construction route of the LEF model will be defined as confirmed and prevailing only when the processes of mantle feed, hydrodynamic transfer of the metallic droplets, and closure of the shell are computed in a single system of equations that yields the values of all four at once under the same set of parameters.

Conclusion B

The existence of synchronous rotation locking at present is a weak dynamic consistency test; demonstrating the preservation and acquisition of the direction vector of the ancient connection requires a full solution of the orbital and tidal evolution after the moment of release.

- **Interpretation of the conclusion:** The astronomical fact that the Moon today turns a single face (the near side) toward the Earth does fit a body that separated with its contact surface facing the parent body, but it is not unique proof, since prolonged tidal friction forces can also produce this geometric locking from alternative initial conditions. The model must demonstrate dynamically, within Paper 4, that the face with which it detached at the moment of release is indeed the face that the orbital system forced it to keep throughout its history.

Conclusion C

The age measures of crustal crystallisation, impact events, and basaltic filling are records of entirely separate geological processes. The order of thermal closure and the mechanical construction duration of the body cannot be decided or interpreted on the basis of basalt age data.

- **Interpretation of the conclusion:** The geochronological age of mineral crystallisation in the primary crust (clock 1), the isotopic age associated with basin-forming impact events (clock 2), and the age of solidification of the dark mafic lava (clock 3) are three independent clocks measuring different events in time. The bound on the order of mechanical closure of the hemispheres must be read exclusively from a comparison of crustal units of similar genetic character on both sides, and not from the dates of the late basalts (such as the Chang'e-5 and Chang'e-6 samples).

Conclusion D

The terrestrial lithospheric dome shell fully and precisely supplies the mass of the modern far-side crust — a balance of 2.85×10^{21} ק"ג against 2.6×10^{21} ק"ג — while the near-side crust is the sial material of the base that was peeled off together with the body. The two are entirely separate items within the global balance equations.

- **Interpretation of the conclusion:** These two masses show a close mathematical agreement within about 10% (updated from earlier calculations that showed an agreement of 7% and a mass of 2.7×10^{21} kg), without applying any arbitrary tuning of free parameters in Equation 5. The near-side crust has a separate geochemical and mechanical source: the

light outer sial layer (SiAl) of the early Earth, rich in silicon and aluminium, on which the body sat and which was peeled off as a solid from above the dense sima layer, and carried along with the Moon at its detachment.

Conclusion E

The mechanical contribution of a thick, light asymmetric far-side crust to the global gravity field pushes the centre of mass vectorially toward the thin side (the near side) under the physical conditions derived. Locking the total linear magnitude and vector direction requires applying a common and uniform density model.

- **Interpretation of the conclusion:** The replacement of dense mantle material by light crustal material in the far hemisphere shifts the centre of mass toward the near side, in full agreement with the geodetic observations. The computed offset range (1.24–3.11 ק"מ in section B.7) will become a rigorous confirming control test only when a single, fixed density model determines simultaneously both the thickness profile and the components of the offset vector, under the constraints of the normalised moment of inertia and the tidal response.

Conclusion F

The carried record of ancient craters, the morphological anomaly of the connection region, and the classification of the three groups of openings stand as independent geodynamic predictions to be tested in a controlled way against competing models of impact, burial and tectonic deformation.

- **Interpretation of the conclusion:** The craters that the lithospheric shell carried with it intact and circular (Paper 7), the updated contact region — centred at a fixed coordinate of 24.1° north, 335.2° east, with the Imbrium basin as the main exit and the Serenitatis basin as the secondary exit (Paper 3) — and the division into three groups of openings must all be tested quantitatively against the alternative interpretations in the scientific literature (craters formed in place by ordinary impact, craters buried under lava flows, and late tectonic deformations of the crust).

Conclusion G

The statistical tests formulated in the present paper evaluate simultaneously and jointly all the mechanical requirements of Paper 3 together with the chronological constraints of the isotopic

clocks. Each test defines precisely the values that would confirm the requirement and the structural conditions that would not fit the version tested.

- **Interpretation of the conclusion:** The model is deliberately placed in a methodological position in which it is open to testing, row by row, in a way that allows an external reader to follow precisely which tests passed with full empirical success (the axis-uniqueness test T5 and the crater-density test T2 within requirement R8; the size-dependent crater-compaction requirement R15; the density of sharp craters in the maria R16; and the order of the geometric centres in the SPA basin within R17), which calculations and balances remain open, and which single future geophysical observation (such as a mismatch in the concentric structure of R8) would bear against the geometry of the dome.

APPENDIX A

Reproducing the centre-of-mass calculation

Mathematical foundations and the mechanics of the integration

- **Purpose of the geodynamic formulation:** The present appendix sets out the analytical basis and the central integration behind Equations (8) and (9) presented in section B.7. The calculation allows direct verification of the way in which an asymmetric crustal distribution — light and thin on the near side, thick and massive on the far side — shifts the position of the Moon's centre of mass relative to its geometric centre, and defines precisely the size of the linear offset and its vector direction.
- **Definition of the layer boundaries in the system:** For a spherical planetary body, in which the dynamic continuity boundary between crust and mantle (Moho) is represented in space by the geometric equation $r = b + A\cos\psi$, a fixed auxiliary parameter is defined:

$$u = \cos\psi$$

where the components are defined physically as follows:

- ψ**:** The exact angular distance from the original geodynamic connection axis.
- b**:** The mean radius of the crust–mantle boundary, equal to the radius of the modern Moon (R_L) minus the mean crustal thickness value (h_0):

$$b = R_L - h_0$$

- A**:** The asymmetry amplitude (Amplitude) by which the density boundary rises vertically toward the connection (and reduces the crustal thickness) and sinks downward far from it on the opposite side.
- **The angular integration variable:** Since the outline of the outer surface keeps a fixed spherical geometry, a Moho boundary rising by A toward the connection dictates a crust thinner by exactly this linear value in the same hemisphere — in full agreement with Equation (3): $h\left(\psi\right) = h_0 - A\cos\psi$. The auxiliary variable u is bounded mathematically in the fixed range:

$$-1 \leq u \leq +1$$

where the value $+1$ represents the central connection axis on the near side, and the value -1 represents the antipodal point opposite it at the centre of the far side.

Cancellation of the symmetric component and the moment equation

- **The rules of mass cancellation:** The global mass moment is defined as the sum (integral) of all the mass elements making up the body, each multiplied by its linear distance along the chosen axis. Dividing this value by the total mass yields the geometric position of the centre of mass. Every structural component that has perfect spherical symmetry — including a uniform homogeneous mantle, a centred central metallic core, or a crust of constant thickness — contributes an identical moment in the two hemispheres. These contributions therefore cancel completely and vanish in the mechanical calculation.
- **The active asymmetric term:** The only component that drives the position of the centre of mass is the asymmetric part of the density distribution — in this case, the dense mafic mantle material that climbs higher toward the surface in the hemisphere where the anorthositic crust is thin (the near side).
- **Definition of the lithostatic density difference:** The internal density addition is defined as the density contrast between the mantle and the crust:

$$\Delta\rho = \rho_m - \rho_c$$

Substituting this component, the combined mass-moment equation is represented as follows:

$$M_L\delta x = \frac{\pi\Delta\rho}{2}\int_{-1}^{1} u\left(b + Au\right)^4 du$$

- **Mechanical interpretation of the terms of the integral (Equation 11):** The term $M_L\delta x$ represents the total mass moment. The integral performs a quantitative summation of the density excess $\Delta\rho$ across the volume enclosed within the raised boundary, with each

volume element weighted by its linear distance along the axis through the weighting factor u (the cosine of the angle from the axis). The polynomial expression $(b+Au)^4$ is derived directly from carrying out a vertical integration of the radius component $\int r^3\,dr$ up to the layer boundary in each spatial direction, while the full angular integration around the axis of symmetry yields the constant multiplicative factor $\pi/2$.

Analytical solution and geometric edge constraints

- **Vanishing of odd powers:** On algebraic expansion of the polynomial $(b+Au)^4$ and its multiplication by the weighting factor u, all the terms with an odd power of the variable u (such as u^1, u^3, u^5) have exactly the same values in the two hemispheres but of opposite mathematical sign ($\pm$). As a consequence, their definite integral over the range between -1 and $+1$ vanishes completely (0).
- **Extraction of the finite-thickness equation:** The even terms remaining after the integration yield the following values:
  - **The term $4b^3Au^2$ yields in the solution of the integral:** $4b^3A\cdot\frac{2}{3}$.
  - **The term $4bA^3u^4$ yields in the solution of the integral:** $4bA^3\cdot\frac{2}{5}$.
- After multiplying the results by the structural factor $\frac{\pi\Delta\rho}{2}$, Equation (9), defined for a layer of finite thickness, is obtained exactly:

$$M_L\delta x = 4\pi\Delta\rho\left(\frac{b^3A}{3}+\frac{bA^3}{5}\right) \Longrightarrow \delta x = \frac{4\pi\Delta\rho}{3M_L}\left(b^3A+\frac{3}{5}bA^3\right)$$

- **Derivation of the thin-shell approximation:** Under the planetary limit constraint in which the crustal thickness and the asymmetry amplitude are entirely negligible relative to the global radius of the body ($A, h_0 \ll R_L$), the mean radius of the boundary (b) converges to the value of the outer lunar radius (R_L), and the second term containing the expression A^3 becomes mathematically negligible. Substituting this constraint reproduces Equation (8) exactly:

$$\delta x \simeq \frac{\Delta\rho}{\bar{\rho}_L}A$$

This formula confirms physically that the centre of mass must shift its position toward the hemisphere with the thin crust, by an amount equal to the linear asymmetry amplitude (A) multiplied by the ratio between the stratigraphic density contrast ($\Delta\rho$) and the total mean density of the body ($\bar{\rho}_L$).

Fixed numerical inputs and the sensitivity constraints of the example

For running the analytical test model, the values of the physical parameters are locked on the basis of the following data:

- **Mean lunar mass (M_L):** 7.35×10^{22} ק"ג.
- **Mean global lunar radius (R_L):** 1.7374×10^{6} מטר (1,737.4 km).
- **Mean basic crustal thickness (h_0):** $45,000$ מטר (45 km).
- **Global asymmetry amplitude (A):** $15,000$ מטר (15 km).

Methodological constraint: These crustal data (a minimum thickness of 30 km at the centre of the near side and a maximum thickness of 60 km on the far side, yielding a mean of 45 km and an amplitude of 15 km as shown in Figure 1) are calibrated example values only, for the purpose of a mechanical sensitivity check. They must not be used as an input file for carrying out a new inversion (Inversion) of the gravity maps of the GRAIL mission. Their sole role is to show mathematically that a crustal profile in this geometric orientation independently produces a centre-of-mass offset lying precisely within the empirical order of magnitude observed on the Moon (the table values in section B.7 yield an offset of 1.24 to 3.11 km for density contrasts of 300 to 750 kg/m³), without claiming to determine at this stage the true density structure of the interior of the body.

Calculation of the spatial vector component of the contact axis

The check of the geometric orientation in section B.7 rests on the calculation of the exact angular distance between the modern map reference point $(\,0^\circ, 0^\circ\,)$ and the geographic location of the updated contact centre in the model. The contact centre is fixed at a coordinate of 24.1° north, 335.2° east (which places it 24.8° west of the prime meridian).

For a point on the surface of a sphere located at geographic latitude φ and at a longitude difference λ relative to the origin point $(\,0, 0\,)$, the global angular distance (γ) is calculated on the basis of spherical trigonometry (the spherical law of cosines) by the following formula:

$$\cos\gamma = \cos\varphi\cos\lambda$$

Substituting the exact coordinates of the lunar contact centre:

$$\cos\gamma = \cos\,(\,24.1^\circ\,)\,\cos\,(\,24.8^\circ\,) \implies \cos\gamma \approx 0.9128 \cdot 0.9077 \approx 0.8286$$

$$\gamma = 34.0^\circ$$

- **Geodynamic implication of the vector:** The solution of the equation locks the angular distance at exactly 34.0° (Equation 12). This finding is of the highest mechanical importance: since the global axis of crustal asymmetry and thickening in the LEF model

passes precisely through the physical contact centre, and lies 34° in angular distance from the modern sub-Earth point facing the Earth, the prediction of the gravitational offset is not limited to the linear mean Earth line. The LEF model requires the calculation of a full spatial offset vector, and the geophysical test will be decided on the basis of the agreement of the angular direction components and not of the length alone. This calculation is not a measurement of the orientation of the Moon at the time of the ancient release, but produces the fixed geometric angle between the two reference points on the present-day modern map.

APPENDIX B

Map of previous research and of the novelty

Classification of the components and the sources in science

The present appendix separates strictly and systematically between the theoretical inputs and data that this paper borrows from previous studies in the scientific literature, and the element of scientific novelty that it adds to the system. The purpose of the separation is to prevent the mistaken attribution of previous ideas to the LEF model (Lunar Elevation & Fission), and conversely — to prevent the imposition of claims that are not part of it.

The classifications in the matrix below move along a methodological continuum from material that is established and proven in science to the independent hypotheses of the model:

- **Accepted physics:** Results and equations from basic textbooks applied here.
- **Previous observation:** Rigorous empirical measurements carried out and published by other researchers.
- **Previous idea:** Theses and theories first proposed by external researchers, attributed to them by name.
- **Model hypothesis:** The unique geodynamic founding hypothesis of the LEF series.
- **This work:** The quantitative, analytical or statistical calculation carried out directly in Paper 12.
- **Open prediction:** A structure or numerical value that the model predicts and expects, which has not yet been tested empirically.

Matrix of the sources of the components and the methodological separation in the model

Geophysical and structural component	Methodological classification	Limits of attribution (what may and may not be attributed to the work)
Calculation of the centre of mass and mass conservation	Accepted physics	Explicit mathematical derivation and direct mechanical application to the pear scenario (Appendix A).
Oxygen, core, crust, basalt and water data	Previous observation	Relies on the measurement files of the cited studies only; there are no new empirical data produced by the author.
Non-uniform heating from a hot Earth	Previous idea — Roy and colleagues	It is strictly forbidden to attribute the physical idea itself of the influence of ancient "Earthshine" radiation (Earthshine) to the LEF origin [S5].
The Procellarum complex as a magmatic and tectonic system	Previous idea — Andrews-Hanna and colleagues	The morphological identification of the rectangular structure belongs to the cited researchers [S4]; its geodynamic interpretation as the remnant of the mechanical connection is a hypothesis of the LEF model that has not yet been tested.
Simultaneous internal feed and rise of the outer shell	Model hypothesis	The integrated mechanical combination that the series of papers is intended to test and measure quantitatively.
Unified thickness–mass–offset balance	This work — consistency test	Development of a closed analytical test on the basis of declared assumptions; it is not to be regarded as a new geodetic or gravitational discovery in science.
Carried crater record and rings as load signs	Open prediction	An independent prediction of the model requiring future identification of sorted populations and the construction of an elastic/plastic mechanical model.

- **Where the weight of the paper lies:** The table defines precisely the focus of the scientific weight of the paper: the geological observations and the physical laws are borrowed and attributed to their original source, and the main contribution of the series is the dynamic combination and integration itself — a deep feed flow and the rise of an outer shell operating together simultaneously in one body — and the definition of the statistical tests capable of deciding it. The two ideas closest to the products of the model, namely the

asymmetric heating of the young Moon by the Earth's radiation [S5] and the characterisation of Procellarum (Procellarum) as a massive system of rifts and magmatic structures [S4], belong to their original researchers; the LEF model incorporates them and interprets the structure of Procellarum as the record of the tectonic connection, a reading that requires separate empirical confirmation.

Further work — openings for future research (collection of the open tasks)

The list below collects in one defined place all the open points and residual calculations of Paper 12. For each item, the mechanical requirement it serves and the measurement protocol that will lead to its decision are defined, in order to provide a clear starting point for external researchers interested in carrying out a consistency check of the model:

1. The statistics of the concentric belts (R8)

- **Current status:** The initial calculation (the September 2026 tests) was run on the four GRAIL crustal-thickness models [S3] and on the global Robbins catalogue [S13] (tests T5 and T2 passed successfully, T1 did not pass because model 1 fell 0.35 km below the existing lower threshold line at 10 km; the recalculation of 15 October 2026 on the updated geometry preserved the ordering in both measures and yielded a difference of 9.9 km in model 1).
- **The open task:**
  - Running test T2 (crater density) on a crater population sorted strictly by stratigraphic age only (pre-Nectarian and Nectarian). The sorting is essential because the model predicts an order for the ancient record carried on the shell, whereas a catalogue including all ages mixes in late impacts that dilute the statistical contrast.
  - Running the global basin tests T3 and T4.
  - Running test T1 (crustal thickness) on models 2–4 under the updated geometry data.
  - Running the same concentric belts directly on the raw Bouguer gravity grids and on the lateral density grids of the high-degree model GRGM1200B (a high-resolution gravity model of the GRAIL mission). The Bouguer grid shows the mass anomalies beneath the surface after subtraction of the topography, and allows identification of deep density changes.

2. Unified multi-system isotopic balance (R1)

- **The open task:** Development of a mathematical model that computes the geochemical composition of a Moon built from a combination of the ancient terrestrial sial crust and deep

mantle material, which underwent processes of mechanical detachment and partial evaporation. The model is required to produce a single combined composition value, which must match simultaneously (at once) the empirical data of the isotope ratios in oxygen (O), titanium (Ti), chromium (Cr), tungsten (W) and volatile elements. Since each element records a separate physical and thermodynamic process, a single mixture solution matching all the systems at once is a significant test, considerably more rigorous than an isolated match in the oxygen system alone.

3. The budget and balance of the metallic mass (R2)

- **The open task:** Construction of a hydrodynamic model of multiphase flow (Multiphase fluid dynamics) that computes the entrainment mechanism of heavy liquid iron droplets sinking within a foamed silicate melt rising vertically, on the basis of the feed rate defined in Paper 3. The calculation will determine quantitatively the size distribution of the droplets, their hydrodynamic settling rate, and the mass fraction that succeeds in being carried upward and collecting into a small lunar core of 1.4% of the mass. The calculation is required to resolve how much of the iron (which tends gravitationally to sink back into the Earth's core) was swept mechanically by the rising flow, in agreement with the strict target that the core sets.

4. The joint solution of thickness and offset (R7)

- **The open task:** Development of a joint global inversion (Joint Inversion) of the crustal thickness values, the density model, the topography, and the full vector components of the centre of mass. The calculation will be carried out directly on the true spatial geometry of the exposure zones and not on the basis of the approximations of a one-dimensional symmetric shell. This task is a mathematical extension of Appendix A: whereas the data of the appendix showed that the order of magnitude of the offset is possible in the example, the joint solution is required to determine whether a single supported crustal density model can produce simultaneously both the size of the linear offset and its inclined vector direction in space.

5. Chronology of two crusts and two ages (R5)

- **The open task:** Carrying out a controlled chronological comparison of primary isotopic crystallisation ages (clock 1) between samples of highland crust (Highlands) on the far side and samples of highland crust on the near side, under the strict constraint of sampling outside the late impact basins. The model predicts and requires that the far-side crust be

chronologically significantly older than the near-side crust. Analysis of the sample material returned by the Chang’e-6 mission (far side) is the first empirical opportunity to carry out this test. Sampling outside the basins is essential in order to avoid material whose clocks were thermally reset by late secondary melting (Impact melt).

6. Geometric characterisation and classification of the openings

- **The open task:** Development of a quantitative sorting and classification model of the 34 mare basins on the near side, separating them into two mechanical groups: ejection openings (which were exposed directly to space and ejected returning terrestrial material) versus conduits/feeders (which transferred molten material upward beneath a sealed cover layer). The classification model will be cross-checked against three empirical data sets: mapping of fields of secondary craters (Secondary craters), measurement of the precise crustal thickness beneath each basin (GRAIL), and the geochemical composition of the earliest lava flows in each basin. Each type of opening must show a different physical signature in these three measures.

7. The sub-crustal structure in the Pacific (R11)

- **The open task:** Running a global structural and tectonic reconstruction of the Earth's deep lithosphere beneath the Pacific, based on terrestrial geophysical data only and entirely independent of the lunar map and of the LEF model. The reconstruction will compare the outlines of the estimated source region (section C.2) and the location of the predicted base of the pear (near the island of Kiritimati) with the tomographic structure, the chemical composition, and the three-dimensional outlines of the Pacific LLVP anomaly (the large low-velocity province in the lowermost mantle) as they are reflected in published global seismic models. The independence of the reconstruction is essential to ensure that the geodynamic comparison is an independent scientific control test, and not a circular repetition of the same initial morphological reading.

Paper 12 is now complete throughout, in its chapters and appendices, without any omission or abridgement of a numerical value, an equation, or a fixed coordinate.

Code availability and the reproduction package

- **Independent reproduction protocol:** All the numerical results, the statistical values (such as the Spearman correlation measures and the differences of the belt means) and the

mass-balance and offset equations detailed throughout the paper are reproduced fully and precisely by the scripts (Scripts) built into the reproduction package of the LEF model.

- **Development environment and libraries:** The source code is developed in a Python 3 environment and relies on the standard scientific libraries NumPy and SciPy for processing the data grids, computing volume integrals and running mathematical inversions.
- **How the script operates:** The software reads directly the empirical input files and the parameters declared in the body of the text, and prints (Output) the final numerical values separately for each section and test (T1–T5).
- **Archiving and access for external researchers:** The code package accompanies the series of papers as supplementary raw material for peer review and independent consistency checking, and it is stored permanently and deposited in the scientific digital repository Zenodo under the official DOI identifier:

$$\mathrm{doi{:}10.5281/zenodo.22743615}$$

This dynamic DOI address is always routed to the most recent edition of the code and version of the system.

References

External sources are marked with the letter S to distinguish them from the requirements R. The sources support the data or the physics attributed to them; they do not validate the LEF model. Sources S2, S3, S7, S10 and S11 were kept from the original manuscript; reading of their full texts was not completed in this pass, and no claim of full bibliographic verification of them is made here.

- [S1] Young, E. D., Kohl, I. E., Warren, P. H., Rubie, D. C., Jacobson, S. A., & Morbidelli, A. (2016). Oxygen isotopic evidence for vigorous mixing during the Moon-forming giant impact. *Science*, 351, 493–496. DOI: 10.1126/science.aad0525. Authors' version: arXiv:1603.04536.
- [S2] Weber, R. C., Lin, P.-Y., Garnero, E. J., Williams, Q., & Lognonné, P. (2011). Seismic detection of the lunar core. *Science*, 331, 309–312. DOI: 10.1126/science.1199375.
- [S3] Wieczorek, M. A., et al. (2013). The crust of the Moon as seen by GRAIL. *Science*, 339, 671–675. DOI: 10.1126/science.1231530.
- [S4] Andrews-Hanna, J. C., et al. (2014). Structure and evolution of the lunar Procellarum region as revealed by GRAIL gravity data. *Nature*, 514, 68–71. DOI: 10.1038/nature13697.

- [S5] Roy, A., Wright, J. T., & Sigurdsson, S. (2014). Earthshine on a young Moon: Explaining the lunar farside highlands. *The Astrophysical Journal Letters*, 788, L42. DOI: 10.1088/2041-8205/788/2/L42. Authors' version: arXiv:1406.2020.
- [S6] Murray, C. D., & Dermott, S. F. (1999). *Solar System Dynamics*. Cambridge University Press. — Source for the orbital and tidal physics, in accordance with the reference framework of Papers 2 and 4.
- [S7] Che, X., et al. (2021). Age and composition of young basalts on the Moon, measured from samples returned by Chang'e-5. *Science*, 374, 887–890. — Source for the basalt dating; not a source for the duration of the connection between the Earth and the Moon.
- [S8] Zhang, Q. W. L., et al. *Lunar farside volcanism 2.8 billion years ago from Chang'e-6 basalts*. *Nature*. Published online November 2024; volume appearance 2025. DOI: 10.1038/s41586-024-08382-0. — The citation details were corrected relative to the original manuscript; no use is made here of an age uncertainty at the level of thousands of years without reading the original dating table.
- [S9] Hauri, E. H., Weinreich, T., Saal, A. E., Rutherford, M. C., & Van Orman, J. A. (2011). High pre-eruptive water contents preserved in lunar melt inclusions. *Science*, 333, 213–215. DOI: 10.1126/science.1204626.
- [S10] Neumann, G. A., Zuber, M. T., Smith, D. E., & Lemoine, F. G. (1996). The lunar crust: Global structure and signature of major basins. *Journal of Geophysical Research*, 101, 16841–16863. — Source of the order of magnitude of the centre-of-mass offset cited in the original manuscript; a full vector check requires the definition of the centre of figure and the offset components of the chosen model.
- [S11] Kleine, T., et al. (2009). Hf–W chronology of the accretion and early evolution of asteroids and terrestrial planets. *Geochimica et Cosmochimica Acta*, 73, 5150–5188. — Source for the chronometric framework; not a confirmation of a 600-million-year construction window.
- [S12] Head, J. W., Fassett, C. I., Kadish, S. J., Smith, D. E., Zuber, M. T., Neumann, G. A., & Mazarico, E. (2010). Global distribution of large lunar craters: implications for resurfacing and impactor populations. *Science*, 329, 1504–1507. — Source for the population of craters of 20 km and larger used in T2.
- [S13] Robbins, S. J. (2019). A new global database of lunar impact craters >1–2 km: 1. Crater locations and sizes, comparisons with published databases, and global analysis. *Journal of Geophysical Research: Planets*, 124, 871–892. DOI: 10.1029/2018JE005592. — Source of the crater catalogue used in T2 (§B.8).

Series manuscripts cited: Duran, Z. Paper 1, *The Model*; Paper 2, *LEF Mechanism in Principle*; Paper 3, *LEF Mechanism in Practice*; Paper 4, *The Orbit*; Paper 5, *The Maria*; Paper 6, *The Craters*; Paper 7, *The Poles*; Paper 8, *South Pole–Aitken*; Paper 9, *The Stripes*; Paper 10, *Mars*; Paper 13, *Research topics* — October 2026. The series manuscripts are the source of the model's definition and calculations, and not external evidence of its correctness.

Declarations

Data and calculations: No new observational catalogue was created here and no new fit to the GRAIL maps was performed. The new analytical calculations, their inputs and their assumptions are given in the body of the paper and in Appendix A. The observational comparisons rest on the sources detailed.

Author: Zohar Duran, Independent Researcher, Ramla, Israel. ORCID: 0009-0005-3194-9387, according to the original manuscript.

Funding and competing interests: According to the author's declaration in the original manuscript, no public or commercial funding was received, and there is a United States provisional patent application No. 64/116,844, filed on 22 July 2026, for an engineering apparatus related to the model. This declaration is reproduced as the author's declaration; the status of the application was not checked here.

© 2026 Zohar Duran. CC BY 4.0. lef-model.com